Differential Equations Workbook Solutions

Systems of differential equations

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MATRIX BASICS

■ 1. Find $3A - (1/2)B + 2C$.

$$A = \begin{bmatrix} 0 & 1 & -\frac{1}{3} \\ 2 & \frac{1}{3} & 4 \\ \frac{2}{3} & -1 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 0 & -4 \\ -2 & 6 & 4 \\ 10 & 0 & 0 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 0 & 3 \\ -2 & 10 & 1 \\ -4 & 3 & 0 \end{bmatrix}$$

Solution:

Because all three matrices have identical dimensions (they are all 3×3 matrices), we can find the sum.

$$3A - \frac{1}{2}B + 2C = 3\begin{bmatrix} 0 & 1 & -\frac{1}{3} \\ 2 & \frac{1}{3} & 4 \\ \frac{2}{3} & -1 & 2 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} 2 & 0 & -4 \\ -2 & 6 & 4 \\ 10 & 0 & 0 \end{bmatrix} + 2\begin{bmatrix} 1 & 0 & 3 \\ -2 & 10 & 1 \\ -4 & 3 & 0 \end{bmatrix}$$

$$3A - \frac{1}{2}B + 2C = \begin{bmatrix} 0 & 3 & -1 \\ 6 & 1 & 12 \\ 2 & -3 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 0 & -2 \\ -1 & 3 & 2 \\ 5 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 6 \\ -4 & 20 & 2 \\ -8 & 6 & 0 \end{bmatrix}$$

$$3A - \frac{1}{2}B + 2C = \begin{bmatrix} 0-1+2 & 3-0+0 & -1-(-2)+6 \\ 6-(-1)+(-4) & 1-3+20 & 12-2+2 \\ 2-5+(-8) & -3-0+6 & 6-0+0 \end{bmatrix}$$

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$$3A - \frac{1}{2}B + 2C = \begin{bmatrix} 1 & 3 & 7 \\ 3 & 18 & 12 \\ -11 & 3 & 6 \end{bmatrix}$$

■ 2. For the 2×2 matrix, compute the determinant.

$$\begin{bmatrix} \frac{3}{2} & 1 \\ 2 & -6 \end{bmatrix}$$

Solution:

The determinant of the 2×2 matrix is given by

$$\begin{vmatrix} \frac{3}{2} & 1 \\ 2 & -6 \end{vmatrix} = \frac{3}{2}(-6) - 1(2)$$

$$\begin{vmatrix} \frac{3}{2} & 1 \\ 2 & -6 \end{vmatrix} = -11$$

■ 3. For the 3×3 matrix, compute the determinant.

$$\begin{bmatrix} -1 & 5 & 0 \\ \frac{1}{2} & -\frac{3}{2} & 1 \\ 4 & 0 & 2 \end{bmatrix}$$

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Solution:

The determinant of the 3×3 matrix is given by

$$\begin{vmatrix} -1 & 5 & 0 \\ \frac{1}{2} & -\frac{3}{2} & 1 \\ 4 & 0 & 2 \end{vmatrix} = - \begin{vmatrix} -\frac{3}{2} & 1 \\ 0 & 2 \end{vmatrix} - 5 \begin{vmatrix} \frac{1}{2} & 1 \\ 4 & 2 \end{vmatrix} + 0 \begin{vmatrix} \frac{1}{2} & -\frac{3}{2} \\ 4 & 0 \end{vmatrix}$$

$$\begin{vmatrix} -1 & 5 & 0 \\ \frac{1}{2} & -\frac{3}{2} & 1 \\ 4 & 0 & 2 \end{vmatrix} = - \left(-\frac{3}{2}(2) - 1(0) \right) - 5 \left(\frac{1}{2}(2) - 1(4) \right)$$

$$\begin{vmatrix} -1 & 5 & 0 \\ \frac{1}{2} & -\frac{3}{2} & 1 \\ 4 & 0 & 2 \end{vmatrix} = - (-3) - 5(1 - 4)$$

$$\begin{vmatrix} -1 & 5 & 0 \\ \frac{1}{2} & -\frac{3}{2} & 1 \\ 4 & 0 & 2 \end{vmatrix} = 18$$

4. Find the Eigenvalues and Eigenvectors of the matrix.

$$A = \begin{bmatrix} 1 & 3 \\ -1 & 5 \end{bmatrix}$$

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Solution:

Since the dimension of the matrix is 2×2, we'll use I_2 to find $A - \lambda I$.

$$A - \lambda I = \begin{bmatrix} 1 & 3 \\ -1 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 3 \\ -1 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & 3 \\ -1 & 5 - \lambda \end{bmatrix}$$

Take the determinant,

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 3 \\ -1 & 5 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (1 - \lambda)(5 - \lambda) - 3(-1)$$

$$|A - \lambda I| = 5 - 6\lambda + \lambda^2 + 3$$

$$|A - \lambda I| = \lambda^2 - 6\lambda + 8$$

then the characteristic equation is

$$\lambda^2 - 6\lambda + 8 = 0$$

$$(\lambda - 2)(\lambda - 4) = 0$$

$$\lambda = 2, 4$$

Therefore, the Eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 4$.

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For $\lambda_1 = 2$, we find

$$A - 2I = \begin{bmatrix} 1-2 & 3 \\ -1 & 5-2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} -1 & 3 \\ -1 & 3 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} -1 & 3 \\ -1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 \\ 1 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 \\ 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 - 3k_2 = 0$$

$$k_1 = 3k_2$$

If we choose $k_2 = 1$, we get $k_1 = 3$, and so the Eigenvector for $\lambda_1 = 2$ is

$$\overrightarrow{k_1} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 4$, we find

$$A - 4I = \begin{bmatrix} 1-4 & 3 \\ -1 & 5-4 \end{bmatrix}$$

$$A - 4I = \begin{bmatrix} -3 & 3 \\ -1 & 1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

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$$\begin{bmatrix} -3 & 3 \\ -1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 - k_2 = 0$$

$$k_1 = k_2$$

If we choose $k_2 = 1$, we get $k_1 = 1$, and so the Eigenvector for $\lambda_2 = 4$ is

$$\overrightarrow{k_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

5. Find the Eigenvalues and Eigenvectors of the matrix.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -4 & -4 & 0 \\ -2 & 1 & 2 \end{bmatrix}$$

Solution:

Since the dimension of the matrix is 3×3, we'll use I_3 to find $A - \lambda I$.

$$A - \lambda I = \begin{bmatrix} 0 & 1 & 0 \\ -4 & -4 & 0 \\ -2 & 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 0 & 1 & 0 \\ -4 & -4 & 0 \\ -2 & 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

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$$A-\lambda I=\begin{bmatrix}-\lambda & 1 & 0\\ -4 & -4-\lambda & 0\\ -2 & 1 & 2-\lambda\end{bmatrix}$$

Take the determinant,

$$|A-\lambda I|=\begin{vmatrix}-\lambda & 1 & 0\\ -4 & -4-\lambda & 0\\ -2 & 1 & 2-\lambda\end{vmatrix}$$

$$|A-\lambda I|=(-\lambda)\begin{vmatrix}-4-\lambda & 0\\ 1 & 2-\lambda\end{vmatrix}-1\begin{vmatrix}-4 & 0\\ -2 & 2-\lambda\end{vmatrix}+0\begin{vmatrix}-4 & -4-\lambda\\ -2 & 1\end{vmatrix}$$

$$|A-\lambda I|=-\lambda((-4-\lambda)(2-\lambda)-0(1))-1((-4)(2-\lambda)-0(-2))$$

$$|A-\lambda I|=\lambda(4+\lambda)(2-\lambda)+4(2-\lambda)$$

$$|A-\lambda I|=(2-\lambda)(\lambda^2+4\lambda+4)$$

then the characteristic equation is

$$(2-\lambda)(\lambda+2)(\lambda+2)=0$$

$$\lambda=2,\ -2,\ -2$$

Therefore, the Eigenvalues are $\lambda_1=2$ and $\lambda_2=\lambda_3=-2$.

For $\lambda_1=2$, we find

$$A-2I=\begin{bmatrix}-2 & 1 & 0\\ -4 & -4-2 & 0\\ -2 & 1 & 2-2\end{bmatrix}$$

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$$A - 2I = \begin{bmatrix} -2 & 1 & 0 \\ -4 & -6 & 0 \\ 2 & 1 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} -2 & 1 & 0 \\ -4 & -6 & 0 \\ 2 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ -4 & -6 & 0 \\ 2 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & -8 & 0 \\ 2 & 1 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & -8 & 0 \\ 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 = 0$$

$$k_2 = 0$$

If we choose $k_3 = 1$, the Eigenvector for $\lambda_1 = 2$ is

$$\overrightarrow{k_1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

For $\lambda_2 = \lambda_3 = -2$, we find

$$A - (-2)I = \begin{bmatrix} -(-2) & 1 & 0 \\ -4 & -4 - (-2) & 0 \\ -2 & 1 & 2 - (-2) \end{bmatrix}$$

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$$A-(-2)I=\begin{bmatrix}2&1&0\\-4&-2&0\\-2&1&4\end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix}2&1&0\\-4&-2&0\\-2&1&4\end{bmatrix}\rightarrow\begin{bmatrix}1&\frac{1}{2}&0\\-4&-2&0\\-2&1&4\end{bmatrix}\rightarrow\begin{bmatrix}1&\frac{1}{2}&0\\0&0&0\\-2&1&4\end{bmatrix}\rightarrow\begin{bmatrix}1&\frac{1}{2}&0\\0&0&0\\0&2&4\end{bmatrix}$$

$$\rightarrow\begin{bmatrix}1&\frac{1}{2}&0\\0&2&4\\0&0&0\end{bmatrix}\rightarrow\begin{bmatrix}1&\frac{1}{2}&0\\0&1&2\\0&0&0\end{bmatrix}\rightarrow\begin{bmatrix}1&0&-1\\0&1&2\\0&0&0\end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1-k_3=0$$

$$k_1=k_3$$

and

$$k_2+2k_3=0$$

$$k_2=-2k_3$$

If we choose $k_3=1$, we get $k_1=1$ and $k_2=-2$, and so the Eigenvector for $\lambda_2=\lambda_3=-2$ is

$$\overrightarrow{k_2}=\begin{bmatrix}1\\-2\\1\end{bmatrix}$$

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■ 6. Find the Eigenvalues and Eigenvectors of the matrix.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 0 & -9 \\ 4 & 1 & 0 \end{bmatrix}$$

Solution:

Since the dimension of the matrix is 3×3, we'll use I_3 to find $A - \lambda I$.

$$A - \lambda I = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 0 & -9 \\ 4 & 1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 0 & -9 \\ 4 & 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 0 & 0 \\ -3 & -\lambda & -9 \\ 4 & 1 & -\lambda \end{bmatrix}$$

Take the determinant,

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 0 & 0 \\ -3 & -\lambda & -9 \\ 4 & 1 & -\lambda \end{vmatrix}$$

$$|A - \lambda I| = (2-\lambda)\begin{vmatrix} -\lambda & -9 \\ 1 & -\lambda \end{vmatrix} - 0\begin{vmatrix} -3 & -9 \\ 4 & -\lambda \end{vmatrix} + 0\begin{vmatrix} -3 & -\lambda \\ 4 & 1 \end{vmatrix}$$

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$$|A - \lambda I| = (2 - \lambda)((-\lambda)(-\lambda) - (-9)(1))$$

$$|A - \lambda I| = (2 - \lambda)(\lambda^2 + 9)$$

then the characteristic equation is

$$(2 - \lambda)(\lambda^2 + 9) = 0$$

$$\lambda = 2, \ \pm 3i$$

Therefore, the Eigenvalues are $\lambda_1 = 2$ and $\lambda_{2,3} = \pm 3i$.

For $\lambda_1 = 2$, we find

$$A - 2I = \begin{bmatrix} 2-2 & 0 & 0 \\ -3 & -2 & -9 \\ 4 & 1 & -2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 0 & 0 & 0 \\ -3 & -2 & -9 \\ 4 & 1 & -2 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 0 & 0 & 0 \\ -3 & -2 & -9 \\ 4 & 1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} -3 & -2 & -9 \\ 4 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{2}{3} & 3 \\ 4 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & \frac{2}{3} & 3 \\ 0 & -\frac{5}{3} & -14 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{2}{3} & 3 \\ 0 & 1 & \frac{42}{5} \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{13}{5} \\ 0 & 1 & \frac{42}{5} \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

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$$k_1 - \frac{13}{5}k_3 = 0$$

$$k_1 = \frac{13}{5}k_3$$

and

$$k_2 + \frac{42}{5}k_3 = 0$$

$$k_2 = -\frac{42}{5}k_3$$

If we choose $k_3 = 5$, we get $k_1 = 13$ and $k_2 = -42$, and so the Eigenvector for $\lambda_1 = 2$ is

$$\overrightarrow{k_1} = \begin{bmatrix} 13 \\ -42 \\ 5 \end{bmatrix}$$

For $\lambda_2 = 3i$, we find

$$A - (3i)I = \begin{bmatrix} 2-3i & 0 & 0 \\ -3 & -3i & -9 \\ 4 & 1 & -3i \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 2-3i & 0 & 0 \\ -3 & -3i & -9 \\ 4 & 1 & -3i \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -3 & -3i & -9 \\ 4 & 1 & -3i \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3i & -9 \\ 4 & 1 & -3i \end{bmatrix}$$

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$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3i & -9 \\ 0 & 1 & -3i \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3i \\ 0 & 1 & -3i \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3i \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 = 0$$

and

$$k_2 - (3i)k_3 = 0$$

$$k_2 = 3ik_3$$

If we choose $k_3 = 1$, we get $k_2 = 3i$, and so the Eigenvector for $\lambda_2 = 3i$ is

$$\overrightarrow{k_2} = \begin{bmatrix} 0 \\ 3i \\ 1 \end{bmatrix}$$

For $\lambda_3 = -3i$, we find

$$A - (-3i)I = \begin{bmatrix} 2+3i & 0 & 0 \\ -3 & 3i & -9 \\ 4 & 1 & 3i \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 2+3i & 0 & 0 \\ -3 & 3i & -9 \\ 4 & 1 & 3i \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -3 & 3i & -9 \\ 4 & 1 & 3i \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3i & -9 \\ 4 & 1 & 3i \end{bmatrix}$$

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$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3i & -9 \\ 0 & 1 & 3i \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3i \\ 0 & 1 & 3i \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3i \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 = 0$$

and

$$k_2 + (3i)k_3 = 0$$

$$k_2 = -3ik_3$$

If we choose $k_3 = 1$, we get $k_2 = -3i$, and so the Eigenvector for $\lambda_3 = -3i$ is

$$\overrightarrow{k_2} = \begin{bmatrix} 0 \\ -3i \\ 1 \end{bmatrix}$$

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BUILDING SYSTEMS

■ 1. Rewrite the system of differential equations in matrix form.

$$x_1' = 3x_1 - 4x_3 + e^t$$

$$x_2' = 5x_2 + x_1 - 13x_3$$

$$x_3' = \sin(t^2) - 3x_1 - 2x_2$$

Solution:

Rewrite the system.

$$x_1' = 3x_1 + 0x_2 - 4x_3 + e^t$$

$$x_2' = 1x_1 + 5x_2 - 13x_3$$

$$x_3' = -3x_1 - 2x_2 + 0x_3 + \sin(t^2)$$

Since we have a nonhomogeneous system, we'll get a matrix equation in the form

$$\vec{x}' = A\vec{x} + F$$

$$\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} 3 & 0 & -4 \\ 1 & 5 & -13 \\ -3 & -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} e^t \\ 0 \\ \sin(t^2) \end{bmatrix}$$

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$$\vec{x}' = \begin{bmatrix} 3 & 0 & -4 \\ 1 & 5 & -13 \\ -3 & -2 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} e^t \\ 0 \\ \sin(t^2) \end{bmatrix}$$

■ 2. Rewrite the system of differential equations with the initial conditions in matrix form.

$$x_1' = -x_1 + 2x_3 - t^2 \qquad x_1(0) = 2$$

$$x_2' = 9x_2 - x_3 \qquad x_2(0) = -3$$

$$x_3' = 2x_2 - 4x_3 + 5\cos t \qquad x_3(0) = 0$$

Solution:

Rewrite the system.

$$x_1' = -1x_1 + 0x_2 + 2x_3 - t^2$$

$$x_2' = 0x_1 + 9x_2 - 1x_3$$

$$x_3' = 0x_1 + 2x_2 - 4x_3 + 5\cos t$$

Since we have a nonhomogeneous system, we'll get a matrix equation in the form

$$\vec{x}' = A\vec{x} + F$$

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$$\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} -1 & 0 & 2 \\ 0 & 9 & -1 \\ 0 & 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} -t^2 \\ 0 \\ 5\cos t \end{bmatrix}$$

$$\vec{x}' = \begin{bmatrix} -1 & 0 & 2 \\ 0 & 9 & -1 \\ 0 & 2 & -4 \end{bmatrix} \vec{x} + \begin{bmatrix} -t^2 \\ 0 \\ 5\cos t \end{bmatrix}$$

The given set of initial conditions can be rewritten together as

$$\vec{x}(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 0 \end{bmatrix}$$

$$\vec{x}(0) = \begin{bmatrix} 2 \\ -3 \\ 0 \end{bmatrix}$$

and so the initial value problem can also be written as

$$\vec{x}' = \begin{bmatrix} -1 & 0 & 2 \\ 0 & 9 & -1 \\ 0 & 2 & -4 \end{bmatrix} \vec{x} + \begin{bmatrix} -t^2 \\ 0 \\ 5\cos t \end{bmatrix} \qquad \vec{x}(0) = \begin{bmatrix} 2 \\ -3 \\ 0 \end{bmatrix}$$

3. Convert the fourth order linear differential equation into a system of differential equations in matrix form.

$$3y^{(4)} + 6y''' - 18y'' - 20t\cos(t^2) = e^t$$

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Solution:

Solve the differential equation for the highest order derivative, $y^{(4)}$.

$$3y^{(4)} + 6y''' - 18y'' - 20t\cos(t^2) = e^t$$

$$3y^{(4)} = e^t + 20t\cos(t^2) - 6y''' + 18y''$$

$$y^{(4)} = \frac{1}{3}e^t + \frac{20}{3}t\cos(t^2) - 2y''' + 6y''$$

Then define

$$x_1(t) = y(t) \qquad x_1'(t) = y'(t) = x_2(t)$$

$$x_2(t) = y'(t) \qquad x_2'(t) = y''(t) = x_3(t)$$

$$x_3(t) = y''(t) \qquad x_3'(t) = y'''(t) = x_4(t)$$

$$x_4(t) = y'''(t) \qquad x_4'(t) = y^{(4)}(t) = \frac{1}{3}e^t + \frac{20}{3}t\cos(t^2) - 2y''' + 6y''$$

$$= 6x_3 - 2x_4 + \frac{20}{3}t\cos(t^2) + \frac{1}{3}e^t$$

Rewriting the nonhomogeneous system as a matrix equation gives

$$\vec{x}' = A\vec{x} + F$$

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 6 & -2 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{20}{3}t\cos(t^2) + \frac{1}{3}e^t \end{bmatrix}$$

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■ 4. Convert the given initial value problem into a system of differential equations and rewrite it as a matrix equation.

$$y''' - 4ty' + 7y - t^2y'' = 2t^3 - 4$$

$$y(0) = -2,\ y'(0) = 1,\ y''(0) = 0$$

Solution:

Solve the differential equation for the highest order derivative, y'''.

$$y''' - t^2y'' - 4ty' + 7y = 2t^3 - 4$$

$$y''' = -7y + 4ty' + t^2y'' + 2t^3 - 4$$

Then define

$$x_1(t) = y(t) \qquad x_1'(t) = y'(t) = x_2(t)$$

$$x_2(t) = y'(t) \qquad x_2'(t) = y''(t) = x_3(t)$$

$$x_3(t) = y''(t) \qquad x_3'(t) = y'''(t) = -7y + 4ty' + t^2y'' + 2t^3 - 4$$

$$= -7x_1 + 4tx_2 + t^2x_3 + 2t^3 - 4$$

Rewriting the nonhomogeneous system as a matrix equation gives

$$\vec{x}' = A\vec{x} + F$$

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$$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -7 & 4t & t^2 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ 2t^3 - 4 \end{bmatrix}$$

Rewrite the given set of initial conditions,

$$x_1(0) = y(0) = -2$$

$$x_2(0) = y'(0) = 1$$

$$x_3(0) = y''(0) = 0$$

and then the initial value problem can be written as

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -7 & 4t & t^2 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ 2t^3 - 4 \end{bmatrix} \qquad \vec{x}(0) = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

■ 5. Convert the third order linear differential equation into a system of differential equations in matrix form.

$$e^t y''' - 5y'' + y' - t^2 y + e^{2t} = 0$$

Solution:

Solve the differential equation for the highest order derivative, y'''.

$$e^t y''' - 5y'' + y' - t^2 y + e^{2t} = 0$$

$$e^t y''' = t^2 y - y' + 5y'' - e^{2t}$$

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$$y''' = t^2e^{-t}y - e^{-t}y' + 5e^{-t}y'' - e^t$$

Then define

$$x_1(t) = y(t) \qquad x_1'(t) = y'(t) = x_2(t)$$

$$x_2(t) = y'(t) \qquad x_2'(t) = y''(t) = x_3(t)$$

$$x_3(t) = y''(t) \qquad x_3'(t) = y'''(t) = t^2e^{-t}y - e^{-t}y' + 5e^{-t}y'' - e^t$$

$$= t^2e^{-t}x_1 - e^{-t}x_2 + 5e^{-t}x_3 - e^t$$

Rewriting the nonhomogeneous system as a matrix equation gives

$$\vec{x}' = A\vec{x} + F$$

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ t^2e^{-t} & -e^{-t} & 5e^{-t} \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ -e^t \end{bmatrix}$$

■ 6. Convert the fourth order linear differential equation with the given initial conditions into a system of differential equations in matrix form.

$$y^{(4)} - (\cos t)y''' + 3e^ty'' - 2y = 0$$

$$y(0) = 3,\ y'(0) = 1,\ y''(0) = -2,\ y'''(0) = 0$$

Solution:

Solve the differential equation for the highest order derivative, $y^{(4)}$.

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$$y^{(4)} - (\cos t)y''' + 3e^t y'' - 2y = 0$$

$$y^{(4)} = 2y - 3e^t y'' + (\cos t)y'''$$

Then define

$$x_1(t) = y(t) \qquad x_1'(t) = y'(t) = x_2(t)$$

$$x_2(t) = y'(t) \qquad x_2'(t) = y''(t) = x_3(t)$$

$$x_3(t) = y''(t) \qquad x_3'(t) = y'''(t) = x_4(t)$$

$$x_4(t) = y'''(t) \qquad x_4'(t) = y^{(4)}(t) = 2y - 3e^t y'' + (\cos t)y'''$$

$$= 2x_1 - 3e^t x_3 + (\cos t)x_4$$

Rewriting the nonhomogeneous system as a matrix equation gives

$$\vec{x}' = A\vec{x} + F$$

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 0 & -3e^t & \cos t \end{bmatrix} \vec{x}$$

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SOLVING SYSTEMS

■ 1. Show that the vector is a solution to the system.

$$x_1' = x_1 + 3x_2 - e^{2t}$$

$$x_2' = 3x_1 + x_2$$

$$\overrightarrow{x_1} = \begin{bmatrix} -3 \\ 3 \end{bmatrix} e^{-2t} + \begin{bmatrix} \frac{1}{8} \\ \frac{3}{8} \end{bmatrix} e^{2t}$$

Solution:

Rewrite the system in matrix form.

$$\overrightarrow{x} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \overrightarrow{x} + \begin{bmatrix} -e^{2t} \\ 0 \end{bmatrix}$$

Rewrite the solution vector as

$$\overrightarrow{x_1} = \begin{bmatrix} -3e^{-2t} \\ 3e^{-2t} \end{bmatrix} + \begin{bmatrix} \frac{1}{8}e^{2t} \\ \frac{3}{8}e^{2t} \end{bmatrix}$$

$$\overrightarrow{x_1} = \begin{bmatrix} -3e^{-2t} + \frac{1}{8}e^{2t} \\ 3e^{-2t} + \frac{3}{8}e^{2t} \end{bmatrix}$$

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The derivative of the solution vector is

$$\vec{x_1}' = \begin{bmatrix} 6e^{-2t} + \frac{1}{4}e^{2t} \\ -6e^{-2t} + \frac{3}{4}e^{2t} \end{bmatrix}$$

Plug the solution and its derivative into the matrix equation.

$$\vec{x} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} -e^{2t} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 6e^{-2t} + \frac{1}{4}e^{2t} \\ -6e^{-2t} + \frac{3}{4}e^{2t} \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -3e^{-2t} + \frac{1}{8}e^{2t} \\ 3e^{-2t} + \frac{3}{8}e^{2t} \end{bmatrix} + \begin{bmatrix} -e^{2t} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 6e^{-2t} + \frac{1}{4}e^{2t} \\ -6e^{-2t} + \frac{3}{4}e^{2t} \end{bmatrix} = \begin{bmatrix} 1\left(-3e^{-2t} + \frac{1}{8}e^{2t}\right) + 3\left(3e^{-2t} + \frac{3}{8}e^{2t}\right) \\ 3\left(-3e^{-2t} + \frac{1}{8}e^{2t}\right) + 1\left(3e^{-2t} + \frac{3}{8}e^{2t}\right) \end{bmatrix} + \begin{bmatrix} -e^{2t} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 6e^{-2t} + \frac{1}{4}e^{2t} \\ -6e^{-2t} + \frac{3}{4}e^{2t} \end{bmatrix} = \begin{bmatrix} 6e^{-2t} + \frac{5}{4}e^{2t} \\ -6e^{-2t} + \frac{3}{4}e^{2t} \end{bmatrix} + \begin{bmatrix} -e^{2t} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 6e^{-2t} + \frac{1}{4}e^{2t} \\ -6e^{-2t} + \frac{3}{4}e^{2t} \end{bmatrix} = \begin{bmatrix} 6e^{-2t} + \frac{1}{4}e^{2t} \\ -6e^{-2t} + \frac{3}{4}e^{2t} \end{bmatrix}$$

Because we found equivalent vectors on both sides of the equation, we've shown that the vector $\vec{x_1}$ is a solution to the system of differential equations.

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■ 2. Show that the vector is a solution to the system.

$$\vec{x}' = \begin{bmatrix} 2 & 1 & 1 \\ -2 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} \vec{x}$$

$$\vec{x_1} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} e^t + \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} e^{2t}$$

Solution:

Rewrite the solution vector as

$$\vec{x_1} = \begin{bmatrix} e^{2t} \\ -e^t - 2e^{2t} \\ e^t + 2e^{2t} \end{bmatrix}$$

The derivative of the solution vector is

$$\vec{x_1}' = \begin{bmatrix} 2e^{2t} \\ -e^t - 4e^{2t} \\ e^t + 4e^{2t} \end{bmatrix}$$

Plug the solution and its derivative into the matrix equation.

$$\vec{x}' = \begin{bmatrix} 2 & 1 & 1 \\ -2 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} \vec{x}$$

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$$\begin{bmatrix} 2e^{2t} \\ -e^t - 4e^{2t} \\ e^t + 4e^{2t} \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ -2 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} e^{2t} \\ -e^t - 2e^{2t} \\ e^t + 2e^{2t} \end{bmatrix}$$

$$\begin{bmatrix} 2e^{2t} \\ -e^t - 4e^{2t} \\ e^t + 4e^{2t} \end{bmatrix} = \begin{bmatrix} 2e^{2t} - e^t - 2e^{2t} + e^t + 2e^{2t} \\ -2e^{2t} - e^t - 2e^{2t} \\ 2e^{2t} - e^t - 2e^{2t} + 2e^t + 4e^{2t} \end{bmatrix}$$

$$\begin{bmatrix} 2e^{2t} \\ -e^t - 4e^{2t} \\ e^t + 4e^{2t} \end{bmatrix} = \begin{bmatrix} 2e^{2t} \\ -e^t - 4e^{2t} \\ e^t + 4e^{2t} \end{bmatrix}$$

Because we found equivalent vectors on both sides of the equation, we've shown that the vector $\vec{x_1}$ is a solution to the system of differential equations.

■ 3. Confirm that the vectors are linearly independent.

$$\vec{x_1} = \begin{bmatrix} 3 \\ -3 \\ 7 \end{bmatrix} e^{3t} \qquad \vec{x_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{6t} \qquad \vec{x_3} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{7t}$$

Solution:

Find the Wronskian of the vector set.

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$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = \begin{vmatrix} 3e^{3t} & 0 & e^{7t} \\ -3e^{3t} & 0 & e^{7t} \\ 7e^{3t} & e^{6t} & e^{7t} \end{vmatrix}$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = 3e^{3t} \begin{vmatrix} 0 & e^{7t} \\ e^{6t} & e^{7t} \end{vmatrix} - 0 \begin{vmatrix} -3e^{3t} & e^{7t} \\ 7e^{3t} & e^{7t} \end{vmatrix} + e^{7t} \begin{vmatrix} -3e^{3t} & 0 \\ 7e^{3t} & e^{6t} \end{vmatrix}$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = 3e^{3t}(0 - e^{13t}) + e^{7t}(-3e^{9t} - 0)$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = -3e^{16t} - 3e^{16t}$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = -6e^{16t}$$

Because $e^{16t} \neq 0$ for any t, the Wronskian is non-zero, which means that the vector set is linearly independent.

■ 4. Verify that the solution vectors satisfy the system of differential equations. Calculate the Wronskian of the solution vectors and write the general solution of the homogeneous system of differential equations.

$$\overrightarrow{x}' = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \overrightarrow{x}$$

$$\{\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{t}, \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} e^{-t}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} e^{2t} \right\}$$

Solution:

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We can verify that each of the vectors $\{\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}\}$ are solutions by substituting them into the matrix equation. First though, we'll need the derivative of each solution vector.

$$\overrightarrow{x_1} = \begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix} \qquad \overrightarrow{x_1}' = \begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix}$$

$$\overrightarrow{x_2} = \begin{bmatrix} -e^{-t} \\ -2e^{-t} \\ e^{-t} \end{bmatrix} \qquad \overrightarrow{x_2}' = \begin{bmatrix} e^{-t} \\ 2e^{-t} \\ -e^{-t} \end{bmatrix}$$

$$\overrightarrow{x_3} = \begin{bmatrix} -e^{2t} \\ e^{2t} \\ e^{2t} \end{bmatrix} \qquad \overrightarrow{x_3}' = \begin{bmatrix} -2e^{2t} \\ 2e^{2t} \\ 2e^{2t} \end{bmatrix}$$

Substitute $\overrightarrow{x_1}$ and its derivative $\overrightarrow{x_1}'$ into the matrix equation.

$$\begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix}$$

$$\begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix} = \begin{bmatrix} e^t + 0 + 0 \\ -e^t + 0 + e^t \\ 0 + 0 + e^t \end{bmatrix}$$

$$\begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix} = \begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix}$$

Substitute $\overrightarrow{x_2}$ and its derivative $\overrightarrow{x_2}'$ into the matrix equation.

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$$\begin{bmatrix} e^{-t} \\ 2e^{-t} \\ -e^{-t} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -e^{-t} \\ -2e^{-t} \\ e^{-t} \end{bmatrix}$$

$$\begin{bmatrix} e^{-t} \\ 2e^{-t} \\ -e^{-t} \end{bmatrix} = \begin{bmatrix} -e^{-t} + 2e^{-t} + 0 \\ e^{-t} + 0 + e^{-t} \\ 0 - 2e^{-t} + e^{-t} \end{bmatrix}$$

$$\begin{bmatrix} e^{-t} \\ 2e^{-t} \\ -e^{-t} \end{bmatrix} = \begin{bmatrix} e^{-t} \\ 2e^{-t} \\ -e^{-t} \end{bmatrix}$$

Substitute $\overrightarrow{x_3}$ and its derivative $\overrightarrow{x_3}'$ into the matrix equation.

$$\begin{bmatrix} -2e^{2t} \\ 2e^{2t} \\ 2e^{2t} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -e^{2t} \\ e^{2t} \\ e^{2t} \end{bmatrix}$$

$$\begin{bmatrix} -2e^{2t} \\ 2e^{2t} \\ 2e^{2t} \end{bmatrix} = \begin{bmatrix} -e^{2t} - e^{2t} + 0 \\ e^{2t} + 0 + e^{2t} \\ 0 + e^{2t} + e^{2t} \end{bmatrix}$$

$$\begin{bmatrix} -2e^{2t} \\ 2e^{2t} \\ 2e^{2t} \end{bmatrix} = \begin{bmatrix} -2e^{2t} \\ 2e^{2t} \\ 2e^{2t} \end{bmatrix}$$

Because we found equivalent vectors on both sides of the equation in each of these three cases, we've shown that all three vectors are solutions to the system of differential equations.

Now let's take the Wronskian of the solution set.

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$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = \begin{vmatrix} e^t & -e^{-t} & -e^{2t} \\ 0 & -2e^{-t} & e^{2t} \\ e^t & e^{-t} & e^{2t} \end{vmatrix}$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = e^t \begin{vmatrix} -2e^{-t} & e^{2t} \\ e^{-t} & e^{2t} \end{vmatrix} - (-e^{-t}) \begin{vmatrix} 0 & e^{2t} \\ e^t & e^{2t} \end{vmatrix} + (-e^{2t}) \begin{vmatrix} 0 & -2e^{-t} \\ e^t & e^{-t} \end{vmatrix}$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = e^t(-2e^{-t}e^{2t} - e^{2t}e^{-t}) + e^{-t}(0 - e^{2t}e^t) - e^{2t}(0 + 2e^{-t}e^t)$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = -2e^te^{-t}e^{2t} - e^te^{2t}e^{-t} - e^{-t}e^{2t}e^t - 2e^{2t}e^{-t}e^t$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = -6e^{2t}$$

Because the Wronskian is non-zero, we can confirm that the vectors in the solution set are linearly independent, which means that the vector set represents a fundamental set of solutions.

Given that this is a fundamental set of solutions, we can write the general solution of the system as

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} e^{2t}$$

■ 5. Verify that the solution vectors satisfy the system of differential equations. Calculate the Wronskian of the solution vectors and write the general solution of the homogeneous system of differential equations.

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$$\vec{x}' = \begin{bmatrix} 6 & 5 \\ -5 & 0 \end{bmatrix} \vec{x}$$

$$\{\vec{x_1}, \vec{x_2}\} = \left\{ \begin{bmatrix} -3\cos(4t) + 4\sin(4t) \\ 5\cos(4t) \end{bmatrix} e^{3t}, \begin{bmatrix} -4\cos(4t) - 3\sin(4t) \\ 5\sin(4t) \end{bmatrix} e^{3t} \right\}$$

Solution:

We can verify that each of the vectors $\{\vec{x_1}, \vec{x_2}\}$ are solutions by substituting them into the matrix equation. First though, we'll need the derivative of each solution vector.

$$\vec{x_1} = \begin{bmatrix} e^{3t}(-3\cos(4t) + 4\sin(4t)) \\ 5e^{3t}\cos(4t) \end{bmatrix} \qquad \vec{x_1}' = \begin{bmatrix} e^{3t}(7\cos(4t) + 24\sin(4t)) \\ 5e^{3t}(3\cos(4t) - 4\sin(4t)) \end{bmatrix}$$

$$\vec{x_2} = \begin{bmatrix} e^{3t}(-4\cos(4t) - 3\sin(4t)) \\ 5e^{3t}\sin(4t) \end{bmatrix} \qquad \vec{x_2}' = \begin{bmatrix} e^{3t}(-24\cos(4t) + 7\sin(4t)) \\ 5e^{3t}(3\sin(4t) + 4\cos(4t)) \end{bmatrix}$$

Substitute $\vec{x_1}$ and its derivative $\vec{x_1}'$ into the matrix equation.

$$\begin{bmatrix} e^{3t}(7\cos(4t) + 24\sin(4t)) \\ 5e^{3t}(3\cos(4t) - 4\sin(4t)) \end{bmatrix} = \begin{bmatrix} 6 & 5 \\ -5 & 0 \end{bmatrix} \begin{bmatrix} e^{3t}(-3\cos(4t) + 4\sin(4t)) \\ 5e^{3t}\cos(4t) \end{bmatrix}$$

$$\begin{bmatrix} e^{3t}(7\cos(4t) + 24\sin(4t)) \\ 5e^{3t}(3\cos(4t) - 4\sin(4t)) \end{bmatrix} = \begin{bmatrix} e^{3t}(7\cos(4t) + 24\sin(4t)) \\ 5e^{3t}(3\cos(4t) - 4\sin(4t)) \end{bmatrix}$$

Substitute $\vec{x_2}$ and its derivative $\vec{x_2}'$ into the matrix equation.

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$$\begin{bmatrix} e^{3t}(-24\cos(4t)+7\sin(4t)) \\ 5e^{3t}(3\sin(4t)+4\cos(4t)) \end{bmatrix} = \begin{bmatrix} 6 & 5 \\ -5 & 0 \end{bmatrix} \begin{bmatrix} e^{3t}(-4\cos(4t)-3\sin(4t)) \\ 5e^{3t}\sin(4t) \end{bmatrix}$$

$$\begin{bmatrix} e^{3t}(-24\cos(4t)+7\sin(4t)) \\ 5e^{3t}(3\sin(4t)+4\cos(4t)) \end{bmatrix} = \begin{bmatrix} e^{3t}(-24\cos(4t)+7\sin(4t)) \\ 5e^{3t}(4\cos(4t)+3\sin(4t)) \end{bmatrix}$$

Because we found equivalent vectors on both sides of the equation in each of these two cases, we've shown that both vectors are solutions to the system of differential equations.

Now let's take the Wronskian of the solution set.

$$W(\overrightarrow{x_1},\overrightarrow{x_2}) = \begin{vmatrix} e^{3t}(-3\cos(4t)+4\sin(4t)) & e^{3t}(-4\cos(4t)-3\sin(4t)) \\ 5e^{3t}\cos(4t) & 5e^{3t}\sin(4t) \end{vmatrix}$$

$$W(\overrightarrow{x_1},\overrightarrow{x_2}) = 5e^{6t}\sin(4t)(-3\cos(4t)+4\sin(4t)) - 5e^{6t}\cos(4t)(-4\cos(4t)-3\sin(4t))$$

$$W(\overrightarrow{x_1},\overrightarrow{x_2}) = 5e^{6t}(-3\cos(4t)\sin(4t)+4\sin^2(4t)+4\cos^2(4t)+3\cos(4t)\sin(4t))$$

$$W(\overrightarrow{x_1},\overrightarrow{x_2}) = 20e^{6t}(\sin^2(4t)+\cos^2(4t))$$

Since $\sin^2(4t)+\cos^2(4t)=1$, we get

$$W(\overrightarrow{x_1},\overrightarrow{x_2}) = 20e^{6t}$$

Because the Wronskian is non-zero, we can confirm that the vectors in the solution set are linearly independent, which means that the vector set represents a fundamental set of solutions.

Given that this is a fundamental set of solutions, we can write the general solution of the system as

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$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2}$$

$$\vec{x} = c_1\begin{bmatrix} -3\cos(4t) + 4\sin(4t) \\ 5\cos(4t) \end{bmatrix} e^{3t} + c_2\begin{bmatrix} -4\cos(4t) - 3\sin(4t) \\ 5\sin(4t) \end{bmatrix} e^{3t}$$

■ 6. Verify that the solution vectors satisfy the system of differential equations. Calculate the Wronskian of the solution vectors and write the general solution of the homogeneous system of differential equations.

$$\vec{x}' = \begin{bmatrix} 5 & -1 & -1 \\ 0 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix} \vec{x}$$

$$\{\vec{x_1}, \vec{x_2}, \vec{x_3}\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{3t}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{5t}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} e^{5t} \right\}$$

Solution:

We can verify that each of the vectors $\{\vec{x_1}, \vec{x_2}, \vec{x_3}\}$ are solutions by substituting them into the matrix equation. First though, we'll need the derivative of each solution vector.

$$\vec{x_1} = \begin{bmatrix} e^{3t} \\ e^{3t} \\ e^{3t} \end{bmatrix} \qquad \vec{x_1}' = \begin{bmatrix} 3e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix}$$

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$$\overrightarrow{x_2} = \begin{bmatrix} e^{5t} \\ 0 \\ 0 \end{bmatrix} \qquad \overrightarrow{x_2}' = \begin{bmatrix} 5e^{5t} \\ 0 \\ 0 \end{bmatrix}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 0 \\ -e^{5t} \\ e^{5t} \end{bmatrix} \qquad \overrightarrow{x_3}' = \begin{bmatrix} 0 \\ -5e^{5t} \\ 5e^{5t} \end{bmatrix}$$

Substitute $\overrightarrow{x_1}$ and its derivative $\overrightarrow{x_1}'$ into the matrix equation.

$$\begin{bmatrix} 3e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix} = \begin{bmatrix} 5 & -1 & -1 \\ 0 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} e^{3t} \\ e^{3t} \\ e^{3t} \end{bmatrix}$$

$$\begin{bmatrix} 3e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix} = \begin{bmatrix} 5e^{3t} - e^{3t} - e^{3t} \\ 4e^{3t} - e^{3t} \\ -e^{3t} + 4e^{3t} \end{bmatrix}$$

$$\begin{bmatrix} 3e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix} = \begin{bmatrix} 3e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix}$$

Substitute $\overrightarrow{x_2}$ and its derivative $\overrightarrow{x_2}'$ into the matrix equation.

$$\begin{bmatrix} 5e^{5t} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 & -1 & -1 \\ 0 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} e^{5t} \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 5e^{5t} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 5e^{5t} - 0 - 0 \\ 0 + 0 - 0 \\ 0 - 0 + 0 \end{bmatrix}$$

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$$\begin{bmatrix} 5e^{5t} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 5e^{5t} \\ 0 \\ 0 \end{bmatrix}$$

Substitute $\overrightarrow{x_3}$ and its derivative $\overrightarrow{x_3}'$ into the matrix equation.

$$\begin{bmatrix} 0 \\ -5e^{5t} \\ 5e^{5t} \end{bmatrix} = \begin{bmatrix} 5 & -1 & -1 \\ 0 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ -e^{5t} \\ e^{5t} \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ -5e^{5t} \\ 5e^{5t} \end{bmatrix} = \begin{bmatrix} e^{5t} - e^{5t} \\ -4e^{5t} - e^{5t} \\ e^{5t} + 4e^{5t} \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ -5e^{5t} \\ 5e^{5t} \end{bmatrix} = \begin{bmatrix} 0 \\ -5e^{5t} \\ 5e^{5t} \end{bmatrix}$$

Because we found equivalent vectors on both sides of the equation in each of these three cases, we've shown that all three vectors are solutions to the system of differential equations.

Now let's take the Wronskian of the solution set.

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = \begin{vmatrix} e^{3t} & e^{5t} & 0 \\ e^{3t} & 0 & -e^{5t} \\ e^{3t} & 0 & e^{5t} \end{vmatrix}$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = e^{3t} \begin{vmatrix} 0 & -e^{5t} \\ 0 & e^{5t} \end{vmatrix} - e^{5t} \begin{vmatrix} e^{3t} & -e^{5t} \\ e^{3t} & e^{5t} \end{vmatrix} + 0 \begin{vmatrix} e^{3t} & 0 \\ e^{3t} & 0 \end{vmatrix}$$

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$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = e^{3t}(0-0) - e^{5t}(e^{3t}e^{5t} + e^{5t}e^{3t})$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = -2e^{13t}$$

Because the Wronskian is non-zero, we can confirm that the vectors in the solution set are linearly independent, which means that the vector set represents a fundamental set of solutions.

Given that this is a fundamental set of solutions, we can write the general solution of the system as

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{5t} + c_3 \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} e^{5t}$$

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DISTINCT REAL EIGENVALUES

■ 1. Solve the system of differential equations.

$$x_1' = 4x_1 + 8x_2$$

$$x_2' = x_1 + 2x_2$$

Solution:

The coefficient matrix is

$$A = \begin{bmatrix} 4 & 8 \\ 1 & 2 \end{bmatrix}$$

and the matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 4 & 8 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 4 - \lambda & 8 \\ 1 & 2 - \lambda \end{bmatrix}$$

Then the determinant is

$$|A - \lambda I| = \begin{vmatrix} 4 - \lambda & 8 \\ 1 & 2 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (4 - \lambda)(2 - \lambda) - (8)(1)$$

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$$|A - \lambda I| = 8 - 6\lambda + \lambda^2 - 8$$

$$|A - \lambda I| = \lambda^2 - 6\lambda$$

and the characteristic equation is

$$\lambda^2 - 6\lambda = 0$$

$$\lambda(\lambda - 6) = 0$$

$$\lambda = 0,\ 6$$

Then for these Eigenvalues, $\lambda_1 = 0$ and $\lambda_2 = 6$, we find

$$A - (0)I = \begin{bmatrix} 4 - 0 & 8 \\ 1 & 2 - 0 \end{bmatrix}$$

$$A - (0)I = \begin{bmatrix} 4 & 8 \\ 1 & 2 \end{bmatrix}$$

and

$$A - 6I = \begin{bmatrix} 4 - 6 & 8 \\ 1 & 2 - 6 \end{bmatrix}$$

$$A - 6I = \begin{bmatrix} -2 & 8 \\ 1 & -4 \end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix} 4 & 8 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 8 \\ 1 & -4 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \qquad \begin{bmatrix} 1 & -4 \\ 1 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & -4 \\ 0 & 0 \end{bmatrix}$$

If we turn these back into systems of equations (in both cases we only need to consider the equation that we get from the first row of each matrix), we get

$$k_1 + 2k_2 = 0 \qquad k_1 - 4k_2 = 0$$

$$k_1 = -2k_2 \qquad k_1 = 4k_2$$

From the first system, we'll choose $k_2 = -1$, which results in $k_1 = 2$. And from the second system, we'll choose $k_2 = 1$, which results in $k_1 = 4$.

$$\overrightarrow{k_1} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

Then the solutions to the system are

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} e^{0t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} e^{6t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Therefore, the general solution to the homogeneous system must be

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$$

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$$\vec{x} = c_1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 4 \\ 1 \end{bmatrix} e^{6t}$$

■ 2. Solve the system of differential equations.

$$x_1' = 7x_1 + 6x_2 + 3x_3$$

$$x_2' = 2x_2 - x_1 + x_3$$

$$x_3' = x_1 + 2x_2 + x_3$$

Solution:

The coefficient matrix is

$$A = \begin{bmatrix} 7 & 6 & 3 \\ -1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

and the matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 7 & 6 & 3 \\ -1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 7 & 6 & 3 \\ -1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

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$$A - \lambda I = \begin{bmatrix} 7-\lambda & 6 & 3 \\ -1 & 2-\lambda & 1 \\ 1 & 2 & 1-\lambda \end{bmatrix}$$

Then the determinant is

$$|A - \lambda I| = \begin{vmatrix} 7-\lambda & 6 & 3 \\ -1 & 2-\lambda & 1 \\ 1 & 2 & 1-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (7-\lambda)\begin{vmatrix} 2-\lambda & 1 \\ 2 & 1-\lambda \end{vmatrix} - 6\begin{vmatrix} -1 & 1 \\ 1 & 1-\lambda \end{vmatrix} + 3\begin{vmatrix} -1 & 2-\lambda \\ 1 & 2 \end{vmatrix}$$

$$|A - \lambda I| = (7-\lambda)((2-\lambda)(1-\lambda) - 1(2)) - 6((-1)(1-\lambda) - 1(1))$$

$$+3((-1)(2) - (2-\lambda)(1))$$

$$|A - \lambda I| = (7-\lambda)(2 - 3\lambda + \lambda^2 - 2) - 6(\lambda - 1 - 1) + 3(-2 - 2 + \lambda)$$

$$|A - \lambda I| = (7-\lambda)\lambda(\lambda - 3) - 6(\lambda - 2) + 3(\lambda - 4)$$

$$|A - \lambda I| = \lambda(7-\lambda)(\lambda - 3) - 6\lambda + 12 + 3\lambda - 12$$

$$|A - \lambda I| = \lambda((7-\lambda)(\lambda - 3) - 3)$$

$$|A - \lambda I| = \lambda(7\lambda - 21 - \lambda^2 + 3\lambda - 3)$$

$$|A - \lambda I| = \lambda(-\lambda^2 + 10\lambda - 24)$$

and the characteristic equation is

$$\lambda(-\lambda^2 + 10\lambda - 24) = 0$$

$$\lambda_1 = 0 \qquad\qquad -\lambda^2 + 10\lambda - 24 = 0$$

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$$\lambda^2 - 10\lambda + 24 = 0$$

$$(\lambda - 6)(\lambda - 4) = 0$$

$$\lambda_{2,3} = 6,\ 4$$

For the Eigenvalue $\lambda_1 = 0$ we find

$$A - (0)I = \begin{bmatrix} 7-0 & 6 & 3 \\ -1 & 2-0 & 1 \\ 1 & 2 & 1-0 \end{bmatrix}$$

$$A - (0)I = \begin{bmatrix} 7 & 6 & 3 \\ -1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 2 & 1 \\ -1 & 2 & 1 \\ 7 & 6 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 2 \\ 7 & 6 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 4 & 2 \\ 0 & -8 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & \frac{1}{2} \\ 0 & -8 & -4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & -8 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this back into a system of equations, we get

$$k_1 = 0$$

and

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$$k_2 + \frac{1}{2}k_3 = 0$$

$$k_2 = -\frac{1}{2}k_3$$

We'll choose $k_3 = 2$, which results in $k_2 = -1$, and so we get the Eigenvalue

$$\overrightarrow{k_1} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

and therefore the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} e^{0t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

For the Eigenvalue $\lambda_2 = 6$ we find

$$A - 6I = \begin{bmatrix} 7-6 & 6 & 3 \\ -1 & 2-6 & 1 \\ 1 & 2 & 1-6 \end{bmatrix}$$

$$A - 6I = \begin{bmatrix} 1 & 6 & 3 \\ -1 & -4 & 1 \\ 1 & 2 & -5 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

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$$\begin{bmatrix} 1 & 6 & 3 \\ 0 & 2 & 4 \\ 1 & 2 & -5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6 & 3 \\ 0 & 2 & 4 \\ 0 & -4 & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6 & 3 \\ 0 & 1 & 2 \\ 0 & -4 & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -9 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this back into a system of equations, we get

$$k_1 - 9k_3 = 0$$

$$k_1 = 9k_3$$

and

$$k_2 + 2k_3 = 0$$

$$k_2 = -2k_3$$

We'll choose $k_3 = 1$, which results in $k_1 = 9$ and $k_2 = -2$, and so we get the Eigenvector

$$\overrightarrow{k_2} = \begin{bmatrix} 9 \\ -2 \\ 1 \end{bmatrix}$$

and therefore the solution vector

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 9 \\ -2 \\ 1 \end{bmatrix} e^{6t}$$

For the Eigenvalue $\lambda_3 = 4$ we find

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$$A - 4I = \begin{bmatrix} 7-4 & 6 & 3 \\ -1 & 2-4 & 1 \\ 1 & 2 & 1-4 \end{bmatrix}$$

$$A - 4I = \begin{bmatrix} 3 & 6 & 3 \\ -1 & -2 & 1 \\ 1 & 2 & -3 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 2 & 1 \\ -1 & -2 & 1 \\ 1 & 2 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & -4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this back into a system of equations, we get

$$k_1 + 2k_2 = 0$$

$$k_1 = -2k_2$$

and

$$k_3 = 0$$

We'll choose $k_2 = -1$, which results in $k_1 = 2$, and so we get the Eigenvector

$$\overrightarrow{k_3} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$$

and therefore the solution vector

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$$\overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} e^{4t}$$

Therefore, the general solution to the homogeneous system must be

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 9 \\ -2 \\ 1 \end{bmatrix} e^{6t} + c_3 \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} e^{4t}$$

■ 3. Solve the system of differential equations, given $x_1(0) = -2$ and $x_2(0) = 19$.

$$x_1' = x_1 + 2x_2$$

$$x_2' = 5x_1 - 2x_2$$

Solution:

The coefficient matrix is

$$A = \begin{bmatrix} 1 & 2 \\ 5 & -2 \end{bmatrix}$$

and the matrix $A - \lambda I$ is

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$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ 5 & -2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & 2 \\ 5 & -2 - \lambda \end{bmatrix}$$

Then the determinant is

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 2 \\ 5 & -2 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (1 - \lambda)(-2 - \lambda) - (2)(5)$$

$$|A - \lambda I| = -2 + \lambda + \lambda^2 - 10$$

$$|A - \lambda I| = \lambda^2 + \lambda - 12$$

and the characteristic equation is

$$\lambda^2 + \lambda - 12 = 0$$

$$(\lambda - 3)(\lambda + 4) = 0$$

$$\lambda_1 = 3 \qquad \lambda_2 = -4$$

For the Eigenvalues $\lambda_1 = 3$ and $\lambda_2 = -4$, we find

$$A - 3I = \begin{bmatrix} 1 - 3 & 2 \\ 5 & -2 - 3 \end{bmatrix}$$

$$A - 3I = \begin{bmatrix} -2 & 2 \\ 5 & -5 \end{bmatrix}$$

and

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$$A-(-4)I=\begin{bmatrix}1+4 & 2\\ 5 & -2+4\end{bmatrix}$$

$$A-(-4)I=\begin{bmatrix}5 & 2\\ 5 & 2\end{bmatrix}$$

Put these matrices into reduced row-echelon form.

$$\begin{bmatrix}-2 & 2\\ 5 & -5\end{bmatrix}$$

$$\begin{bmatrix}5 & 2\\ 5 & 2\end{bmatrix}$$

$$\begin{bmatrix}1 & -1\\ 5 & -5\end{bmatrix}$$

$$\begin{bmatrix}5 & 2\\ 0 & 0\end{bmatrix}$$

$$\begin{bmatrix}1 & -1\\ 0 & 0\end{bmatrix}$$

$$\begin{bmatrix}1 & \frac{2}{5}\\ 0 & 0\end{bmatrix}$$

If we turn these back into systems of equations, we get

$$k_1-k_2=0$$

$$k_1+\frac{2}{5}k_2=0$$

$$k_1=k_2$$

$$k_1=-\frac{2}{5}k_2$$

From the first system, we'll choose $k_2=1$, which results in $k_1=1$. And from the second system, we'll choose $k_2=-5$, which results in $k_1=2$.

$$\overrightarrow{k_1}=\begin{bmatrix}1\\ 1\end{bmatrix}$$

$$\overrightarrow{k_2}=\begin{bmatrix}2\\ -5\end{bmatrix}$$

Then the solutions to the system are

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$$\vec{x_1} = \vec{k_1} e^{\lambda_1 t} \qquad \vec{x_2} = \vec{k_2} e^{\lambda_2 t}$$

$$\vec{x_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} \qquad \vec{x_2} = \begin{bmatrix} 2 \\ -5 \end{bmatrix} e^{-4t}$$

Therefore, the general solution to the homogeneous system must be

$$\vec{x} = c_1 \vec{x_1} + c_2 \vec{x_2}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 2 \\ -5 \end{bmatrix} e^{-4t}$$

Rewrite the general solution,

$$\vec{x} = \begin{bmatrix} c_1 e^{3t} & 2c_2 e^{-4t} \\ c_1 e^{3t} & -5c_2 e^{-4t} \end{bmatrix}$$

then substitute the initial condition,

$$\vec{x}(0) = \begin{bmatrix} -2 \\ 19 \end{bmatrix}$$

to get

$$\begin{bmatrix} -2 \\ 19 \end{bmatrix} = \begin{bmatrix} c_1 e^{3(0)} & 2c_2 e^{-4(0)} \\ c_1 e^{3(0)} & -5c_2 e^{-4(0)} \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ 19 \end{bmatrix} = \begin{bmatrix} c_1 & 2c_2 \\ c_1 & -5c_2 \end{bmatrix}$$

This gives the system of equations

$$c_1 + 2c_2 = -2$$

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$$c_1 - 5c_2 = 19$$

We can rewrite the system as

$$c_1 = -2 - 2c_2$$

$$c_1 = 19 + 5c_2$$

to get

$$-2 - 2c_2 = 19 + 5c_2$$

$$7c_2 = -21$$

$$c_2 = -3$$

and

$$c_1 = -2 - 2(-3)$$

$$c_1 = -2 + 6$$

$$c_1 = 4$$

The solution of the system of differential equations with the given initial conditions is

$$\vec{x} = 4\begin{bmatrix}1\\1\end{bmatrix}e^{3t} - 3\begin{bmatrix}2\\-5\end{bmatrix}e^{-4t}$$

$$\vec{x} = \begin{bmatrix}4\\4\end{bmatrix}e^{3t} + \begin{bmatrix}-6\\15\end{bmatrix}e^{-4t}$$

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■ 4. Convert the differential equation into a system of equations, then use the system to solve the initial value problem, given $y(0) = 2$ and $y'(0) = 6$.

$$y'' - 6y' - 7y = 0$$

Solution:

Solve the original differential equation for $y''(t)$.

$$y'' - 6y' - 7y = 0$$

$$y'' = 6y' + 7y$$

We can start by defining

$$x_1(t) = y(t)$$

$$x_2(t) = y'(t)$$

Then if we take the derivatives of $x_1(t)$ and $x_2(t)$, we get

$$x_1'(t) = y'(t) = x_2(t)$$

$$x_2'(t) = y''(t) = 6y'(t) + 7y(t) = 6x_2(t) + 7x_1(t)$$

Therefore, we have the system of differential equations

$$x_1' = x_2$$

$$x_2' = 7x_1 + 6x_2$$

The coefficient matrix is

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$$A = \begin{bmatrix} 0 & 1 \\ 7 & 6 \end{bmatrix}$$

and the matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 0 & 1 \\ 7 & 6 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -\lambda & 1 \\ 7 & 6 - \lambda \end{bmatrix}$$

Then the determinant is

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 1 \\ 7 & 6 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = -\lambda(6 - \lambda) - 1(7)$$

$$|A - \lambda I| = -6\lambda + \lambda^2 - 7$$

$$|A - \lambda I| = \lambda^2 - 6\lambda - 7$$

So the characteristic equation is

$$\lambda^2 - 6\lambda - 7 = 0$$

$$(\lambda - 7)(\lambda + 1) = 0$$

$$\lambda_1 = 7 \qquad \lambda_2 = -1$$

Then for these Eigenvalues, $\lambda_1 = 7$ and $\lambda_2 = -1$, we find

$$A - 7I = \begin{bmatrix} -7 & 1 \\ 7 & 6 - 7 \end{bmatrix}$$

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$$A - 7I = \begin{bmatrix} -7 & 1 \\ 7 & -1 \end{bmatrix}$$

and

$$A - (-1)I = \begin{bmatrix} 1 & 1 \\ 7 & 6+1 \end{bmatrix}$$

$$A - (-1)I = \begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix}$$

Put these matrices into reduced row-echelon form.

$$\begin{bmatrix} -7 & 1 \\ 7 & -1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{7} \\ 7 & -1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{1}{7} \\ 0 & 0 \end{bmatrix}$$

If we turn these back into systems of equations, we get

$$k_1 - \frac{1}{7}k_2 = 0 \qquad k_1 + k_2 = 0$$

$$k_1 = \frac{1}{7}k_2 \qquad k_1 = -k_2$$

From the first system, we'll choose $k_2 = 7$, which results in $k_1 = 1$. And from the second system, we'll choose $k_2 = -1$, which results in $k_1 = 1$. So the Eigenvectors are

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$$\vec{k_1} = \begin{bmatrix} 1 \\ 7 \end{bmatrix} \qquad \vec{k_2} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Then the solutions to the system are

$$\vec{x_1} = \vec{k_1} e^{\lambda_1 t} \qquad \vec{x_2} = \vec{k_2} e^{\lambda_2 t}$$

$$\vec{x_1} = \begin{bmatrix} 1 \\ 7 \end{bmatrix} e^{7t} \qquad \vec{x_2} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t}$$

Therefore, the general solution to the homogeneous system must be

$$\vec{x} = c_1 \vec{x_1} + c_2 \vec{x_2}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 7 \end{bmatrix} e^{7t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t}$$

Rewrite the initial conditions $y(0) = 2$ and $y'(0) = 6$ as

$$x_1(0) = y(0) = 2$$

$$x_2(0) = y'(0) = 6$$

With these two values, we can say

$$\vec{x}(0) = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

and then substitute this into the general solution.

$$\begin{bmatrix} 2 \\ 6 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 7 \end{bmatrix} e^{7(0)} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-(0)}$$

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$$\begin{bmatrix} 2 \\ 6 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 7 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} c_1 + c_2 \\ 7c_1 - c_2 \end{bmatrix}$$

We get the system of equations

$$c_1 + c_2 = 2$$

$$7c_1 - c_2 = 6$$

Solving this system, we'll get

$$c_1 = 1$$

$$c_2 = 1$$

Therefore, the solution to the system is

$$\overrightarrow{x} = \begin{bmatrix} 1 \\ 7 \end{bmatrix} e^{7t} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t}$$

And the solution $y(t)$ to the initial value problem is

$$y(t) = x_1(t)$$

$$y(t) = e^{7t} + e^{-t}$$

■ 5. Convert the differential equation into a system of equations, then solve the system.

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$$y''' - 2y'' - 9y' + 18y = 0$$

Solution:

Solve the original differential equation for $y'''(t)$.

$$y''' = 2y'' + 9y' - 18y$$

We can start by defining

$$x_1(t) = y(t)$$

$$x_2(t) = y'(t)$$

$$x_3(t) = y''(t)$$

Then if we take the derivatives of $x_1(t)$, $x_2(t)$, and $x_3(t)$ we get

$$x_1'(t) = y'(t) = x_2(t)$$

$$x_2'(t) = y''(t) = x_3(t)$$

$$x_3'(t) = y'''(t) = 2y'' + 9y' - 18y = 2x_3 + 9x_2 - 18x_1$$

Therefore, we have the system of differential equations

$$x_1' = x_2$$

$$x_2' = x_3$$

$$x_3' = -18x_1 + 9x_2 + 2x_3$$

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The coefficient matrix is

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -18 & 9 & 2 \end{bmatrix}$$

and the matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ -18 & 9 & 2-\lambda \end{bmatrix}$$

Then the determinant is

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ -18 & 9 & 2-\lambda \end{vmatrix}$$

$$|A - \lambda I| = -\lambda \begin{vmatrix} -\lambda & 1 \\ 9 & 2-\lambda \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 \\ -18 & 2-\lambda \end{vmatrix} + 0 \begin{vmatrix} 0 & -\lambda \\ -18 & 9 \end{vmatrix}$$

$$|A - \lambda I| = -\lambda((-\lambda)(2-\lambda) - 1(9)) - 1(0(2-\lambda) - 1(-18)) + 0((0(9) - (-\lambda)(-18))$$

$$|A - \lambda I| = -\lambda(\lambda^2 - 2\lambda - 9) - 18$$

$$|A - \lambda I| = -\lambda^3 + 2\lambda^2 + 9\lambda - 18$$

$$|A - \lambda I| = -\lambda^2(\lambda - 2) + 9(\lambda - 2)$$

$$|A - \lambda I| = (\lambda - 2)(9 - \lambda^2)$$

So the characteristic equation is

$$(\lambda - 2)(9 - \lambda^2) = 0$$

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$$(\lambda - 2)(3 - \lambda)(3 + \lambda) = 0$$

$$\lambda_1 = 2 \qquad \lambda_2 = 3 \qquad \lambda_3 = -3$$

For the Eigenvalue $\lambda_1 = 2$ we find

$$A - 2I = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 1 \\ -18 & 9 & 2-2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 1 \\ -18 & 9 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & -2 & 1 \\ -18 & 9 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{4} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

Turning this back into a system of equations, we get

$$k_1 - \frac{1}{4}k_3 = 0$$

$$k_1 = \frac{1}{4}k_3$$

and

$$k_2 - \frac{1}{2}k_3 = 0$$

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$$k_2 = \frac{1}{2}k_3$$

We'll choose $k_3 = 4$, which results in $k_1 = 1$ and $k_2 = 2$, and therefore we get the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} e^{2t}$$

For the Eigenvalue $\lambda_2 = 3$ we find

$$A - 3I = \begin{bmatrix} -3 & 1 & 0 \\ 0 & -3 & 1 \\ -18 & 9 & 2-3 \end{bmatrix}$$

$$A - 3I = \begin{bmatrix} -3 & 1 & 0 \\ 0 & -3 & 1 \\ -18 & 9 & -1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & -3 & 1 \\ -18 & 9 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & -3 & 1 \\ 0 & 3 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & -\frac{1}{3} \\ 0 & 3 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{9} \\ 0 & 1 & -\frac{1}{3} \\ 0 & 3 & -1 \end{bmatrix}$$

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$$\rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{9} \\ 0 & 1 & -\frac{1}{3} \\ 0 & 0 & 0 \end{bmatrix}$$

Turning this back into a system of equations, we get

$$k_1 - \frac{1}{9}k_3 = 0$$

$$k_1 = \frac{1}{9}k_3$$

and

$$k_2 - \frac{1}{3}k_3 = 0$$

$$k_2 = \frac{1}{3}k_3$$

We'll choose $k_3 = 9$, which results in $k_1 = 1$ and $k_2 = 3$, and therefore we get the Eigenvector

$$\overrightarrow{k_2} = \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} e^{3t}$$

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For the Eigenvalue $\lambda_3 = -3$ we find

$$A-(-3)I=\begin{bmatrix}3 & 1 & 0\\ 0 & 3 & 1\\ -18 & 9 & 2+3\end{bmatrix}$$

$$A-(-3)I=\begin{bmatrix}3 & 1 & 0\\ 0 & 3 & 1\\ -18 & 9 & 5\end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix}1 & \frac{1}{3} & 0\\ 0 & 3 & 1\\ -18 & 9 & 5\end{bmatrix}\rightarrow\begin{bmatrix}1 & \frac{1}{3} & 0\\ 0 & 3 & 1\\ 0 & 15 & 5\end{bmatrix}\rightarrow\begin{bmatrix}1 & \frac{1}{3} & 0\\ 0 & 1 & \frac{1}{3}\\ 0 & 15 & 5\end{bmatrix}$$

$$\rightarrow\begin{bmatrix}1 & 0 & -\frac{1}{9}\\ 0 & 1 & \frac{1}{3}\\ 0 & 15 & 5\end{bmatrix}\rightarrow\begin{bmatrix}1 & 0 & -\frac{1}{9}\\ 0 & 1 & \frac{1}{3}\\ 0 & 0 & 0\end{bmatrix}$$

Turning this back into a system of equations, we get

$$k_1-\frac{1}{9}k_3=0$$

$$k_1=\frac{1}{9}k_3$$

and

$$k_2+\frac{1}{3}k_3=0$$

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$$k_2 = -\frac{1}{3}k_3$$

We'll choose $k_3 = 9$, which results in $k_1 = 1$ and $k_2 = -3$, and therefore we get the Eigenvector

$$\overrightarrow{k_3} = \begin{bmatrix} 1 \\ -3 \\ 9 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 1 \\ -3 \\ 9 \end{bmatrix} e^{-3t}$$

So the general solution is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} 1 \\ -3 \\ 9 \end{bmatrix} e^{-3t}$$

And the solution $y(t)$ to the original differential equation is

$$y(t) = c_1 e^{2t} + c_2 e^{3t} + c_3 e^{-3t}$$

■ 6. Solve the initial value problem, given $y(0) = 2$, $y'(0) = 6$, and $y''(0) = 16$.

$$y''' - 6y'' + 11y' - 6y = 0$$

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Solution:

Solve the original differential equation for y'''.

$$y''' = 6y'' - 11y' + 6y$$

We can start by defining

$$x_1(t) = y(t)$$

$$x_2(t) = y'(t)$$

$$x_3(t) = y''(t)$$

Then if we take the derivatives of $x_1(t)$, $x_2(t)$, and $x_3(t)$ we get

$$x_1'(t) = y'(t) = x_2(t)$$

$$x_2'(t) = y''(t) = x_3(t)$$

$$x_3'(t) = y'''(t) = 6y'' - 11y' + 6y = 6x_3 - 11x_2 + 6x_1$$

Therefore, we have the system of differential equations

$$x_1' = x_2$$

$$x_2' = x_3$$

$$x_3' = 6x_1 - 11x_2 + 6x_3$$

The coefficient matrix is

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$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix}$$

and the matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 6 & -11 & 6 - \lambda \end{bmatrix}$$

Then the determinant is

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 6 & -11 & 6 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = -\lambda \begin{vmatrix} -\lambda & 1 \\ -11 & 6 - \lambda \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 \\ 6 & 6 - \lambda \end{vmatrix} + 0 \begin{vmatrix} 0 & -\lambda \\ 6 & -11 \end{vmatrix}$$

$$|A - \lambda I| = -\lambda(-\lambda(6 - \lambda) - 1(-11)) - 1(0(6 - \lambda) - 1(6)) + 0(0(-11) - (-\lambda)(6))$$

$$|A - \lambda I| = -\lambda(\lambda^2 - 6\lambda + 11) + 6$$

$$|A - \lambda I| = -\lambda^3 + 6\lambda^2 - 11\lambda + 6$$

So the characteristic equation is

$$-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$(\lambda - 1)(\lambda - 2)(3 - \lambda) = 0$$

$$\lambda_1 = 1 \qquad \lambda_2 = 2 \qquad \lambda_3 = 3$$

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For the Eigenvalue $\lambda_1 = 1$ we find

$$A - (1)I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 6 & -11 & 6-1 \end{bmatrix}$$

$$A - (1)I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 6 & -11 & 5 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 1 \\ 6 & -11 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 1 \\ 0 & -5 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & -5 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & -5 & 5 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Turning this back into a system of equations, we get

$$k_1 - k_3 = 0$$

$$k_1 = k_3$$

and

$$k_2 - k_3 = 0$$

$$k_2 = k_3$$

We'll choose $k_3 = 1$, which results in $k_1 = 1$ and $k_2 = 1$, and therefore we get the Eigenvector

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$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{t}$$

For the Eigenvalue $\lambda_2 = 2$ we find

$$A - 2I = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 1 \\ 6 & -11 & 6-2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 1 \\ 6 & -11 & 4 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & -2 & 1 \\ 6 & -11 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & -2 & 1 \\ 0 & -8 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{1}{2} \\ 0 & -8 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{4} \\ 0 & 1 & -\frac{1}{2} \\ 0 & -8 & 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{4} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

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Turning this back into a system of equations, we get

$$k_1 - \frac{1}{4}k_3 = 0$$

$$k_1 = \frac{1}{4}k_3$$

and

$$k_2 - \frac{1}{2}k_3 = 0$$

$$k_2 = \frac{1}{2}k_3$$

We'll choose $k_3 = 4$, which results in $k_1 = 1$ and $k_2 = 2$, and therefore we get the Eigenvector

$$\overrightarrow{k_2} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} e^{2t}$$

For the Eigenvalue $\lambda_3 = 3$ we find

$$A - 3I = \begin{bmatrix} -3 & 1 & 0 \\ 0 & -3 & 1 \\ 6 & -11 & 6-3 \end{bmatrix}$$

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$$A - 3I = \begin{bmatrix} -3 & 1 & 0 \\ 0 & -3 & 1 \\ 6 & -11 & 3 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & -3 & 1 \\ 6 & -11 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & -3 & 1 \\ 0 & -9 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & -\frac{1}{3} \\ 0 & -9 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{9} \\ 0 & 1 & -\frac{1}{3} \\ 0 & -9 & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{9} \\ 0 & 1 & -\frac{1}{3} \\ 0 & 0 & 0 \end{bmatrix}$$

Turning this back into a system of equations, we get

$$k_1 - \frac{1}{9}k_3 = 0$$

$$k_1 = \frac{1}{9}k_3$$

and

$$k_2 - \frac{1}{3}k_3 = 0$$

$$k_2 = \frac{1}{3}k_3$$

We'll choose $k_3 = 9$, which results in $k_1 = 1$ and $k_2 = 3$, and therefore we get the Eigenvector

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$$\overrightarrow{k_3} = \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} e^{3t}$$

So the general solution is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} e^{2t} + c_3 \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} e^{3t}$$

The initial conditions can be rewritten as

$$\overrightarrow{x}(0) = \begin{bmatrix} 2 \\ 6 \\ 16 \end{bmatrix}$$

so we'll plug this into the general solution to get

$$\begin{bmatrix} 2 \\ 6 \\ 16 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{(0)} + c_2 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} e^{2(0)} + c_3 \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} e^{3(0)}$$

$$\begin{bmatrix} 2 \\ 6 \\ 16 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix}$$

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So we get the system of equations

$$c_1 + c_2 + c_3 = 2$$

$$c_1 + 2c_2 + 3c_3 = 6$$

$$c_1 + 4c_2 + 9c_3 = 16$$

After solving the system, we get

$$c_1 = -1$$

$$c_2 = 2$$

$$c_3 = 1$$

The solution is therefore

$$\vec{x} = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} e^t + \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix} e^{2t} + \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix} e^{3t}$$

Therefore, the solution to the initial value problem is

$$y(t) = -e^t + 2e^{2t} + e^{3t}$$

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EQUAL REAL EIGENVALUES WITH MULTIPLICITY TWO

■ 1. Solve the system of the differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \vec{x}$$

Solution:

We'll find $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 1 & 1 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 2-\lambda \end{bmatrix}$$

and then we'll find its determinant.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 & 1 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 2-\lambda \end{vmatrix}$$

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$$|A-\lambda I| = (1-\lambda)\begin{vmatrix} 1-\lambda & 2 \\ 0 & 2-\lambda \end{vmatrix} - 1\begin{vmatrix} 0 & 2 \\ 0 & 2-\lambda \end{vmatrix} + 1\begin{vmatrix} 0 & 1-\lambda \\ 0 & 0 \end{vmatrix}$$

$$|A-\lambda I| = (1-\lambda)[(1-\lambda)(2-\lambda) - 2(0)]$$

$$-[(0)(2-\lambda) - 2(0)] + [(0)(0) - (1-\lambda)(0)]$$

$$|A-\lambda I| = (1-\lambda)(1-\lambda)(2-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(1-\lambda)(1-\lambda)(2-\lambda) = 0$$

$$\lambda_1 = \lambda_2 = 1 \qquad \lambda_3 = 2$$

We'll handle $\lambda_3 = 2$ first, starting with finding $A - \lambda_3 I$.

$$A - 2I = \begin{bmatrix} 1-2 & 1 & 1 \\ 0 & 1-2 & 2 \\ 0 & 0 & 2-2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -1 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 - 3k_3 = 0$$

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$$k_1 = 3k_3$$

and

$$k_2 - 2k_3 = 0$$

$$k_2 = 2k_3$$

If we choose $k_3 = 1$, we find $k_1 = 3$ and $k_2 = 2$ and therefore we get the Eigenvector

$$\overrightarrow{k_3} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} e^{2t}$$

Then for the repeated Eigenvalues $\lambda_1 = \lambda_2 = 1$, we find

$$A - (1)I = \begin{bmatrix} 1-1 & 1 & 1 \\ 0 & 1-1 & 2 \\ 0 & 0 & 2-1 \end{bmatrix}$$

$$A - (1)I = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

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If we turn this matrix into a system of equations, we get

$$k_2 = 0$$

$$k_3 = 0$$

Then there's only one linearly independent Eigenvector for $\lambda_1 = \lambda_2 = 1$, and we'll choose

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t$$

Because we only find one Eigenvector for the two Eigenvalues $\lambda_1 = \lambda_2 = 1$, we have to use $\overrightarrow{k_1} = (1,0,0)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

This matrix equation gives $p_2 + p_3 = 1$ and $p_3 = 0$, and therefore $p_2 = 1$. We'll choose $p_1 = 0$, so

$$\overrightarrow{p} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

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Then the third solution is

$$\vec{x_2} = \vec{k_1} t e^{\lambda_1 t} + \vec{p} e^{\lambda_1 t} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^t$$

Then the general solution is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^t \right) + c_3 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} e^{2t}$$

2. Solve the system of the differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix} \vec{x}$$

Solution:

We'll find $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

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$$A - \lambda I = \begin{bmatrix} 1-\lambda & 1 & 1 \\ 2 & 2-\lambda & 2 \\ 3 & 3 & 3-\lambda \end{bmatrix}$$

and then we'll find its determinant.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 & 1 \\ 2 & 2-\lambda & 2 \\ 3 & 3 & 3-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)\begin{vmatrix} 2-\lambda & 2 \\ 3 & 3-\lambda \end{vmatrix} - 1\begin{vmatrix} 2 & 2 \\ 3 & 3-\lambda \end{vmatrix} + 1\begin{vmatrix} 2 & 2-\lambda \\ 3 & 3 \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)[(2-\lambda)(3-\lambda) - (2)(3)]$$

$$-[(2)(3-\lambda) - (2)(3)] + [(2)(3) - (2-\lambda)(3)]$$

$$|A - \lambda I| = (1-\lambda)(6 - 5\lambda + \lambda^2 - 6) - (6 - 2\lambda - 6) + (6 - 6 + 3\lambda)$$

$$|A - \lambda I| = (1-\lambda)(-5\lambda + \lambda^2) + 2\lambda + 3\lambda$$

$$|A - \lambda I| = -5\lambda + \lambda^2 + 5\lambda^2 - \lambda^3 + 5\lambda$$

$$|A - \lambda I| = 6\lambda^2 - \lambda^3$$

$$|A - \lambda I| = \lambda^2(6 - \lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$\lambda^2(6 - \lambda) = 0$$

$$\lambda_1 = 6 \qquad \lambda_2 = \lambda_3 = 0$$

We'll handle $\lambda_1 = 6$ first, starting with finding $A - \lambda_1 I$.

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$$A-6I=\begin{bmatrix}1-6 & 1 & 1\\ 2 & 2-6 & 2\\ 3 & 3 & 3-6\end{bmatrix}$$

$$A-6I=\begin{bmatrix}-5 & 1 & 1\\ 2 & -4 & 2\\ 3 & 3 & -3\end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix}1 & -\frac{1}{5} & -\frac{1}{5}\\ 2 & -4 & 2\\ 3 & 3 & -3\end{bmatrix}\rightarrow\begin{bmatrix}1 & -\frac{1}{5} & -\frac{1}{5}\\ 0 & -\frac{18}{5} & \frac{12}{5}\\ 3 & 3 & -3\end{bmatrix}\rightarrow\begin{bmatrix}1 & -\frac{1}{5} & -\frac{1}{5}\\ 0 & -\frac{18}{5} & \frac{12}{5}\\ 0 & \frac{18}{5} & -\frac{12}{5}\end{bmatrix}\rightarrow\begin{bmatrix}1 & -\frac{1}{5} & -\frac{1}{5}\\ 0 & 1 & -\frac{2}{3}\\ 0 & \frac{18}{5} & -\frac{12}{5}\end{bmatrix}$$

$$\rightarrow\begin{bmatrix}1 & 0 & -\frac{1}{3}\\ 0 & 1 & -\frac{2}{3}\\ 0 & \frac{18}{5} & -\frac{12}{5}\end{bmatrix}\rightarrow\begin{bmatrix}1 & 0 & -\frac{1}{3}\\ 0 & 1 & -\frac{2}{3}\\ 0 & 0 & 0\end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1-\frac{1}{3}k_3=0$$

$$k_1=\frac{1}{3}k_3$$

and

$$k_2-\frac{2}{3}k_3=0$$

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$$k_2 = \frac{2}{3}k_3$$

If we choose $k_3 = 3$, we find $k_1 = 1$ and $k_2 = 2$ and therefore we get the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} e^{6t}$$

Then for the repeated Eigenvalues $\lambda_2 = \lambda_3 = 0$, we find

$$A - (0)I = \begin{bmatrix} 1-0 & 1 & 1 \\ 2 & 2-0 & 2 \\ 3 & 3 & 3-0 \end{bmatrix}$$

$$A - (0)I = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 3 & 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 + k_2 + k_3 = 0$$

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$$k_1 = -k_2 - k_3$$

We can choose $k_2 = 0$ and $k_3 = -1$ to get $k_1 = 1$ and therefore

$$\vec{k_2} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

But we can also choose $k_2 = 1$ and $k_3 = -1$ to get $k_1 = 0$ and therefore

$$\vec{k_3} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

These linearly independent Eigenvectors give the solution vectors

$$\vec{x_2} = \vec{k_2} e^{\lambda_2 t} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\vec{x_3} = \vec{k_3} e^{\lambda_3 t} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

Then the general solution is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} e^{6t} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

■ 3. Find the general solution to the system of differential equations.

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$$\vec{x}' = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 0 & 4 \\ -1 & 1 & 0 \end{bmatrix} \vec{x}$$

Solution:

We'll find $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 0 & 4 \\ -1 & 1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 0 & 4 \\ -1 & 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3-\lambda & -1 & 2 \\ 2 & -\lambda & 4 \\ -1 & 1 & -\lambda \end{bmatrix}$$

and then we'll find its determinant.

$$|A - \lambda I| = \begin{vmatrix} 3-\lambda & -1 & 2 \\ 2 & -\lambda & 4 \\ -1 & 1 & -\lambda \end{vmatrix}$$

$$|A - \lambda I| = (3-\lambda)\begin{vmatrix} -\lambda & 4 \\ 1 & -\lambda \end{vmatrix} - (-1)\begin{vmatrix} 2 & 4 \\ -1 & -\lambda \end{vmatrix} + 2\begin{vmatrix} 2 & -\lambda \\ -1 & 1 \end{vmatrix}$$

$$|A - \lambda I| = (3-\lambda)[(-\lambda)(-\lambda) - (4)(1)]$$

$$+[(2)(-\lambda) - (4)(-1)] + 2[(2)(1) - (-\lambda)(-1)]$$

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$$|A - \lambda I| = (3 - \lambda)(\lambda^2 - 4) + (-2\lambda + 4) + 2(2 - \lambda)$$

$$|A - \lambda I| = (3 - \lambda)(\lambda^2 - 4) - 2(\lambda - 2) - 2(\lambda - 2)$$

$$|A - \lambda I| = (\lambda - 2)[(3 - \lambda)(\lambda + 2) - 4]$$

$$|A - \lambda I| = (\lambda - 2)(-\lambda^2 + \lambda + 2)$$

$$|A - \lambda I| = -(\lambda - 2)^2(\lambda + 1)$$

Solve the characteristic equation for the Eigenvalues.

$$-(\lambda - 2)^2(\lambda + 1) = 0$$

$$\lambda_1 = -1 \qquad \lambda_2 = \lambda_3 = 2$$

We'll handle $\lambda_1 = -1$ first, starting with finding $A - \lambda_1 I$.

$$A - (-1)I = \begin{bmatrix} 3+1 & -1 & 2 \\ 2 & 1 & 4 \\ -1 & 1 & 1 \end{bmatrix}$$

$$A - (-1)I = \begin{bmatrix} 4 & -1 & 2 \\ 2 & 1 & 4 \\ -1 & 1 & 1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -\frac{1}{4} & \frac{1}{2} \\ 2 & 1 & 4 \\ -1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{4} & \frac{1}{2} \\ 0 & \frac{3}{2} & 3 \\ -1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{4} & \frac{1}{2} \\ 0 & \frac{3}{2} & 3 \\ 0 & \frac{3}{4} & \frac{3}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{4} & \frac{1}{2} \\ 0 & 1 & 2 \\ 0 & \frac{3}{4} & \frac{3}{2} \end{bmatrix}$$

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$$\rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & \frac{3}{4} & \frac{3}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 + k_3 = 0$$

$$k_1 = -k_3$$

and

$$k_2 + 2k_3 = 0$$

$$k_2 = -2k_3$$

If we choose $k_3 = -1$, we find $k_1 = 1$ and $k_2 = 2$ and therefore we get the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} e^{-t}$$

Then for the repeated Eigenvalues $\lambda_2 = \lambda_3 = 2$, we find

$$A - 2I = \begin{bmatrix} 3-2 & -1 & 2 \\ 2 & -2 & 4 \\ -1 & 1 & -2 \end{bmatrix}$$

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$$A - 2I = \begin{bmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ -1 & 1 & -2 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 0 & 0 \\ -1 & 1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 - k_2 + 2k_3 = 0$$

$$k_1 = k_2 - 2k_3$$

We can choose $k_2 = 1$ and $k_3 = 0$ to get $k_1 = 1$ and therefore

$$\vec{k_2} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

But we can also choose $k_2 = 0$ and $k_3 = -1$ to get $k_1 = 2$ and therefore

$$\vec{k_3} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

These linearly independent Eigenvectors give the solution vectors

$$\vec{x_2} = \vec{k_2} e^{\lambda_2 t} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

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$$\overrightarrow{x_3} = \overrightarrow{k_3}e^{\lambda_3 t} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} e^{2t}$$

Then the general solution is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t} + c_3 \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} e^{2t}$$

■ 4. Find the general solution to the system of differential equations.

$$\overrightarrow{x}' = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \overrightarrow{x}$$

Solution:

We'll find $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

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$$A - \lambda I = \begin{bmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & 3 \\ 0 & 0 & 1-\lambda \end{bmatrix}$$

and then we'll find its determinant.

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & 3 \\ 0 & 0 & 1-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (2-\lambda)\begin{vmatrix} 2-\lambda & 3 \\ 0 & 1-\lambda \end{vmatrix} - (-1)\begin{vmatrix} -1 & 3 \\ 0 & 1-\lambda \end{vmatrix} + 0\begin{vmatrix} -1 & 2-\lambda \\ 0 & 0 \end{vmatrix}$$

$$|A - \lambda I| = (2-\lambda)[(2-\lambda)(1-\lambda) - 3(0)]$$

$$+[(-1)(1-\lambda) - 3(0)] + 0[(-1)(0) - (2-\lambda)(0)]$$

$$|A - \lambda I| = (2-\lambda)(2-\lambda)(1-\lambda) - (1-\lambda)$$

$$|A - \lambda I| = (1-\lambda)[(2-\lambda)(2-\lambda) - 1]$$

$$|A - \lambda I| = (1-\lambda)(4 - 4\lambda + \lambda^2 - 1)$$

$$|A - \lambda I| = (1-\lambda)(\lambda^2 - 4\lambda + 3)$$

$$|A - \lambda I| = (1-\lambda)(\lambda - 3)(\lambda - 1)$$

Solve the characteristic equation for the Eigenvalues.

$$-(\lambda - 1)(\lambda - 1)(\lambda - 3) = 0$$

$$\lambda_1 = \lambda_2 = 1 \qquad \lambda_3 = 3$$

We'll handle $\lambda_3 = 3$ first, starting with finding $A - \lambda_3 I$.

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$$A - 3I = \begin{bmatrix} 2-3 & -1 & 0 \\ -1 & 2-3 & 3 \\ 0 & 0 & 1-3 \end{bmatrix}$$

$$A - 3I = \begin{bmatrix} -1 & -1 & 0 \\ -1 & -1 & 3 \\ 0 & 0 & -2 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 1 & 0 \\ -1 & -1 & 3 \\ 0 & 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 + k_2 = 0$$

$$k_1 = -k_2$$

and

$$k_3 = 0$$

If we choose $k_2 = -1$, we find $k_1 = 1$ and therefore we get the Eigenvector

$$\overrightarrow{k_3} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{-t}$$

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Then for the repeated Eigenvalues $\lambda_1 = \lambda_2 = 1$, we find

$$A - (1)I = \begin{bmatrix} 2-1 & -1 & 0 \\ -1 & 2-1 & 3 \\ 0 & 0 & 1-1 \end{bmatrix}$$

$$A - (1)I = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 - k_2 = 0$$

$$k_1 = k_2$$

and

$$k_3 = 0$$

We can choose $k_2 = 1$ to get $k_1 = 1$ and therefore the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

and the solution vector

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$$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t} = \begin{bmatrix}1\\1\\0\end{bmatrix}e^t$$

Because we only find one Eigenvector for the two Eigenvalues $\lambda_1 = \lambda_2 = 1$, we have to use $\overrightarrow{k_1} = (1,1,0)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

$$\begin{bmatrix}1 & -1 & 0\\-1 & 1 & 3\\0 & 0 & 0\end{bmatrix}\begin{bmatrix}p_1\\p_2\\p_3\end{bmatrix} = \begin{bmatrix}1\\1\\0\end{bmatrix}$$

This matrix equation gives $p_1 - p_2 = 1$ and $-p_1 + p_2 + 3p_3 = 1$. If we solve this system, we find

$$\overrightarrow{p} = \begin{bmatrix}1\\0\\\frac{2}{3}\end{bmatrix}$$

Then the third solution is

$$\overrightarrow{x_2} = \overrightarrow{k_1}te^{\lambda_1 t} + \overrightarrow{p}e^{\lambda_1 t} = \begin{bmatrix}1\\1\\0\end{bmatrix}te^t + \begin{bmatrix}1\\0\\\frac{2}{3}\end{bmatrix}e^t$$

Then the general solution is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

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$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 1 \\ 0 \\ \frac{2}{3} \end{bmatrix} e^t \right) + c_3 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{3t}$$

■ 5. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 5 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 5 \end{bmatrix} \vec{x}$$

Solution:

We'll find $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 5 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 5 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 5 - \lambda & 0 & 0 \\ 1 & 2 - \lambda & 0 \\ 1 & 0 & 5 - \lambda \end{bmatrix}$$

and then we'll find its determinant.

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$$|A-\lambda I| = \begin{vmatrix} 5-\lambda & 0 & 0 \\ 1 & 2-\lambda & 0 \\ 1 & 0 & 5-\lambda \end{vmatrix}$$

$$|A-\lambda I| = (5-\lambda)\begin{vmatrix} 2-\lambda & 0 \\ 0 & 5-\lambda \end{vmatrix} - 0\begin{vmatrix} 1 & 0 \\ 1 & 5-\lambda \end{vmatrix} + 0\begin{vmatrix} 1 & 2-\lambda \\ 1 & 0 \end{vmatrix}$$

$$|A-\lambda I| = (5-\lambda)[(2-\lambda)(5-\lambda) - 0(0)] - 0 + 0$$

$$|A-\lambda I| = (5-\lambda)(2-\lambda)(5-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(5-\lambda)(2-\lambda)(5-\lambda) = 0$$

$$\lambda_1 = \lambda_2 = 5 \qquad \lambda_3 = 2$$

We'll handle $\lambda_3 = 2$ first, starting with finding $A - \lambda_3 I$.

$$A - 2I = \begin{bmatrix} 5-2 & 0 & 0 \\ 1 & 2-2 & 0 \\ 1 & 0 & 5-2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

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$$k_1 = 0$$

$$k_3 = 0$$

We can choose $k_2 = 1$ to get the Eigenvector

$$\overrightarrow{k_3} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

Then for the repeated Eigenvalues $\lambda_1 = \lambda_2 = 5$, we find

$$A - 5I = \begin{bmatrix} 5-5 & 0 & 0 \\ 1 & 2-5 & 0 \\ 1 & 0 & 5-5 \end{bmatrix}$$

$$A - 5I = \begin{bmatrix} 0 & 0 & 0 \\ 1 & -3 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 = 0$$

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$$k_2 = 0$$

We can choose $k_3 = 1$ to get the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{5t}$$

Because we only find one Eigenvector for the two Eigenvalues $\lambda_1 = \lambda_2 = 5$, we have to use $\overrightarrow{k_1} = (0,0,1)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & -3 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

This matrix equation gives $p_1 - 3p_2 = 0$ and $p_1 = 1$. If we solve this system, we find

$$\overrightarrow{p} = \begin{bmatrix} 1 \\ \frac{1}{3} \\ 0 \end{bmatrix}$$

Then the third solution is

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$$\overrightarrow{x_2} = \overrightarrow{k_1} te^{\lambda_1 t} + \overrightarrow{p} e^{\lambda_1 t} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} te^{5t} + \begin{bmatrix} 1 \\ \frac{1}{3} \\ 0 \end{bmatrix} e^{5t}$$

Then the general solution is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{5t} + c_2 \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} te^{5t} + \begin{bmatrix} 1 \\ \frac{1}{3} \\ 0 \end{bmatrix} e^{5t} \right) + c_3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

■ 6. Find the general solution to the system of differential equations.

$$\overrightarrow{x}' = \begin{bmatrix} 1 & 0 & 4 \\ 0 & -3 & 0 \\ 4 & 0 & 1 \end{bmatrix} \overrightarrow{x}$$

Solution:

We'll find $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 1 & 0 & 4 \\ 0 & -3 & 0 \\ 4 & 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 0 & 4 \\ 0 & -3 & 0 \\ 4 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

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$$A - \lambda I = \begin{bmatrix} 1-\lambda & 0 & 4 \\ 0 & -3-\lambda & 0 \\ 4 & 0 & 1-\lambda \end{bmatrix}$$

and then we'll find its determinant.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 4 \\ 0 & -3-\lambda & 0 \\ 4 & 0 & 1-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)\begin{vmatrix} -3-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} - 0\begin{vmatrix} 0 & 0 \\ 4 & 1-\lambda \end{vmatrix} + 4\begin{vmatrix} 0 & -3-\lambda \\ 4 & 0 \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)[(-3-\lambda)(1-\lambda) - 0(0)] - 0 + 4[0(0) - (-3-\lambda)(4)]$$

$$|A - \lambda I| = -(1-\lambda)(3+\lambda)(1-\lambda) + 16(3+\lambda)$$

$$|A - \lambda I| = -(3+\lambda)[(1-\lambda)^2 - 16]$$

$$|A - \lambda I| = -(3+\lambda)(1-2\lambda+\lambda^2 - 16)$$

$$|A - \lambda I| = -(3+\lambda)(\lambda^2 - 2\lambda - 15)$$

$$|A - \lambda I| = -(\lambda+3)(\lambda+3)(\lambda-5)$$

Solve the characteristic equation for the Eigenvalues.

$$-(\lambda+3)(\lambda+3)(\lambda-5) = 0$$

$$\lambda_1 = 5 \qquad \lambda_2 = \lambda_3 = -3$$

We'll handle $\lambda_1 = 5$ first, starting with finding $A - \lambda_1 I$.

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$$A - 5I = \begin{bmatrix} 1-5 & 0 & 4 \\ 0 & -3-5 & 0 \\ 4 & 0 & 1-5 \end{bmatrix}$$

$$A - 5I = \begin{bmatrix} -4 & 0 & 4 \\ 0 & -8 & 0 \\ 4 & 0 & -4 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & -8 & 0 \\ 4 & 0 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & -8 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 - k_3 = 0$$

$$k_1 = k_3$$

and

$$k_2 = 0$$

We can choose $k_3 = 1$ to get $k_1 = 1$ and therefore the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{5t}$$

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Then for the repeated Eigenvalues $\lambda_2 = \lambda_3 = -3$, we find

$$A + 3I = \begin{bmatrix} 1+3 & 0 & 4 \\ 0 & -3+3 & 0 \\ 4 & 0 & 1+3 \end{bmatrix}$$

$$A + 3I = \begin{bmatrix} 4 & 0 & 4 \\ 0 & 0 & 0 \\ 4 & 0 & 4 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 4 & 0 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get

$$k_1 + k_3 = 0$$

$$k_1 = -k_3$$

We can choose $k_2 = 0$ and $k_3 = -1$ to get the Eigenvector

$$\vec{k_2} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

and the solution vector

$$\vec{x_2} = \vec{k_2} e^{\lambda_2 t} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} e^{-3t}$$

But we could also choose $k_2 = 1$ and $k_3 = -1$ to get the Eigenvector

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$$\overrightarrow{k_3} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} e^{-3t}$$

Then the general solution is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} e^{-3t} + c_3 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} e^{-3t}$$

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EQUAL REAL EIGENVALUES WITH MULTIPLICITY THREE

■ 1. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \vec{x}$$

Solution:

We'll need to find the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 1-\lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 1-\lambda \end{vmatrix}$$

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$$|A-\lambda I| = (1-\lambda)\begin{vmatrix} 1-\lambda & 2 \\ 0 & 1-\lambda \end{vmatrix} - 2\begin{vmatrix} 0 & 2 \\ 0 & 1-\lambda \end{vmatrix} + 3\begin{vmatrix} 0 & 1-\lambda \\ 0 & 0 \end{vmatrix}$$

$$|A-\lambda I| = (1-\lambda)((1-\lambda)(1-\lambda)-(2)(0))$$

$$-2((0)(1-\lambda)-(2)(0)) + 3((0)(0)-(1-\lambda)(0))$$

$$|A-\lambda I| = (1-\lambda)(1-\lambda)(1-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(1-\lambda)(1-\lambda)(1-\lambda) = 0$$

$$\lambda_1 = \lambda_2 = \lambda_3 = 1$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = \lambda_3 = 1$, we find

$$A-(1)I = \begin{bmatrix} 1-1 & 2 & 3 \\ 0 & 1-1 & 2 \\ 0 & 0 & 1-1 \end{bmatrix}$$

$$A-(1)I = \begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 0 & 1 & \frac{3}{2} \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & \frac{3}{2} \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this matrix into a system of equations, we get $k_2 = 0$ and $k_3 = 0$. We can choose any value for k_1. If we choose $k_1 = 1$, we find

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$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Other than the trivial solution $\overrightarrow{k_1} = (0,0,0)$, any other (k_1, k_2, k_3) set we find from $(k_1, k_2, k_3) = (k_1,0,0)$ will result in a vector that's linearly dependent with the $\overrightarrow{k_1} = (1,0,0)$ vector we already found. So we can say that the Eigenvalue $\lambda_1 = \lambda_2 = \lambda_3 = 1$ produces only one Eigenvector.

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t$$

Because we only find one Eigenvector for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = 1$, we have to use $\overrightarrow{k_1} = (1,0,0)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p} = \overrightarrow{k_1}$$

$$\begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives $2p_2 + 3p_3 = 1$ and $2p_3 = 0$, which means we find $p_2 = 1/2$ and $p_3 = 0$, and we can pick any value for p_1 so we'll choose $p_1 = 0$.

$$\overrightarrow{p} = \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

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Then our second solution will be

$$\vec{x_2} = \vec{k_1} te^{\lambda_1 t} + \vec{p} e^{\lambda_1 t}$$

$$\vec{x_2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{t} + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} e^{t}$$

Now we'll use $\vec{k_1} = (1,0,0)$ and $\vec{p} = (0,1/2,0)$ to find a third solution.

$$(A - \lambda_1 I)\vec{q} = \vec{p}$$

$$\begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives $2q_2 + 3q_3 = 0$ and $2q_3 = 1/2$, which means we find $q_2 = -3/8$ and $q_3 = 1/4$, and we can pick any value for q_1, so we'll pick $q_1 = 0$.

$$\vec{q} = \begin{bmatrix} 0 \\ -\frac{3}{8} \\ \frac{1}{4} \end{bmatrix}$$

Then our third solution will be

$$\vec{x_3} = \vec{k_1} \frac{t^2}{2} e^{\lambda_1 t} + \vec{p} te^{\lambda_1 t} + \vec{q} e^{\lambda_1 t}$$

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$$\overrightarrow{x_3} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{t^2}{2} e^t + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ -\frac{3}{8} \\ \frac{1}{4} \end{bmatrix} e^t$$

So the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} e^t \right)$$

$$+c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{t^2}{2} e^t + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ -\frac{3}{8} \\ \frac{1}{4} \end{bmatrix} e^t \right)$$

■ 2. Find the general solution to the system of differential equations.

$$\overrightarrow{x}' = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \overrightarrow{x}$$

Solution:

We'll need to find the matrix $A - \lambda I$,

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$$A - \lambda I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 0 & 1 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 1-\lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 1 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 1-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)\begin{vmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} - 0\begin{vmatrix} 0 & 2 \\ 0 & 1-\lambda \end{vmatrix} + 1\begin{vmatrix} 0 & 1-\lambda \\ 0 & 0 \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)((1-\lambda)(1-\lambda) - (0)(0))$$

$$-0((0)(1-\lambda) - (2)(0)) + ((0)(0) - (1-\lambda)(0))$$

$$|A - \lambda I| = (1-\lambda)(1-\lambda)(1-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(1-\lambda)(1-\lambda)(1-\lambda) = 0$$

$$\lambda_1 = \lambda_2 = \lambda_3 = 1$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = \lambda_3 = 1$, we find

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$$A-(1)I=\begin{bmatrix}1-1 & 0 & 1\\ 0 & 1-1 & 2\\ 0 & 0 & 1-1\end{bmatrix}$$

$$A-(1)I=\begin{bmatrix}0 & 0 & 1\\ 0 & 0 & 2\\ 0 & 0 & 0\end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix}0 & 0 & 1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

This matrix gives us the equation $k_3 = 0$, and we can pick an values for k_1 and k_2, so we could choose $k_1 = 1$ and $k_2 = 0$.

$$\overrightarrow{k_1}=\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}$$

We could also choose $k_1 = 0$ and $k_2 = 1$.

$$\overrightarrow{k_2}=\begin{bmatrix}0\\ 1\\ 0\end{bmatrix}$$

The solution vectors associated with these Eigenvectors are

$$\overrightarrow{x_1}=\overrightarrow{k_1}e^{\lambda_1 t}=\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}e^t$$

$$\overrightarrow{x_2}=\overrightarrow{k_2}e^{\lambda_2 t}=\begin{bmatrix}0\\ 1\\ 0\end{bmatrix}e^t$$

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Because we only find two linearly independent Eigenvectors for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = 1$, we have to use $\overrightarrow{k_1} = (1,0,0)$ and $\overrightarrow{k_2} = (0,1,0)$ to find a third solution.

$$(A - \lambda_1 I)\overrightarrow{p} = a_1\overrightarrow{k_1} + a_2\overrightarrow{k_2}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$p_3 = a_1$$

$$2p_3 = a_2$$

Choosing $p_3 = 1$ gives $a_1 = 1$ and $a_2 = 2$. We can choose any value for p_1 and p_2, so we'll pick $p_1 = 0$ and $p_2 = 0$ to get

$$\overrightarrow{p} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

and

$$a_1\overrightarrow{k_1} + a_2\overrightarrow{k_2} = 1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

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$$a_1\vec{k_1} + a_2\vec{k_2} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

Then our third solution will be

$$\vec{x_3} = (a_1\vec{k_1} + a_2\vec{k_2})te^{\lambda_3 t} + \vec{p}e^{\lambda_3 t}$$

$$\vec{x_3} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^t$$

and the general solution to the homogeneous system is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^t + c_3 \left(\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^t \right)$$

■ 3. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \vec{x}$$

Solution:

We'll need to find the matrix $A - \lambda I$,

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$$A - \lambda I = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3-\lambda & 0 & 0 \\ 0 & 3-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 3-\lambda & 0 & 0 \\ 0 & 3-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (3-\lambda)(3-\lambda)(3-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(3-\lambda)(3-\lambda)(3-\lambda) = 0$$

$$\lambda_1 = \lambda_2 = \lambda_3 = 3$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = \lambda_3 = 3$, we find

$$A - 3I = \begin{bmatrix} 3-3 & 0 & 0 \\ 0 & 3-3 & 0 \\ 0 & 0 & 3-3 \end{bmatrix}$$

$$A - 3I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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We can choose any values for k_1, k_2, and k_3, so we'll use 1, 0, and 0 in all three combinations to get the three linearly independent Eigenvectors

$$\vec{k_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \vec{k_2} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{ and } \vec{k_3} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

and then the solution vectors

$$\vec{x_1} = \vec{k_1} e^{\lambda_1 t} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{3t}$$

$$\vec{x_2} = \vec{k_2} e^{\lambda_2 t} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{3t}$$

$$\vec{x_3} = \vec{k_3} e^{\lambda_3 t} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{3t}$$

Then the general solution to the homogeneous system is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{3t}$$

■ 4. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 0 & 2 & 6 \\ -2 & -4 & -5 \\ 0 & 0 & -2 \end{bmatrix} \vec{x}$$

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Solution:

We'll need to find the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 0 & 2 & 6 \\ -2 & -4 & -5 \\ 0 & 0 & -2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 0 & 2 & 6 \\ -2 & -4 & -5 \\ 0 & 0 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -\lambda & 2 & 6 \\ -2 & -4 - \lambda & -5 \\ 0 & 0 & -2 - \lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 2 & 6 \\ -2 & -4 - \lambda & -5 \\ 0 & 0 & -2 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (-\lambda) \begin{vmatrix} -4 - \lambda & -5 \\ 0 & -2 - \lambda \end{vmatrix} - 2 \begin{vmatrix} -2 & -5 \\ 0 & -2 - \lambda \end{vmatrix} + 6 \begin{vmatrix} -2 & -4 - \lambda \\ 0 & 0 \end{vmatrix}$$

$$|A - \lambda I| = (-\lambda)((-4 - \lambda)(-2 - \lambda) - (-5)(0))$$

$$-2((-2)(-2 - \lambda) - (-5)(0)) + 6((-2)(0) - (-4 - \lambda)(0))$$

$$|A - \lambda I| = (-\lambda)(-4 - \lambda)(-2 - \lambda) + 4(-2 - \lambda)$$

$$|A - \lambda I| = (-2 - \lambda)[(-\lambda)(-4 - \lambda) + 4]$$

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$$|A - \lambda I| = (-2 - \lambda)(\lambda^2 + 4\lambda + 4)$$

$$|A - \lambda I| = (-2 - \lambda)(\lambda + 2)^2$$

$$|A - \lambda I| = -(2 + \lambda)^3$$

Solve the characteristic equation for the Eigenvalues.

$$-(2 + \lambda)^3 = 0$$

$$\lambda_1 = \lambda_2 = \lambda_3 = -2$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = \lambda_3 = -2$, we find

$$A - (-2)I = \begin{bmatrix} 2 & 2 & 6 \\ -2 & -4+2 & -5 \\ 0 & 0 & -2+2 \end{bmatrix}$$

$$A - (-2)I = \begin{bmatrix} 2 & 2 & 6 \\ -2 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 1 & 3 \\ -2 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Turning the matrix into a system of equations gives

$$k_1 + k_2 = 0$$

$$k_1 = -k_2$$

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and

$$k_3 = 0$$

We can choose $k_2 = -1$, which gives $k_1 = 1$ and therefore the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} e^{-2t}$$

Because we only find one linearly independent Eigenvector for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = -2$, we have to use $\overrightarrow{k_1} = (1, -1, 0)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p} = \overrightarrow{k_1}$$

$$\begin{bmatrix} 2 & 2 & 6 \\ -2 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$2p_1 + 2p_2 + 6p_3 = 1$$

$$-2p_1 - 2p_2 - 5p_3 = -1$$

We can choose any value for p_3, so we'll choose $p_3 = 0$ to simplify the system to just $2p_1 + 2p_2 = 1$. We'll choose $p_1 = 1$ and $p_2 = -1/2$ to get

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$$\vec{p} = \begin{bmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{bmatrix}$$

Then our second solution will be

$$\vec{x_2} = \vec{k_1} t e^{\lambda_1 t} + \vec{p} e^{\lambda_1 t}$$

$$\vec{x_2} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} t e^{-2t} + \begin{bmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{bmatrix} e^{-2t}$$

Now we'll use $\vec{k_1} = (1, -1, 0)$ and $\vec{p} = (1, -1/2, 0)$ to find a third solution.

$$(A - \lambda_1 I)\vec{q} = \vec{p}$$

$$\begin{bmatrix} 2 & 2 & 6 \\ -2 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$2q_1 + 2q_2 + 6q_3 = 1$$

$$-2q_1 - 2q_2 - 5q_3 = -\frac{1}{2}$$

We can choose any value for q_3, so we'll choose $q_3 = 1/2$ to simplify the system to just $q_1 + q_2 = -1$. Then we'll choose $q_1 = -1$ and $q_2 = 0$ to get the Eigenvector

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$$\vec{q} = \begin{bmatrix} -1 \\ 0 \\ \frac{1}{2} \end{bmatrix}$$

Then the solution vector will be

$$\vec{x_3} = \vec{k_1}\frac{t^2}{2}e^{\lambda_1 t} + \vec{p}te^{\lambda_1 t} + \vec{q}e^{\lambda_1 t}$$

$$\vec{x_3} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}\frac{t^2}{2}e^{-2t} + \begin{bmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{bmatrix}te^{-2t} + \begin{bmatrix} -1 \\ 0 \\ \frac{1}{2} \end{bmatrix}e^{-2t}$$

and the general solution to the homogeneous system is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}e^{-2t} + c_2\left(\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}te^{-2t} + \begin{bmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{bmatrix}e^{-2t}\right)$$

$$+c_3\left(\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}\frac{t^2}{2}e^{-2t} + \begin{bmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{bmatrix}te^{-2t} + \begin{bmatrix} -1 \\ 0 \\ \frac{1}{2} \end{bmatrix}e^{-2t}\right)$$

■ 5. Find the general solution to the system of differential equations.

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$$\vec{x}' = \begin{bmatrix} -3 & 2 & 0 \\ -2 & 1 & 0 \\ -2 & 2 & -1 \end{bmatrix} \vec{x}$$

Solution:

We'll need to find the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} -3 & 2 & 0 \\ -2 & 1 & 0 \\ -2 & 2 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -3 & 2 & 0 \\ -2 & 1 & 0 \\ -2 & 2 & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -3 - \lambda & 2 & 0 \\ -2 & 1 - \lambda & 0 \\ -2 & 2 & -1 - \lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} -3 - \lambda & 2 & 0 \\ -2 & 1 - \lambda & 0 \\ -2 & 2 & -1 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (-3 - \lambda) \begin{vmatrix} 1 - \lambda & 0 \\ 2 & -1 - \lambda \end{vmatrix} - 2 \begin{vmatrix} -2 & 0 \\ -2 & -1 - \lambda \end{vmatrix} + 0 \begin{vmatrix} -2 & 1 - \lambda \\ -2 & 2 \end{vmatrix}$$

$$|A - \lambda I| = (-3 - \lambda)((1 - \lambda)(-1 - \lambda) - (2)(0))$$

$$-2((-2)(-1 - \lambda) - (2)(0)) + 0((-2)(2) - (1 - \lambda)(-2))$$

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$$|A - \lambda I| = (-3 - \lambda)(1 - \lambda)(-1 - \lambda) + 4(-1 - \lambda)$$

$$|A - \lambda I| = -(1 + \lambda)^3$$

Solve the characteristic equation for the Eigenvalues.

$$-(1 + \lambda)(1 + \lambda)(1 + \lambda) = 0$$

$$\lambda_1 = \lambda_2 = \lambda_3 = -1$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = \lambda_3 = -1$, we find

$$A - (-1)I = \begin{bmatrix} -3 + 1 & 2 & 0 \\ -2 & 1 + 1 & 0 \\ -2 & 2 & -1 + 1 \end{bmatrix}$$

$$A - (-1)I = \begin{bmatrix} -2 & 2 & 0 \\ -2 & 2 & 0 \\ -2 & 2 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} -2 & 2 & 0 \\ -2 & 2 & 0 \\ -2 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -2 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

We can rewrite the matrix as the equation

$$k_1 - k_2 = 0$$

$$k_1 = k_2$$

We can choose any value for k_3, so we'll pick $k_3 = 0$ and $k_2 = 1$ to get $k_1 = 1$ and the Eigenvector

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$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

But we could also choose $k_3 = 1$ and $k_2 = 1$ to get $k_1 = 1$ and the Eigenvector

$$\overrightarrow{k_2} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

The Eigenvectors give us two solution vectors,

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{-t}$$

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{-t}$$

Because we only find two linearly independent Eigenvectors for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = -1$, we have to use $\overrightarrow{k_1} = (1,1,0)$ and $\overrightarrow{k_2} = (1,1,1)$ to find a third solution.

$$(A - \lambda_1 I)\overrightarrow{p_1} = a_1\overrightarrow{k_1} + a_2\overrightarrow{k_2}$$

$$\begin{bmatrix} -2 & 2 & 0 \\ -2 & 2 & 0 \\ -2 & 2 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 2 & 0 \\ -2 & 2 & 0 \\ -2 & 2 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 \\ a_1 + a_2 \\ a_2 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

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$$-2p_1 + 2p_2 = a_1 + a_2$$

$$-2p_1 + 2p_2 = a_2$$

So, $a_1 + a_2 = a_2$, or $a_1 = 0$. The system simplifies to just $-2p_1 + 2p_2 = a_2$, and we can choose $p_1 = 1$, $p_2 = 0$, and $p_3 = 0$, and we get $a_2 = -2$.

$$\vec{p} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

and

$$a_1\vec{k_1} + a_2\vec{k_2} = 0 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$a_1\vec{k_1} + a_2\vec{k_2} = \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix}$$

Then the solution vector will be

$$\vec{x_3} = (a_1\vec{k_1} + a_2\vec{k_2})te^{\lambda_3 t} + \vec{p}e^{\lambda_3 t}$$

$$\vec{x_3} = \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix} te^{-t} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t}$$

and the general solution to the homogeneous system is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{-t} + c_3 \left(\begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix} te^{-t} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t} \right)$$

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■ 6. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \vec{x}$$

Solution:

We'll need to find the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 1 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (2-\lambda) \begin{vmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 \\ 0 & 2-\lambda \end{vmatrix} + 0 \begin{vmatrix} 0 & 2-\lambda \\ 0 & 0 \end{vmatrix}$$

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$$|A - \lambda I| = (2 - \lambda)((2 - \lambda)(2 - \lambda) - (1)(0))$$

$$-((0)(2 - \lambda) - (1)(0)) + 0((0)(0) - (2 - \lambda)(0))$$

$$|A - \lambda I| = (2 - \lambda)(2 - \lambda)(2 - \lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(2 - \lambda)(2 - \lambda)(2 - \lambda) = 0$$

$$\lambda_1 = \lambda_2 = \lambda_3 = 2$$

Then for these Eigenvalues, $\lambda_1 = \lambda_2 = \lambda_3 = 2$, we find

$$A - 2I = \begin{bmatrix} 2 - 2 & 1 & 0 \\ 0 & 2 - 2 & 1 \\ 0 & 0 & 2 - 2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

We get the system of equations

$$k_2 = 0$$

$$k_3 = 0$$

We can choose any value for k_1, so we'll pick $k_1 = 1$ and we'll get the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

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and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{2t}$$

Because we only find one linearly independent Eigenvector for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = 2$, we have to use $\overrightarrow{k_1} = (1,0,0)$ to find a second solution.

$$(A - \lambda_1 I)\overrightarrow{p} = \overrightarrow{k_1}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$p_2 = 1$$

$$p_3 = 0$$

We can choose any value for p_1, so we'll pick $p_1 = 0$ to get the Eigenvector

$$\overrightarrow{p} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_2} = \overrightarrow{k_1} t e^{\lambda_1 t} + \overrightarrow{p} e^{\lambda_1 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{2t} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

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Now we'll use $\vec{k_1} = (1,0,0)$ and $\vec{p} = (0,1,0)$ to find a third solution.

$$(A - \lambda_1 I)\vec{q} = \vec{p}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$q_2 = 0$$

$$q_3 = 1$$

We can choose any value for q_1, so we'll pick $q_1 = 0$ to get the Eigenvector

$$\vec{q} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

and the solution vector

$$\vec{x_3} = \vec{k_1}\frac{t^2}{2}e^{\lambda_1 t} + \vec{p}te^{\lambda_1 t} + \vec{q}e^{\lambda_1 t}$$

$$\vec{x_3} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\frac{t^2}{2}e^{2t} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}te^{2t} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}e^{2t}$$

Then the general solution to the homogeneous system is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

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$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{2t} + c_2 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{2t} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t} \right)$$

$$+ c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{t^2}{2} e^{2t} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} te^{2t} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t} \right)$$

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COMPLEX EIGENVALUES

■ 1. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 2 & 4 \\ -2 & -2 \end{bmatrix} \vec{x}$$

Solution:

The determinant $|A - \lambda I|$ is

$$\begin{vmatrix} 2-\lambda & 4 \\ -2 & -2-\lambda \end{vmatrix} = (2-\lambda)(-2-\lambda) - (4)(-2)$$

$$\begin{vmatrix} 2-\lambda & 4 \\ -2 & -2-\lambda \end{vmatrix} = \lambda^2 - 4 + 8$$

$$\begin{vmatrix} 2-\lambda & 4 \\ -2 & -2-\lambda \end{vmatrix} = \lambda^2 + 4$$

Solve the characteristic equation for the Eigenvalues.

$$\lambda^2 + 4 = 0$$

$$\lambda = \pm 2i$$

Then for these complex conjugate Eigenvalues, $\lambda_1 = 2i$ and $\lambda_2 = -2i$, we find

$$A - (2i)I = \begin{bmatrix} 2-2i & 4 \\ -2 & -2-2i \end{bmatrix} \qquad A - (-2i)I = \begin{bmatrix} 2+2i & 4 \\ -2 & -2+2i \end{bmatrix}$$

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Put these matrices into reduced row-echelon form.

$$\begin{bmatrix} 2-2i & 4 \\ -2 & -2-2i \end{bmatrix} \qquad \begin{bmatrix} 2+2i & 4 \\ -2 & -2+2i \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{2}{1-i} \\ -2 & -2-2i \end{bmatrix} \qquad \begin{bmatrix} 1 & \frac{2}{1+i} \\ -2 & -2+2i \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{2}{1-i} \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & \frac{2}{1+i} \\ 0 & 0 \end{bmatrix}$$

If we turn these matrices back into systems of equations, we get

$$k_1 + \left(\frac{2}{1-i}\right)k_2 = 0 \qquad k_1 + \left(\frac{2}{1+i}\right)k_2 = 0$$

$$k_1 = \left(-\frac{2}{1-i}\right)k_2 \qquad k_1 = \left(-\frac{2}{1+i}\right)k_2$$

Choosing $k_2 = i - 1$ gives $k_1 = 2$ in the first system, and choosing $k_2 = -(1+i)$ gives $k_1 = 2$ in the second system, which gives the Eigenvectors

$$\vec{k_1} = \begin{bmatrix} 2 \\ i-1 \end{bmatrix} \qquad \vec{k_2} = \begin{bmatrix} 2 \\ -i-1 \end{bmatrix}$$

and the solution vectors

$$\vec{x_1} = \vec{k_1}e^{\lambda_1 t} \qquad \vec{x_2} = \vec{k_2}e^{\lambda_2 t}$$

$$\vec{x_1} = \begin{bmatrix} 2 \\ i-1 \end{bmatrix} e^{(2i)t} \qquad \vec{x_2} = \begin{bmatrix} 2 \\ -i-1 \end{bmatrix} e^{(-2i)t}$$

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Then the general solution will be

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2}$$

$$\vec{x} = c_1\begin{bmatrix} 2 \\ i-1 \end{bmatrix} e^{(2i)t} + c_2\begin{bmatrix} 2 \\ -i-1 \end{bmatrix} e^{(-2i)t}$$

■ 2. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} -6 & 4 \\ -8 & 2 \end{bmatrix}\vec{x}$$

Solution:

The determinant $|A - \lambda I|$ is

$$\begin{vmatrix} -6-\lambda & 4 \\ -8 & 2-\lambda \end{vmatrix} = (-6-\lambda)(2-\lambda) - (4)(-8)$$

$$\begin{vmatrix} -6-\lambda & 4 \\ -8 & 2-\lambda \end{vmatrix} = -12 + 6\lambda - 2\lambda + \lambda^2 + 32$$

$$\begin{vmatrix} -6-\lambda & 4 \\ -8 & 2-\lambda \end{vmatrix} = \lambda^2 + 4\lambda + 20$$

Solve the characteristic equation for the Eigenvalues.

$$\lambda^2 + 4\lambda + 20 = 0$$

$$(\lambda + 2)^2 + 16 = 0$$

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$$\lambda + 2 = \pm 4i$$

$$\lambda = -2 \pm 4i$$

Then for these complex conjugate Eigenvalues, $\lambda_1 = -2 + 4i$ and $\lambda_2 = -2 - 4i$, we find

$$A - (-2 + 4i)I = \begin{bmatrix} -6 - (-2 + 4i) & 4 \\ -8 & 2 - (-2 + 4i) \end{bmatrix}$$

$$A - (-2 + 4i)I = \begin{bmatrix} -4 - 4i & 4 \\ -8 & 4 - 4i \end{bmatrix}$$

and

$$A - (-2 - 4i)I = \begin{bmatrix} -6 - (-2 - 4i) & 4 \\ -8 & 2 - (-2 - 4i) \end{bmatrix}$$

$$A - (-2 - 4i)I = \begin{bmatrix} -4 + 4i & 4 \\ -8 & 4 + 4i \end{bmatrix}$$

Put these matrices into reduced row-echelon form.

$$\begin{bmatrix} -4 - 4i & 4 \\ -8 & 4 - 4i \end{bmatrix}$$

$$\begin{bmatrix} -4 + 4i & 4 \\ -8 & 4 + 4i \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{1}{-1 - i} \\ -8 & 4 - 4i \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{1}{-1 + i} \\ -8 & 4 + 4i \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{1}{-1 - i} \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{1}{-1 + i} \\ 0 & 0 \end{bmatrix}$$

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If we turn these matrices back into systems of equations, we get

$$k_1 + \left(\frac{1}{-1-i}\right)k_2 = 0 \qquad k_1 + \left(\frac{1}{-1+i}\right)k_2 = 0$$

$$k_1 = \left(\frac{1}{1+i}\right)k_2 \qquad k_1 = \left(\frac{1}{1-i}\right)k_2$$

Choosing $k_2 = 1 + i$ gives $k_1 = 1$ in the first system, and choosing $k_2 = 1 - i$ gives $k_1 = 1$ in the second system, which gives the Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 1+i \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 1-i \end{bmatrix}$$

and the solution vectors

$$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2}e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 1+i \end{bmatrix} e^{(2+4i)t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ 1-i \end{bmatrix} e^{(-2-4i)t}$$

Then the general solution will be

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 1+i \end{bmatrix} e^{(-2+4i)t} + c_2 \begin{bmatrix} 1 \\ 1-i \end{bmatrix} e^{(-2-4i)t}$$

■ 3. Find the general solution to the system of differential equations.

$$\overrightarrow{x}' = \begin{bmatrix} 1 & -1 \\ 5 & 5 \end{bmatrix} \overrightarrow{x}$$

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Solution:

The determinant $|A - \lambda I|$ is

$$\begin{vmatrix} 1-\lambda & -1 \\ 5 & 5-\lambda \end{vmatrix} = (1-\lambda)(5-\lambda) - (-1)(5)$$

$$\begin{vmatrix} 1-\lambda & -1 \\ 5 & 5-\lambda \end{vmatrix} = 5 - \lambda - 5\lambda + \lambda^2 + 5$$

$$\begin{vmatrix} 1-\lambda & -1 \\ 5 & 5-\lambda \end{vmatrix} = \lambda^2 - 6\lambda + 10$$

Solve the characteristic equation for the Eigenvalues.

$$\lambda^2 - 6\lambda + 10 = 0$$

$$(\lambda - 3)^2 + 1 = 0$$

$$\lambda = 3 \pm i$$

Then for the complex conjugate Eigenvalues, $\lambda = 3 \pm i$, we find

$$A - (3+i)I = \begin{bmatrix} 1-(3+i) & -1 \\ 5 & 5-(3+i) \end{bmatrix}$$

$$A - (3+i)I = \begin{bmatrix} -2-i & -1 \\ 5 & 2-i \end{bmatrix}$$

and

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$$A-(3-i)I=\begin{bmatrix}1-(3-i) & -1\\ 5 & 5-(3-i)\end{bmatrix}$$

$$A-(3-i)I=\begin{bmatrix}-2+i & -1\\ 5 & 2+i\end{bmatrix}$$

Put these matrices into reduced row-echelon form.

$$\begin{bmatrix}-2-i & -1\\ 5 & 2-i\end{bmatrix}$$

$$\begin{bmatrix}-2+i & -1\\ 5 & 2+i\end{bmatrix}$$

$$\begin{bmatrix}1 & \frac{1}{2+i}\\ 5 & 2-i\end{bmatrix}$$

$$\begin{bmatrix}1 & \frac{1}{2-i}\\ 5 & 2+i\end{bmatrix}$$

$$\begin{bmatrix}1 & \frac{1}{2+i}\\ 0 & 0\end{bmatrix}$$

$$\begin{bmatrix}1 & \frac{1}{2-i}\\ 0 & 0\end{bmatrix}$$

If we turn these matrices back into systems of equations, we get

$$k_1+\left(\frac{1}{2+i}\right)k_2=0$$

$$k_1+\left(\frac{1}{2-i}\right)k_2=0$$

$$k_1=-\left(\frac{1}{2+i}\right)k_2$$

$$k_1=-\left(\frac{1}{2-i}\right)k_2$$

Choosing $k_2=-(2+i)$ gives $k_1=1$ in the first system, and choosing $k_2=-(2-i)$ gives $k_1=1$ in the second system, which gives the Eigenvectors

$$\overrightarrow{k_1}=\begin{bmatrix}1\\ -(2+i)\end{bmatrix}$$

$$\overrightarrow{k_2}=\begin{bmatrix}1\\ -(2-i)\end{bmatrix}$$

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and the solution vectors

$$\vec{x_1} = \vec{k_1} e^{\lambda_1 t} \qquad \vec{x_2} = \vec{k_2} e^{\lambda_2 t}$$

$$\vec{x_1} = \left(\begin{bmatrix} 1 \\ -2 \end{bmatrix} \cos t + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \sin t \right) e^{3t} \qquad \vec{x_2} = \left(\begin{bmatrix} 0 \\ -1 \end{bmatrix} \cos t + \begin{bmatrix} 1 \\ -2 \end{bmatrix} \sin t \right) e^{3t}$$

Therefore, the general solution will be

$$\vec{x} = c_1 \vec{x_1} + c_2 \vec{x_2}$$

$$\vec{x} = c_1 \left(\begin{bmatrix} 1 \\ -2 \end{bmatrix} \cos t + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \sin t \right) e^{3t}$$

$$+ c_2 \left(\begin{bmatrix} 0 \\ -1 \end{bmatrix} \cos t + \begin{bmatrix} 1 \\ -2 \end{bmatrix} \sin t \right) e^{3t}$$

■ 4. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 6 & -10 \\ 5 & -4 \end{bmatrix} \vec{x}$$

Solution:

The determinant $|A - \lambda I|$ is

$$\begin{vmatrix} 6-\lambda & -10 \\ 5 & -4-\lambda \end{vmatrix} = (6-\lambda)(-4-\lambda) - (5)(-10)$$

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$$\begin{vmatrix} 6-\lambda & -10 \\ 5 & -4-\lambda \end{vmatrix} = -24 - 2\lambda + \lambda^2 + 50$$

$$\begin{vmatrix} 6-\lambda & -10 \\ 5 & -4-\lambda \end{vmatrix} = (\lambda - 1)^2 + 25$$

Solve the characteristic equation for the Eigenvalues.

$$(\lambda - 1)^2 + 25 = 0$$

$$\lambda = 1 \pm 5i$$

Then for these complex conjugate Eigenvalues, $\lambda_1 = 1 + 5i$ and $\lambda_2 = 1 - 5i$, we find

$$A - (1+5i)I = \begin{bmatrix} 6-(1+5i) & -10 \\ 5 & -4-(1+5i) \end{bmatrix}$$

$$A - (1+5i)I = \begin{bmatrix} 5-5i & -10 \\ 5 & -5-5i \end{bmatrix}$$

and

$$A - (1-5i)I = \begin{bmatrix} 6-(1-5i) & -10 \\ 5 & -4-(1-5i) \end{bmatrix}$$

$$A - (1-5i)I = \begin{bmatrix} 5+5i & -10 \\ 5 & -5+5i \end{bmatrix}$$

Put these matrices into reduced row-echelon form.

$$\begin{bmatrix} 5-5i & -10 \\ 5 & -5-5i \end{bmatrix} \qquad \begin{bmatrix} 5+5i & -10 \\ 5 & -5+5i \end{bmatrix}$$

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$$\begin{bmatrix} 1 & -\frac{2}{1-i} \\ 5 & -5-5i \end{bmatrix} \qquad \begin{bmatrix} 1 & -\frac{2}{1+i} \\ 5 & -5+5i \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{2}{1-i} \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & -\frac{2}{1+i} \\ 0 & 0 \end{bmatrix}$$

If we turn matrices back into systems of equations, we get

$$k_1 + \left(-\frac{2}{1-i}\right) k_2 = 0 \qquad k_1 + \left(-\frac{2}{1+i}\right) k_2 = 0$$

$$k_1 = \left(\frac{2}{1-i}\right) k_2 \qquad k_1 = \left(\frac{2}{1+i}\right) k_2$$

Choosing $k_2 = 1 - i$ gives $k_1 = 2$ in the first system, and choosing $k_2 = 1 + i$ gives $k_1 = 2$ in the second system, which gives the Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 2 \\ 1-i \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 2 \\ 1+i \end{bmatrix}$$

and the solution vectors

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 2 \\ 1-i \end{bmatrix} e^{(1+5i)t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 2 \\ 1+i \end{bmatrix} e^{(1-5i)t}$$

Then the general solution will be

$$\overrightarrow{x} = c_1 \overrightarrow{x_1} + c_2 \overrightarrow{x_2}$$

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$$\vec{x} = c_1 \begin{bmatrix} 2 \\ 1-i \end{bmatrix} e^{(1+5i)t} + c_2 \begin{bmatrix} 2 \\ 1+i \end{bmatrix} e^{(1-5i)t}$$

■ 5. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} -2 & 0 & 0 \\ 1 & 2 & 5 \\ 2 & -1 & 4 \end{bmatrix} \vec{x}$$

Solution:

The determinant $|A - \lambda I|$ is

$$|A - \lambda I| = \begin{vmatrix} -2-\lambda & 0 & 0 \\ 1 & 2-\lambda & 5 \\ 2 & -1 & 4-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (-2-\lambda)\begin{vmatrix} 2-\lambda & 5 \\ -1 & 4-\lambda \end{vmatrix} - 0\begin{vmatrix} 1 & 5 \\ 2 & 4-\lambda \end{vmatrix} + 0\begin{vmatrix} 1 & 2-\lambda \\ 2 & -1 \end{vmatrix}$$

$$|A - \lambda I| = (-2-\lambda)((2-\lambda)(4-\lambda)+5)$$

$$|A - \lambda I| = (-2-\lambda)((\lambda-3)^2+4)$$

Solve the characteristic equation for the Eigenvalues.

$$(-2-\lambda)((\lambda-3)^2+4) = 0$$

$$\lambda_1 = -2 \qquad \lambda_{2,3} = 3 \pm 2i$$

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Then for $\lambda_1 = -2$, we find

$$A-(-2)I=\begin{bmatrix}-2+2 & 0 & 0\\ 1 & 2+2 & 5\\ 2 & -1 & 4+2\end{bmatrix}$$

$$A-(-2)I=\begin{bmatrix}0 & 0 & 0\\ 1 & 4 & 5\\ 2 & -1 & 6\end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix}1 & 4 & 5\\ 2 & -1 & 6\\ 0 & 0 & 0\end{bmatrix}\rightarrow\begin{bmatrix}1 & 4 & 5\\ 0 & -9 & -4\\ 0 & 0 & 0\end{bmatrix}\rightarrow\begin{bmatrix}1 & 4 & 5\\ 0 & 1 & \frac{4}{9}\\ 0 & 0 & 0\end{bmatrix}\rightarrow\begin{bmatrix}1 & 0 & \frac{29}{9}\\ 0 & 1 & \frac{4}{9}\\ 0 & 0 & 0\end{bmatrix}$$

If we turn this matrix back into a system of equations, we get

$$k_1+\frac{29}{9}k_3=0$$

$$k_1=-\frac{29}{9}k_3$$

and

$$k_2+\frac{4}{9}k_3=0$$

$$k_2=-\frac{4}{9}k_3$$

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If we choose $k_3 = -9$, we find $k_1 = 29$ and $k_2 = 4$, and therefore we find the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 29 \\ 4 \\ -9 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 29 \\ 4 \\ -9 \end{bmatrix} e^{-2t}$$

Then for the complex Eigenvalue $\lambda_2 = 3 + 2i$, we find

$$A - (3+2i)I = \begin{bmatrix} -2-(3+2i) & 0 & 0 \\ 1 & 2-(3+2i) & 5 \\ 2 & -1 & 4-(3+2i) \end{bmatrix}$$

$$A - (3+2i)I = \begin{bmatrix} -5-2i & 0 & 0 \\ 1 & -1-2i & 5 \\ 2 & -1 & 1-2i \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1-2i & 5 \\ 2 & -1 & 1-2i \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1-2i & 5 \\ 2 & -1 & 1-2i \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1-2i & 5 \\ 0 & -1 & 1-2i \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1-2i \\ 0 & -1-2i & 5 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1+2i \\ 0 & -1-2i & 5 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1+2i \\ 0 & 0 & 0 \end{bmatrix}$$

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If we turn this matrix back into a system of equations, we get

$$k_1 = 0$$

and

$$k_2 + (-1 + 2i)k_3 = 0$$

$$k_2 = (1 - 2i)k_3$$

Choosing $k_3 = 1$ gives $k_2 = 1 - 2i$, which gives the Eigenvector

$$\overrightarrow{k_2} = \begin{bmatrix} 0 \\ 1 - 2i \\ 1 \end{bmatrix}$$

and the solution vectors

$$\overrightarrow{x_2} = \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \cos(2t) - \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix} \sin(2t) \right) e^{3t}$$

$$\overrightarrow{x_3} = \left(\begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix} \cos(2t) + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \sin(2t) \right) e^{3t}$$

Therefore, the general solution will be

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 29 \\ 4 \\ -9 \end{bmatrix} e^{-2t} + c_2 \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \cos(2t) - \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix} \sin(2t) \right) e^{3t}$$

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$$+c_3\left(\begin{bmatrix}0\\-2\\0\end{bmatrix}\cos(2t)+\begin{bmatrix}0\\1\\1\end{bmatrix}\sin(2t)\right)e^{3t}$$

■ 6. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix}1 & 2 & 3\\0 & -1 & 2\\0 & -5 & 1\end{bmatrix}\vec{x}$$

Solution:

The determinant $|A-\lambda I|$ is

$$|A-\lambda I| = \begin{vmatrix}1-\lambda & 2 & 3\\0 & -1-\lambda & 2\\0 & -5 & 1-\lambda\end{vmatrix}$$

$$|A-\lambda I| = (1-\lambda)\begin{vmatrix}-1-\lambda & 2\\-5 & 1-\lambda\end{vmatrix} - 2\begin{vmatrix}0 & 2\\0 & 1-\lambda\end{vmatrix} + 3\begin{vmatrix}0 & -1-\lambda\\0 & -5\end{vmatrix}$$

$$|A-\lambda I| = (1-\lambda)((-1-\lambda)(1-\lambda)+10)$$

$$|A-\lambda I| = (1-\lambda)(\lambda^2+9)$$

Solve the characteristic equation for the Eigenvalues.

$$\lambda_1 = 1 \qquad \lambda_{2,3} = \pm 3i$$

Then for $\lambda_1 = 1$, we find

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$$A-(1)I=\begin{bmatrix}1-1 & 2 & 3\\ 0 & -1-1 & 2\\ 0 & -5 & 1-1\end{bmatrix}$$

$$A-(1)I=\begin{bmatrix}0 & 2 & 3\\ 0 & -2 & 2\\ 0 & -5 & 0\end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix}0 & 1 & 0\\ 0 & 2 & 3\\ 0 & -2 & 2\end{bmatrix}\rightarrow\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 3\\ 0 & -2 & 2\end{bmatrix}\rightarrow\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 3\\ 0 & 0 & 2\end{bmatrix}\rightarrow\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}$$

If we turn this matrix back into a systems of equations, we get

$$k_2=0$$

$$k_3=0$$

If we choose $k_1=1$, we get the Eigenvector

$$\overrightarrow{k_1}=\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1}=\overrightarrow{k_1}e^{\lambda_1 t}$$

$$\overrightarrow{x_1}=\begin{bmatrix}1\\ 0\\ 0\end{bmatrix}e^{t}$$

Then for the complex Eigenvalue $\lambda_2=3i$, we find

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$$A-(3i)I=\begin{bmatrix}1-3i & 2 & 3\\ 0 & -1-3i & 2\\ 0 & -5 & 1-3i\end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix}1 & \frac{2}{1-3i} & \frac{3}{1-3i}\\ 0 & -1-3i & 2\\ 0 & -5 & 1-3i\end{bmatrix}\rightarrow\begin{bmatrix}1 & \frac{1+3i}{5} & \frac{3(1+3i)}{10}\\ 0 & -1-3i & 2\\ 0 & -5 & 1-3i\end{bmatrix}\rightarrow\begin{bmatrix}1 & \frac{1+3i}{5} & \frac{3(1+3i)}{10}\\ 0 & -5 & 1-3i\\ 0 & -1-3i & 2\end{bmatrix}$$

$$\rightarrow\begin{bmatrix}1 & \frac{1+3i}{5} & \frac{3(1+3i)}{10}\\ 0 & 1 & \frac{-1+3i}{5}\\ 0 & -1-3i & 2\end{bmatrix}\rightarrow\begin{bmatrix}1 & \frac{1+3i}{5} & \frac{3(1+3i)}{10}\\ 0 & 1 & \frac{-1+3i}{5}\\ 0 & 0 & 0\end{bmatrix}\rightarrow\begin{bmatrix}1 & 0 & \frac{7+9i}{10}\\ 0 & 1 & \frac{-1+3i}{5}\\ 0 & 0 & 0\end{bmatrix}$$

If we turn this matrix back into a system of equations, we get

$$k_1+\left(\frac{7+9i}{10}\right)k_3=0$$

$$k_1=-\left(\frac{7+9i}{10}\right)k_3$$

and

$$k_2+\left(\frac{-1+3i}{5}\right)k_3=0$$

$$k_2=\left(\frac{1-3i}{5}\right)k_3$$

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Choosing $k_3 = 10$ gives $k_1 = -7 - 9i$ and $k_2 = 2 - 6i$, which gives us the Eigenvector

$$\vec{k_2} = \begin{bmatrix} -7 - 9i \\ 2 - 6i \\ 10 \end{bmatrix}$$

and the solution vector

$$\vec{x_2} = \begin{bmatrix} -7 \\ 2 \\ 10 \end{bmatrix} \cos(3t) - \begin{bmatrix} -9 \\ -6 \\ 0 \end{bmatrix} \sin(3t)$$

$$\vec{x_3} = \begin{bmatrix} -9 \\ -6 \\ 0 \end{bmatrix} \cos(3t) + \begin{bmatrix} -7 \\ 2 \\ 10 \end{bmatrix} \sin(3t)$$

Therefore, the general solution will be

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} -7 \\ 2 \\ 10 \end{bmatrix} \cos(3t) - \begin{bmatrix} -9 \\ -6 \\ 0 \end{bmatrix} \sin(3t) \right)$$

$$+ c_3 \left(\begin{bmatrix} -9 \\ -6 \\ 0 \end{bmatrix} \cos(3t) + \begin{bmatrix} -7 \\ 2 \\ 10 \end{bmatrix} \sin(3t) \right)$$

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PHASE PORTRAITS FOR DISTINCT REAL EIGENVALUES

■ 1. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -4 & 4 \\ -2 & 5 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} -4 - \lambda & 4 \\ -2 & 5 - \lambda \end{bmatrix}$$

and the characteristic equation

$$(-4 - \lambda)(5 - \lambda) - (4)(-2) = 0$$

$$\lambda^2 - \lambda - 12 = 0$$

$$(\lambda - 4)(\lambda + 3) = 0$$

$$\lambda = -3, 4$$

gives the Eigenvalues $\lambda_1 = -3$ and $\lambda_2 = 4$, and their associated Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

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The Eigenvector $\overrightarrow{k_1} = (4,1)$ lies along the line $y = (1/4)x$, and the Eigenvector $\overrightarrow{k_2} = (1,2)$ lies along the line $y = 2x$, so we'll sketch these lines to represent the Eigenvectors in our phase portrait.

The Eigenvalue associated with $\overrightarrow{k_1} = (4,1)$ is $\lambda = -3$, which means the direction along that trajectory is toward the origin. The Eigenvalue associated with $\overrightarrow{k_2} = (1,2)$ is $\lambda_2 = 4$, which means the direction along that trajectory is away from the origin.

Because the Eigenvalues have opposite signs, we're dealing with an unstable saddle point that attracts trajectories in some places and repels them in others.

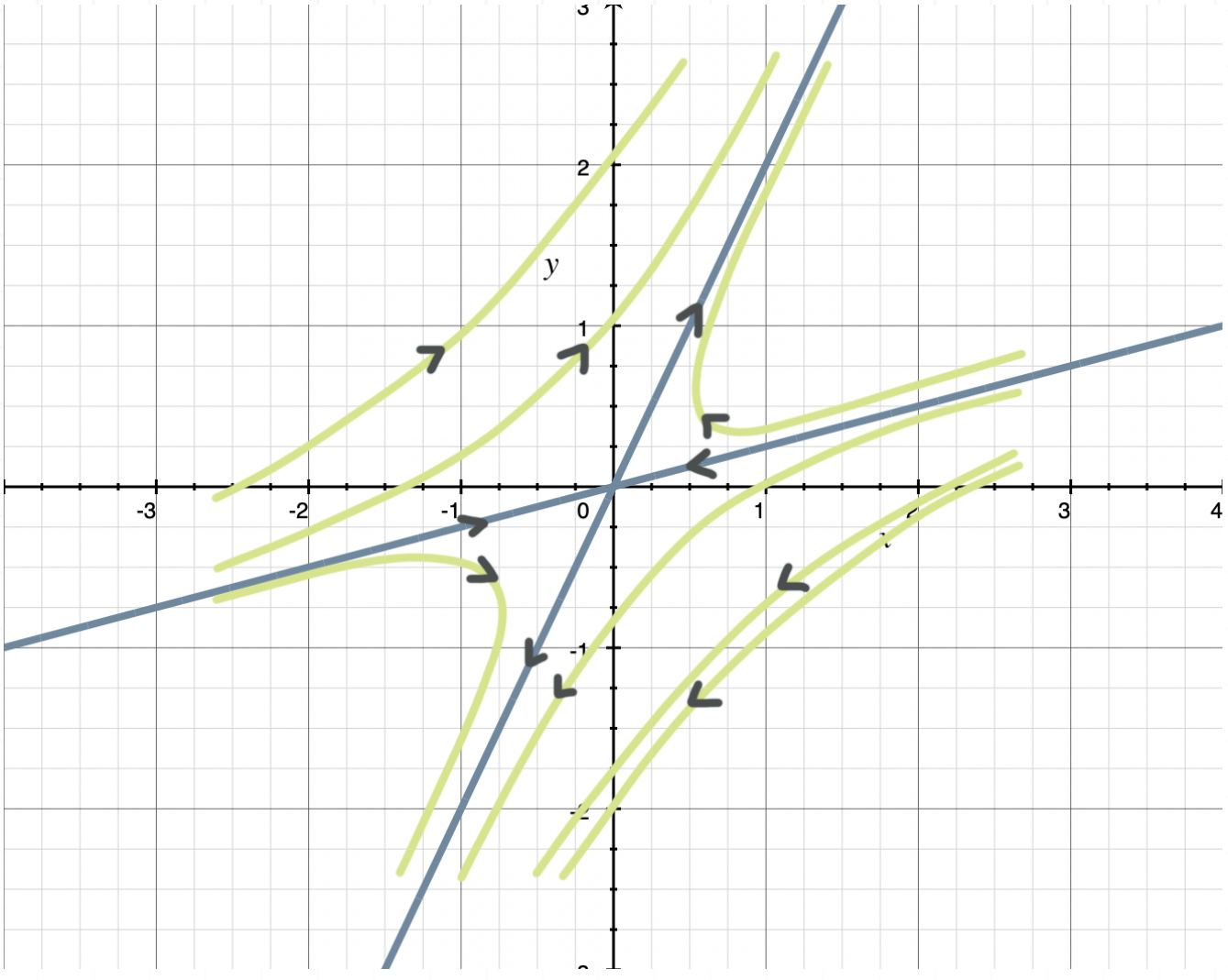
To apply the $t \to \pm\infty$ test, we'll use the general solution to the system,

$$\overrightarrow{x} = c_1 \begin{bmatrix} 4 \\ 1 \end{bmatrix} e^{-3t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{4t}$$

Because e^{-3t} dominates e^{4t} as $t \to -\infty$, the $\overrightarrow{k_2} = (1,2)$ vector drops away first, meaning that our trajectories are going to "start" parallel to $\overrightarrow{k_1} = (4,1)$. On the other end, e^{-3t} goes to 0 faster than e^{4t} as $t \to \infty$, so the $\overrightarrow{k_1} = (4,1)$ vector will drop away first, meaning that our trajectories are going to "end" parallel to $\overrightarrow{k_2} = (1,2)$.

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■ 2. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 2 & 1 \\ 4 & -1 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & 1 \\ 4 & -1 - \lambda \end{bmatrix}$$

and the characteristic equation

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$$(2-\lambda)(-1-\lambda)-(1)(4)=0$$

$$\lambda^2-\lambda-6=0$$

$$(\lambda-3)(\lambda+2)=0$$

$$\lambda=3,\ -2$$

gives the Eigenvalues $\lambda_1=3$ and $\lambda_2=-2$, and their associated Eigenvectors

$$\overrightarrow{k_1}=\begin{bmatrix}1\\1\end{bmatrix} \qquad \overrightarrow{k_2}=\begin{bmatrix}1\\-4\end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1}=(1,1)$ lies along the line $y=x$, and the Eigenvector $\overrightarrow{k_2}=(1,-4)$ lies along the line $y=-4x$, so we'll sketch these lines to represent the Eigenvectors in our phase portrait.

The Eigenvalue associated with $\overrightarrow{k_1}=(1,1)$ is $\lambda=3$, which means the direction along that trajectory is away from the origin. The Eigenvalue associated with $\overrightarrow{k_2}=(1,-4)$ is $\lambda_2=-2$, which means the direction along that trajectory is toward the origin.

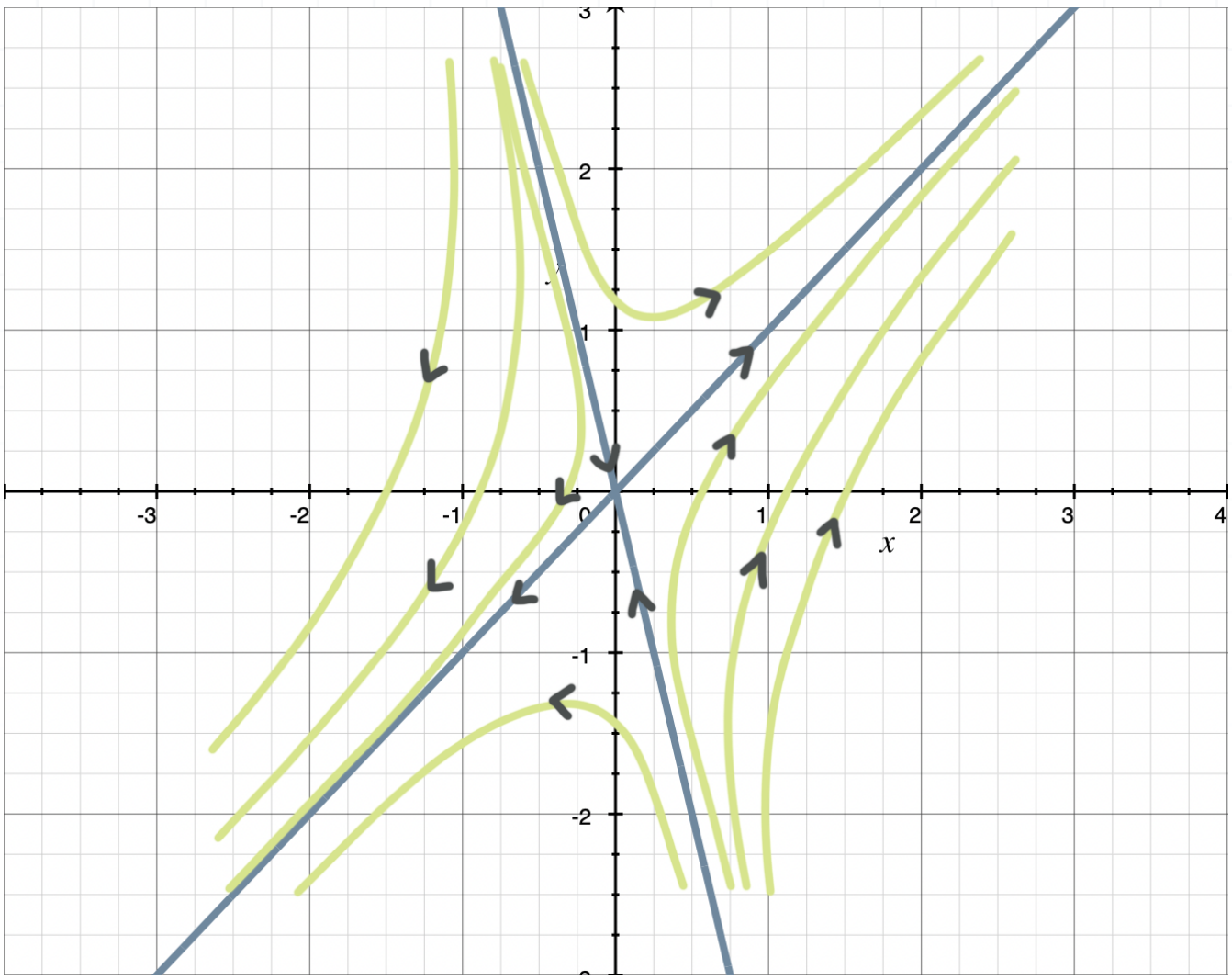
Because the Eigenvalues have opposite signs, we're dealing with an unstable saddle point that attracts trajectories in some places and repels them in others.

To apply the $t\to\pm\infty$ test, we'll use the general solution to the system,

$$\overrightarrow{x}=c_1\begin{bmatrix}1\\1\end{bmatrix}e^{3t}+c_2\begin{bmatrix}1\\-4\end{bmatrix}e^{-2t}$$

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Because e^{-2t} dominates e^{3t} as $t \to -\infty$, the $\overrightarrow{k_1} = (1,1)$ vector drops away first, meaning that our trajectories are going to "start" parallel to $\overrightarrow{k_2} = (1,-4)$. On the other end, e^{-2t} goes to 0 faster than e^{3t} as $t \to \infty$, so the $\overrightarrow{k_2} = (1,-4)$ vector will drop away first, meaning that our trajectories are going to "end" parallel to $\overrightarrow{k_1} = (1,1)$.



3. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

Solution:

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The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & 1 \\ 4 & 2 - \lambda \end{bmatrix}$$

and the characteristic equation

$$(2 - \lambda)(2 - \lambda) - (1)(4) = 0$$

$$\lambda^2 - 4\lambda = 0$$

$$\lambda(\lambda - 4) = 0$$

$$\lambda = 0,\ 4$$

gives the Eigenvalues $\lambda_1 = 0$ and $\lambda_2 = 4$, and their associated Eigenvectors

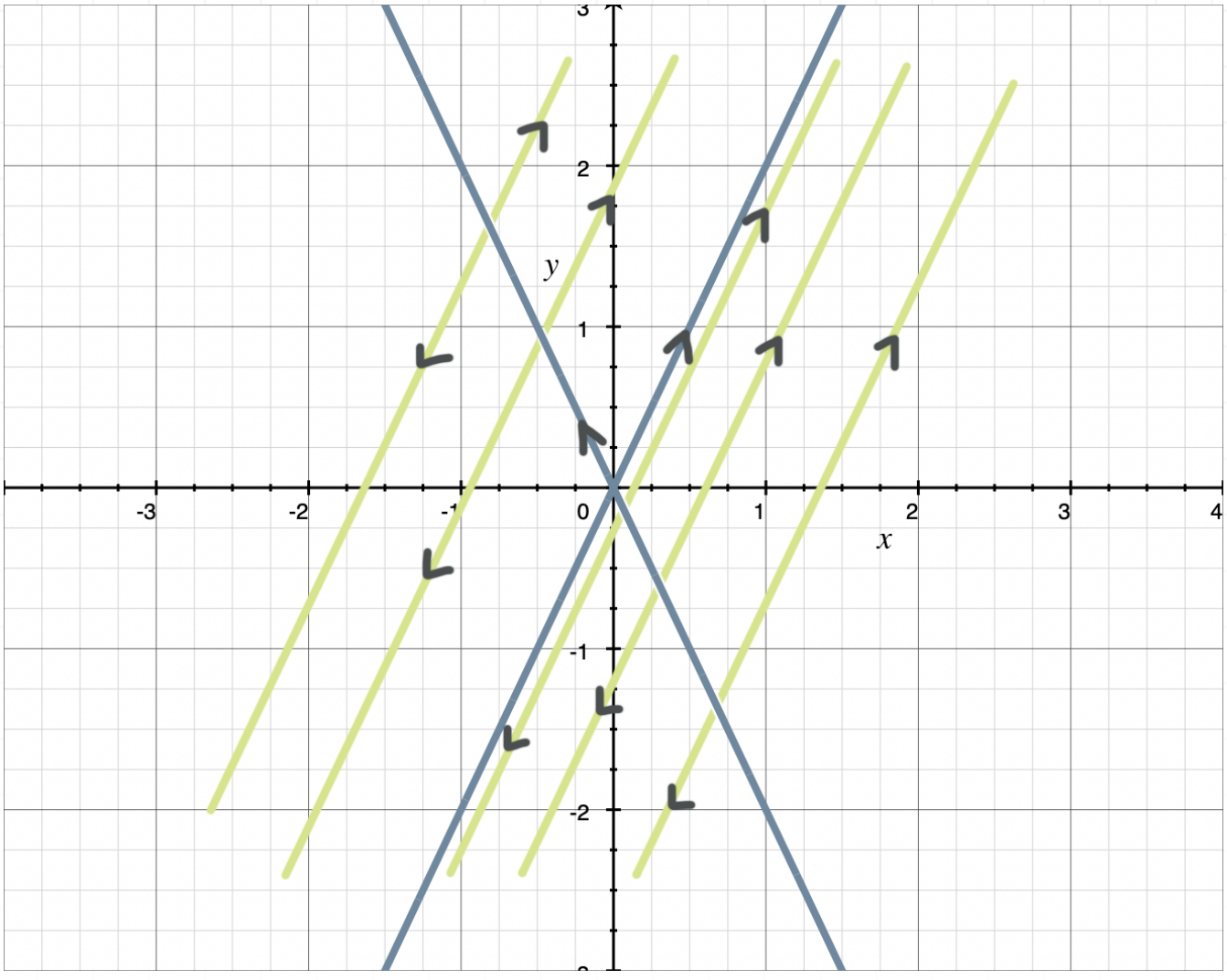
$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1} = (1, -2)$ lies along the line $y = -2x$, and the Eigenvector $\overrightarrow{k_2} = (1,2)$ lies along the line $y = 2x$, so we'll sketch these lines to represent the Eigenvectors in our phase portrait.

The Eigenvalue associated with $\overrightarrow{k_1} = (1, -2)$ is $\lambda = 0$, which means equilibrium exists along the entire line $\overrightarrow{k_1} = (1, -2)$. The Eigenvalue associated with $\overrightarrow{k_2} = (1,2)$ is $\lambda_2 = 4$, which means the direction along that trajectory is away from the origin.

Because the non-zero Eigenvalue is positive, we're dealing with an unstable line of equilibrium along $\overrightarrow{k_1} = (1, -2)$ that repels all trajectories.

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■ 4. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 5 - \lambda & 1 \\ 4 & 2 - \lambda \end{bmatrix}$$

and the characteristic equation

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$$(5-\lambda)(2-\lambda)-(1)(4)=0$$

$$\lambda^2-7\lambda+6=0$$

$$(\lambda-6)(\lambda-1)=0$$

$$\lambda=6,\ 1$$

gives the Eigenvalues $\lambda_1=6$ and $\lambda_2=1$, and their associated Eigenvectors

$$\overrightarrow{k_1}=\begin{bmatrix}1\\1\end{bmatrix} \qquad \overrightarrow{k_2}=\begin{bmatrix}1\\-4\end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1}=(1,1)$ lies along the line $y=x$, and the Eigenvector $\overrightarrow{k_2}=(1,-4)$ lies along the line $y=-4x$, so we'll sketch these lines to represent the Eigenvectors in our phase portrait.

The Eigenvalue associated with $\overrightarrow{k_1}=(1,1)$ is $\lambda=6$, which means the direction along that trajectory is away from the origin. The Eigenvalue associated with $\overrightarrow{k_2}=(1,-4)$ is $\lambda_2=1$, which means the direction along that trajectory is also away from the origin.

Because both Eigenvalues are positive, we're dealing with an unstable repeller node that repels all trajectories.

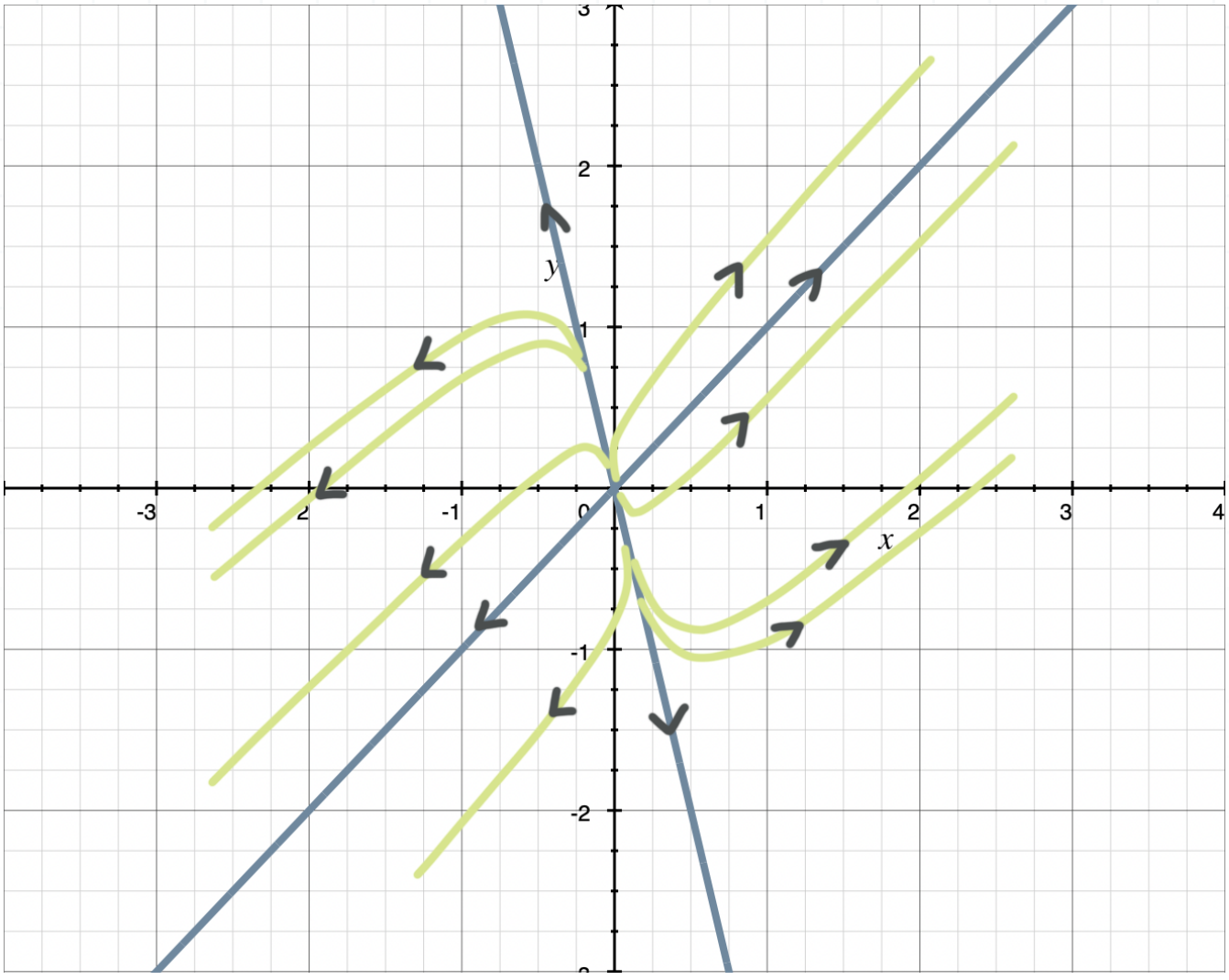
To apply the $t\to\pm\infty$ test, we'll use the general solution to the system,

$$\overrightarrow{x}=c_1\begin{bmatrix}1\\1\end{bmatrix}e^{6t}+c_2\begin{bmatrix}1\\-4\end{bmatrix}e^{t}$$

Because e^t dominates e^{6t} as $t\to-\infty$, the $\overrightarrow{k_1}=(1,1)$ vector drops away first, meaning that our trajectories are going to "start" parallel to $\overrightarrow{k_2}=(1,-4)$.

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On the other end, e^{6t} dominates e^{t} as $t \to \infty$, so the $\overrightarrow{k_2} = (1, -4)$ vector will drop away first, meaning that our trajectories are going to "end" parallel to $\overrightarrow{k_1} = (1,1)$.



5. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -2 & 4 \\ 0 & 0 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

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$$A - \lambda I = \begin{bmatrix} -2 - \lambda & 4 \\ 0 & -\lambda \end{bmatrix}$$

and the characteristic equation

$$(-2 - \lambda)(-\lambda) - (4)(0) = 0$$

$$\lambda^2 + 2\lambda = 0$$

$$\lambda(\lambda + 2) = 0$$

$$\lambda = 0, \ -2$$

gives the Eigenvalues $\lambda_1 = 0$ and $\lambda_2 = -2$, and their associated Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

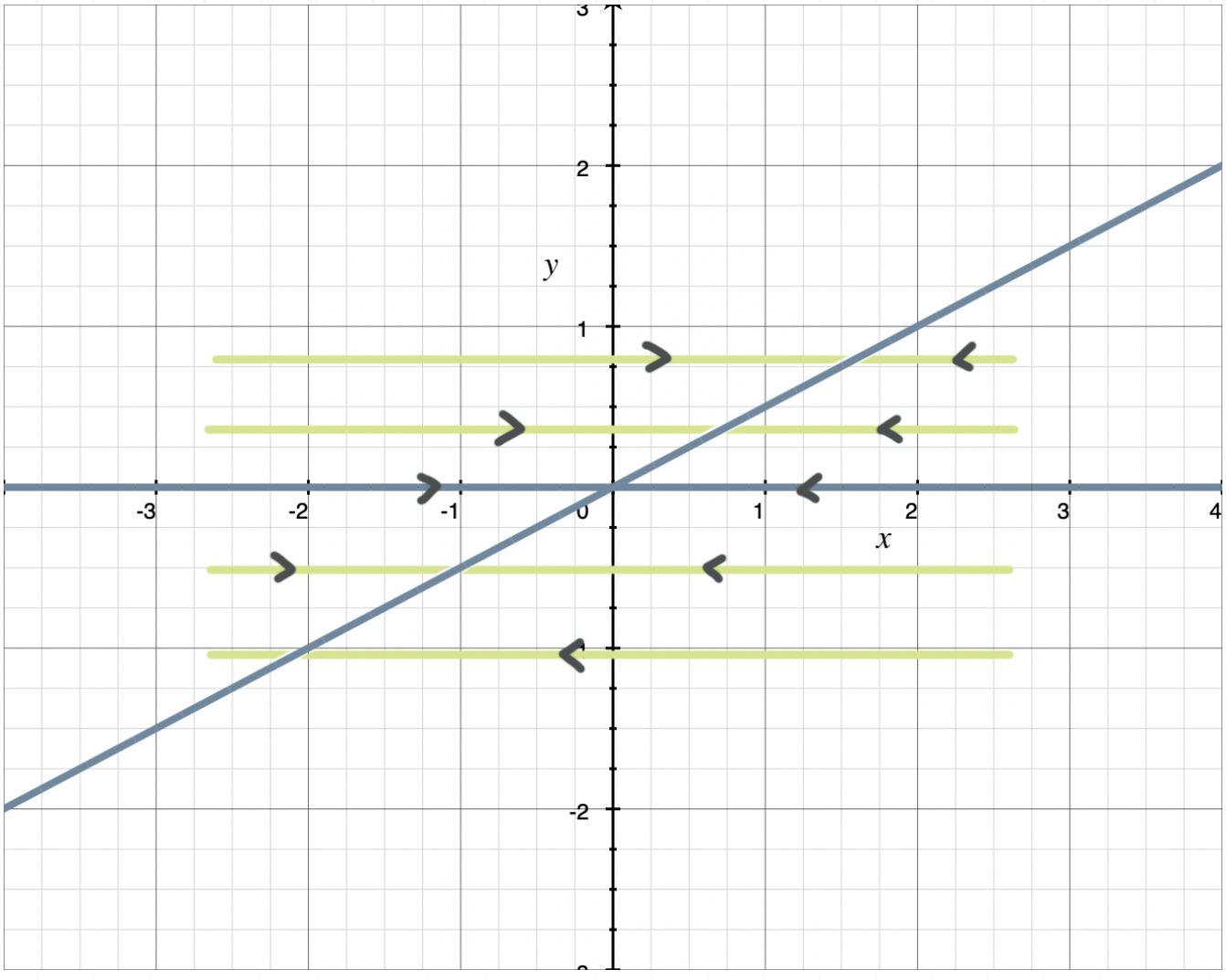
The Eigenvector $\overrightarrow{k_1} = (2,1)$ lies along the line $y = (1/2)x$, and the Eigenvector $\overrightarrow{k_2} = (1,0)$ lies along the line $y = 0$, so we'll sketch these lines to represent the Eigenvectors in our phase portrait.

The Eigenvalue associated with $\overrightarrow{k_1} = (2,1)$ is $\lambda = 0$, which means equilibrium will exist everywhere along the line $y = (1/2)x$. The Eigenvalue associated with $\overrightarrow{k_2} = (1,0)$ is $\lambda_2 = -2$, which means the direction along that trajectory is away from the origin.

Because the non-zero Eigenvalue is negative, we're dealing with a stable line of equilibrium that attracts all trajectories.

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6. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -3 & -1 \\ 0 & -4 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} -3 - \lambda & -1 \\ 0 & -4 - \lambda \end{bmatrix}$$

and the characteristic equation

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$$(-3-\lambda)(-4-\lambda)-(-1)(0)=0$$

$$\lambda^2+7\lambda+12=0$$

$$(\lambda+4)(\lambda+3)=0$$

$$\lambda=-3,\ -4$$

gives the Eigenvalues $\lambda_1=-3$ and $\lambda_2=-4$, and their associated Eigenvectors

$$\overrightarrow{k_1}=\begin{bmatrix}1\\0\end{bmatrix} \qquad \overrightarrow{k_2}=\begin{bmatrix}1\\1\end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1}=(1,0)$ lies along the line $y=0$, and the Eigenvector $\overrightarrow{k_2}=(1,1)$ lies along the line $y=x$, so we'll sketch these lines to represent the Eigenvectors in our phase portrait.

The Eigenvalue associated with $\overrightarrow{k_1}=(1,0)$ is $\lambda=-3$, which means the direction along that trajectory is toward the origin. The Eigenvalue associated with $\overrightarrow{k_2}=(1,1)$ is $\lambda_2=-4$, which means the direction along that trajectory is also toward the origin.

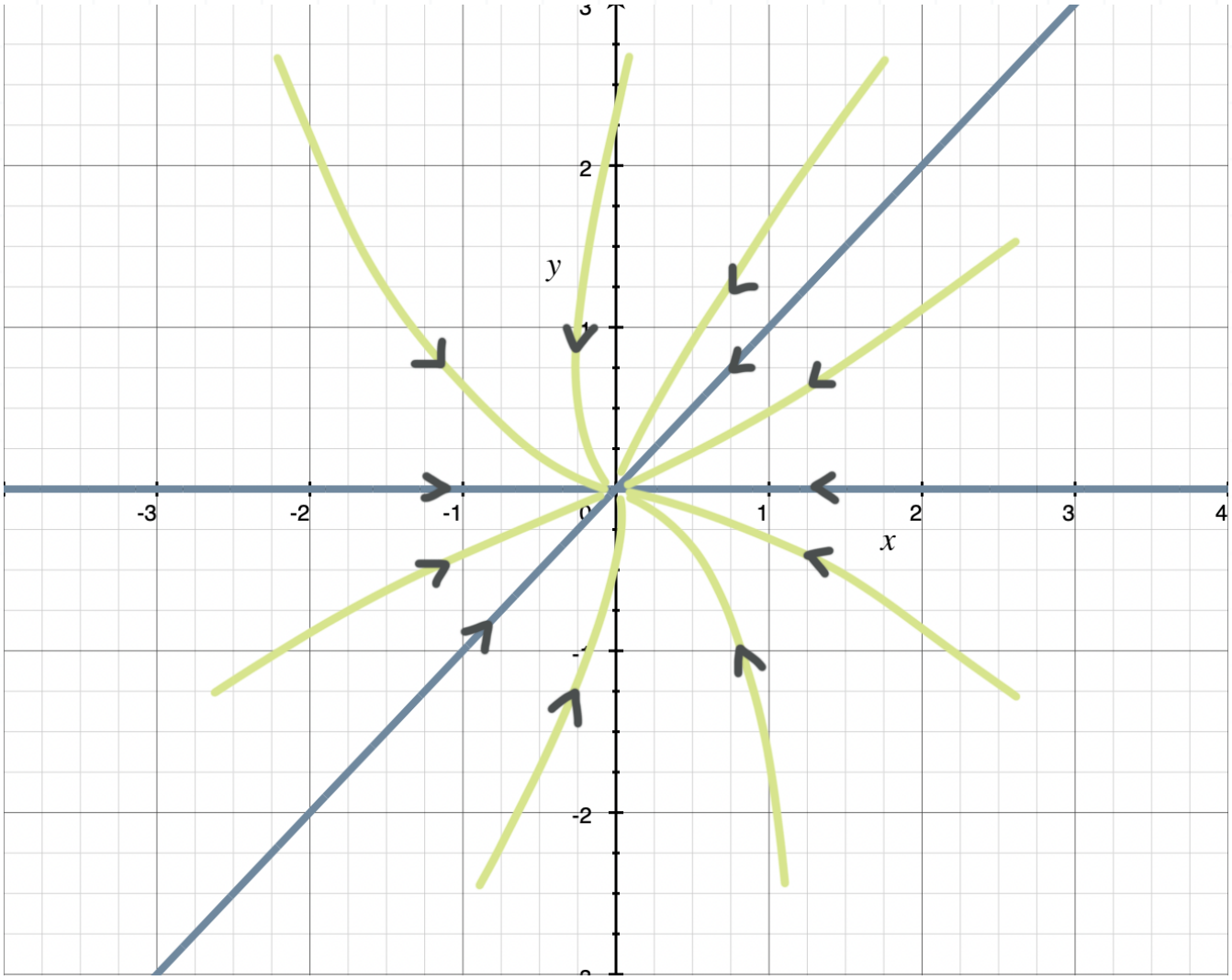
Because both the Eigenvalues are negative, we're dealing with an asymptotically stable node that attracts all trajectories.

To apply the $t\to\pm\infty$ test, we'll use the general solution to the system,

$$\overrightarrow{x}=c_1\begin{bmatrix}1\\0\end{bmatrix}e^{-3t}+c_2\begin{bmatrix}1\\1\end{bmatrix}e^{-4t}$$

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Because e^{-4t} dominates e^{-3t} as $t \to -\infty$, the $\overrightarrow{k_1} = (1,0)$ vector drops away first, meaning that our trajectories are going to "start" parallel to $\overrightarrow{k_2} = (1,1)$. On the other end, e^{-4t} goes to 0 faster than e^{-3t} as $t \to \infty$, so the $\overrightarrow{k_2} = (1,1)$ vector will drop away first, meaning that our trajectories are going to "end" parallel to $\overrightarrow{k_1} = (1,0)$.



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PHASE PORTRAITS FOR EQUAL REAL EIGENVALUES

■ 1. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 6 & 2 \\ -8 & -2 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 6 - \lambda & 2 \\ -8 & -2 - \lambda \end{bmatrix}$$

and the characteristic equation

$$(6 - \lambda)(-2 - \lambda) - (2)(-8) = 0$$

$$\lambda^2 - 4\lambda + 4 = 0$$

$$(\lambda - 2)(\lambda - 2) = 0$$

$$\lambda = 2, 2$$

gives the Eigenvalues $\lambda_1 = \lambda_2 = 2$, and the associated Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

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The Eigenvector $\overrightarrow{k_1} = (1, -2)$ lies along the line $y = -2x$, so we'll sketch this line in our phase portrait. The Eigenvalue associated with $\overrightarrow{k_1} = (1, -2)$ is $\lambda = 2$, which means the direction along that trajectory is away from the origin. Because the Eigenvalue is positive, we're dealing with an unstable repeller node that repels all trajectories.

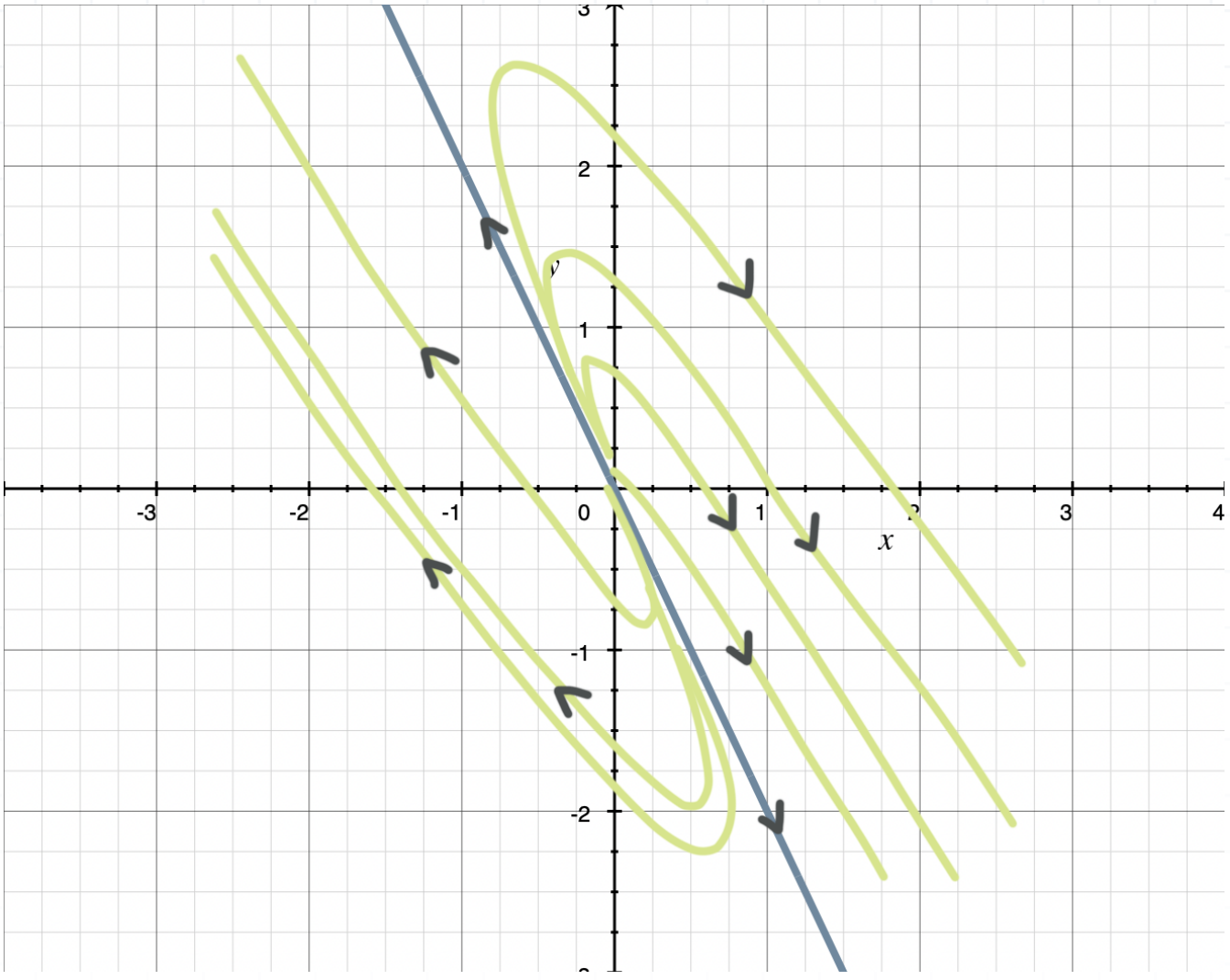
If we test the vector $\overrightarrow{x} = (1,0)$ in the matrix equation associated with the system, we get

$$\overrightarrow{x}' = \begin{bmatrix} 6 & 2 \\ -8 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\overrightarrow{x}' = \begin{bmatrix} 6 \\ -8 \end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(6, -8)$ (toward the third quadrant), which means that the phase portrait must look something like

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■ 2. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -3 & -4 \\ 4 & 5 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} -3 - \lambda & -4 \\ 4 & 5 - \lambda \end{bmatrix}$$

and the characteristic equation

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$$(-3-\lambda)(5-\lambda)-(-4)(4)=0$$

$$\lambda^2-2\lambda+1=0$$

$$(\lambda-1)(\lambda-1)=0$$

$$\lambda=1,\ 1$$

gives the Eigenvalues $\lambda_1=\lambda_2=1$, and the associated Eigenvector

$$\overrightarrow{k_1}=\begin{bmatrix}1\\-1\end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1}=(1,-1)$ lies along the line $y=-x$, so we'll sketch this line in our phase portrait. The Eigenvalue associated with $\overrightarrow{k_1}=(1,-1)$ is $\lambda=1$, which means the direction along that trajectory is away from the origin. Because the Eigenvalue is positive, we're dealing with an unstable repeller node that repels all trajectories.

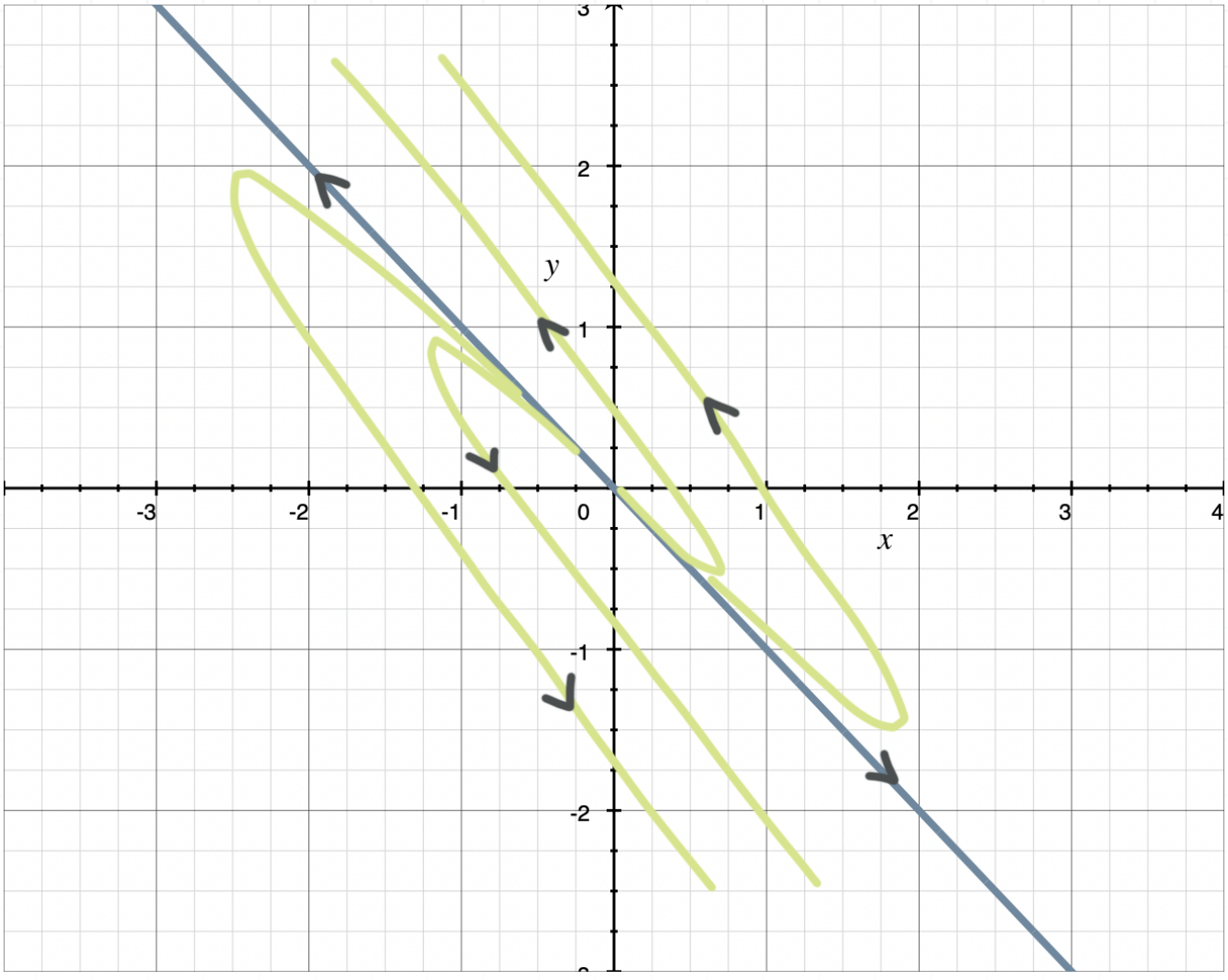
If we test the vector $\overrightarrow{x}=(1,0)$ in the matrix equation associated with the system, we get

$$\overrightarrow{x}'=\begin{bmatrix}-3 & -4\\ 4 & 5\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}$$

$$\overrightarrow{x}'=\begin{bmatrix}-3\\4\end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(-3,4)$ (toward the second quadrant), which means that the phase portrait must look something like

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■ 3. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -1 & 9 \\ -1 & -7 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} -1 - \lambda & 9 \\ -1 & -7 - \lambda \end{bmatrix}$$

and the characteristic equation

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$$(-1-\lambda)(-7-\lambda)-(9)(-1)=0$$

$$\lambda^2+8\lambda+16=0$$

$$(\lambda+4)(\lambda+4)=0$$

$$\lambda=-4,\ -4$$

gives the Eigenvalues $\lambda_1=\lambda_2=-4$, and the associated Eigenvector

$$\overrightarrow{k_1}=\begin{bmatrix}3\\-1\end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1}=(3,-1)$ lies along the line $y=(-1/3)x$, so we'll sketch this line in our phase portrait. The Eigenvalue associated with $\overrightarrow{k_1}=(3,-1)$ is $\lambda=-4$, which means the direction along that trajectory is toward the origin. Because the Eigenvalue is negative, we're dealing with an asymptotically stable attractor node that attracts all trajectories.

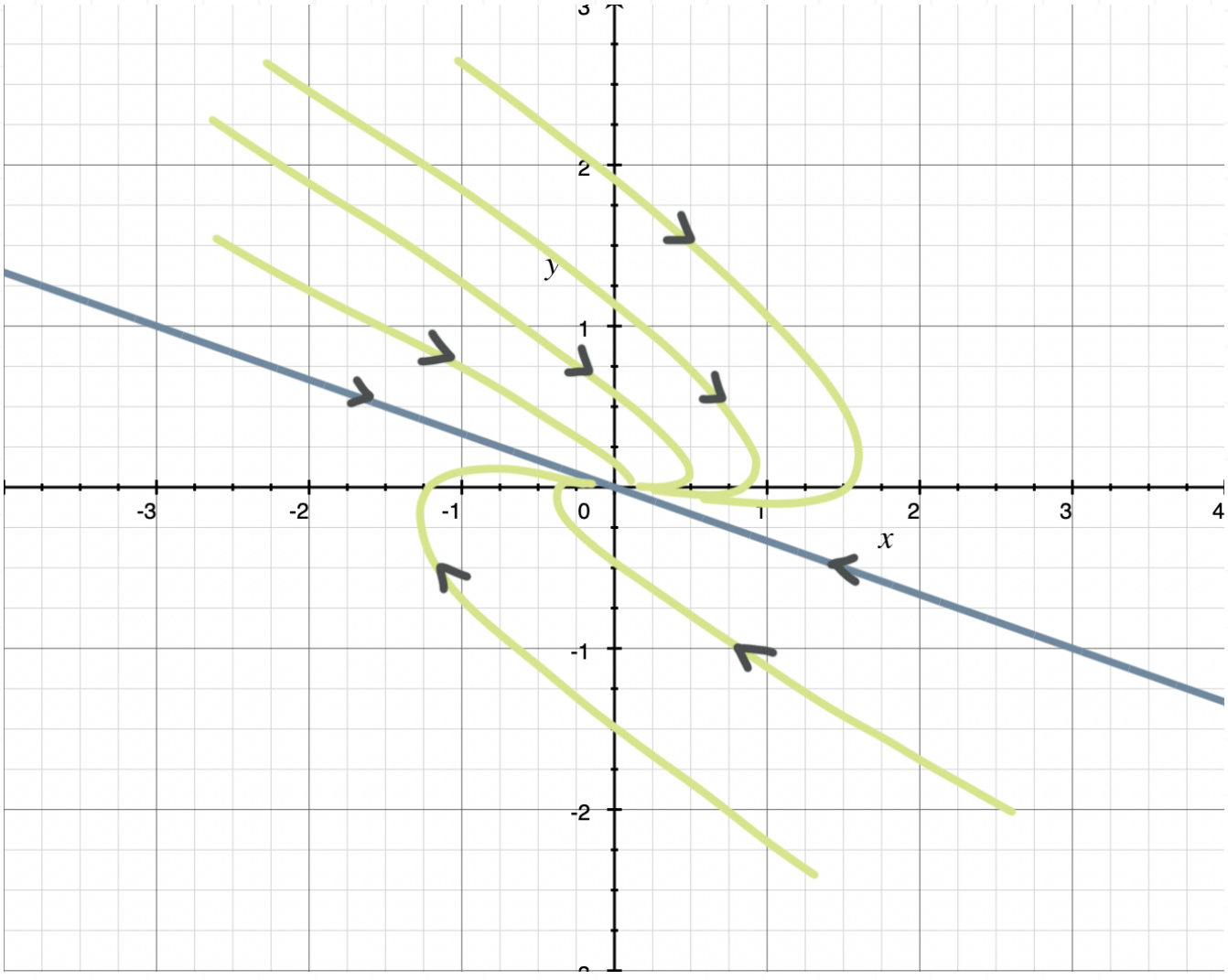
If we test the vector $\overrightarrow{x}=(1,0)$ in the matrix equation associated with the system, we get

$$\overrightarrow{x}'=\begin{bmatrix}-1 & 9\\-1 & -7\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}$$

$$\overrightarrow{x}'=\begin{bmatrix}-1\\-1\end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(-1,-1)$ (toward the third quadrant), which means that the phase portrait must look something like

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■ 4. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 10 & 4 \\ -4 & 2 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 10 - \lambda & 4 \\ -4 & 2 - \lambda \end{bmatrix}$$

and the characteristic equation

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$$(10-\lambda)(2-\lambda)-(4)(-4)=0$$

$$\lambda^2-12\lambda+36=0$$

$$(\lambda-6)(\lambda-6)=0$$

$$\lambda=6,\ 6$$

gives the Eigenvalues $\lambda_1=\lambda_2=6$, and the associated Eigenvector

$$\overrightarrow{k_1}=\begin{bmatrix}1\\-1\end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1}=(1,-1)$ lies along the line $y=-x$, so we'll sketch this line in our phase portrait. The Eigenvalue associated with $\overrightarrow{k_1}=(1,-1)$ is $\lambda=6$, which means the direction along that trajectory is away from the origin. Because the Eigenvalue is positive, we're dealing with an unstable repeller node that repels all trajectories.

If we test the vector $\overrightarrow{x}=(1,0)$ in the matrix equation associated with the system, we get

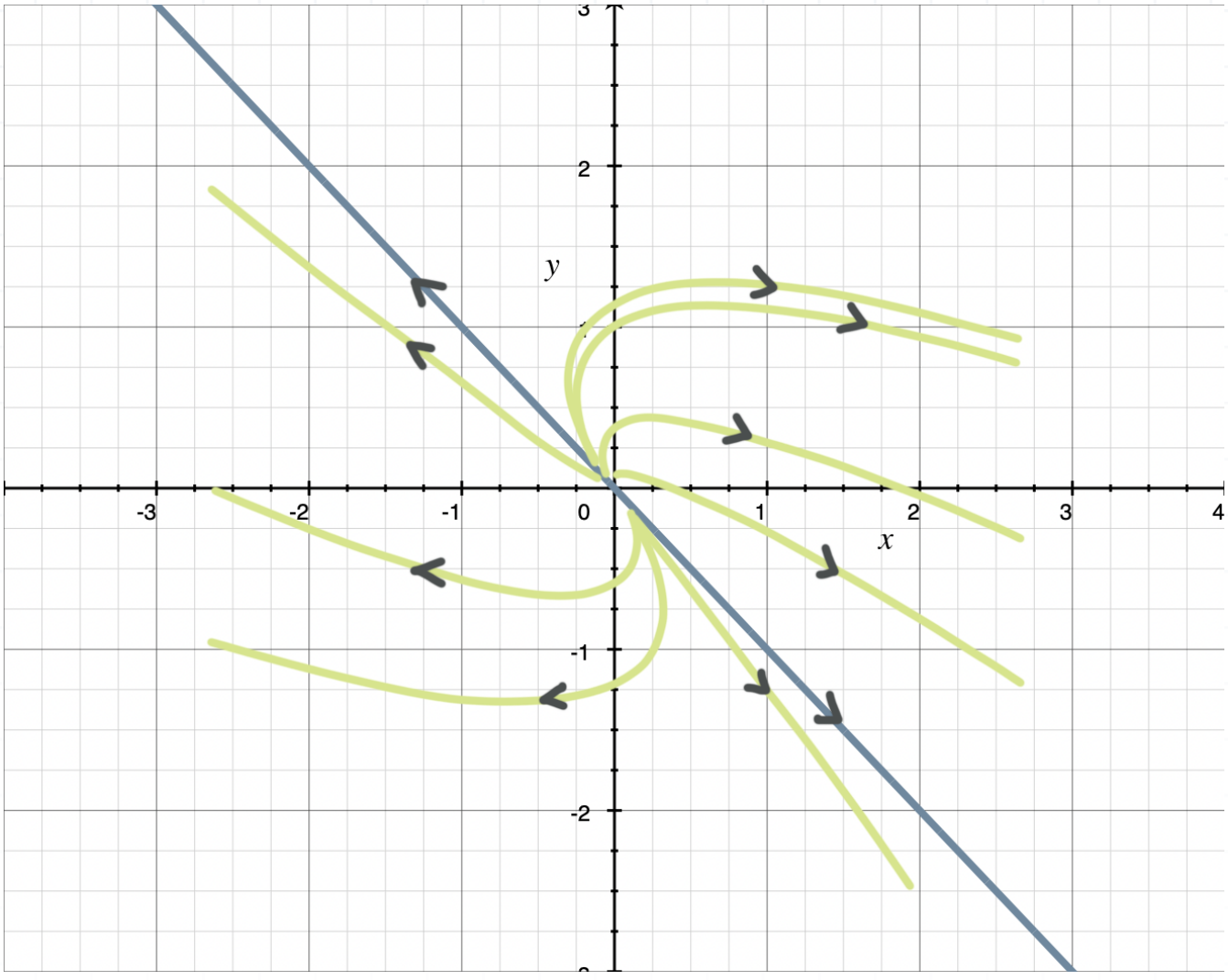
$$\overrightarrow{x}'=\begin{bmatrix}10&4\\-4&2\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}$$

$$\overrightarrow{x}'=\begin{bmatrix}10\\-4\end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(10,-4)$ (toward the fourth quadrant), which means that the phase portrait must look something like

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5. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 16 & -5 \\ 5 & 6 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 16 - \lambda & -5 \\ 5 & 6 - \lambda \end{bmatrix}$$

and the characteristic equation

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$$(16-\lambda)(6-\lambda)-(-5)(5)=0$$

$$\lambda^2-22\lambda+121=0$$

$$(\lambda-11)(\lambda-11)=0$$

$$\lambda=11,\ 11$$

gives the Eigenvalues $\lambda_1=\lambda_2=11$, and the associated Eigenvector

$$\overrightarrow{k_1}=\begin{bmatrix}1\\1\end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1}=(1,1)$ lies along the line $y=x$, so we'll sketch this line in our phase portrait. The Eigenvalue associated with $\overrightarrow{k_1}=(1,1)$ is $\lambda=11$, which means the direction along that trajectory is away from the origin. Because the Eigenvalue is positive, we're dealing with an unstable repeller node that repels all trajectories.

If we test the vector $\overrightarrow{x}=(1,0)$ in the matrix equation associated with the system, we get

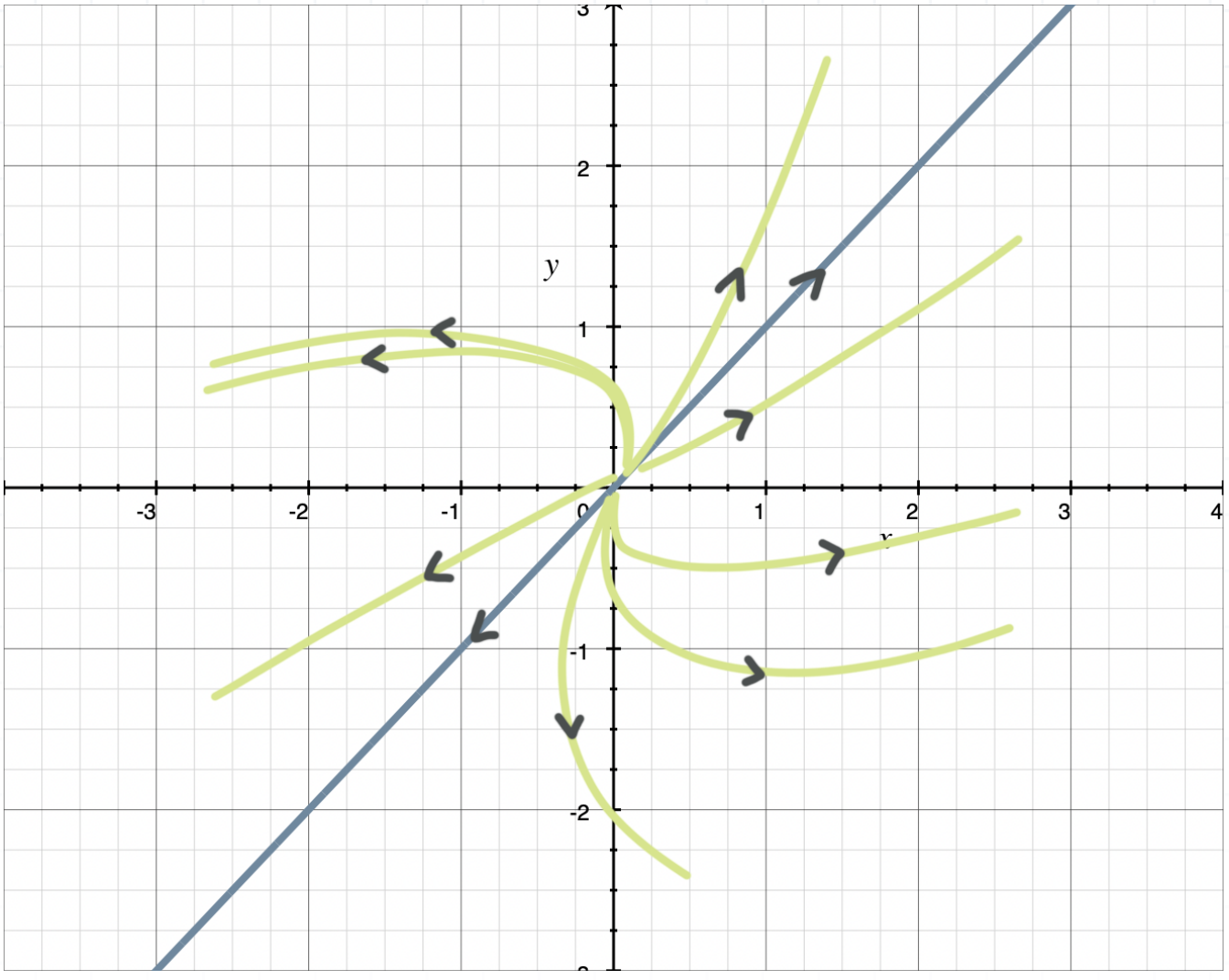
$$\overrightarrow{x}'=\begin{bmatrix}16&-5\\5&6\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}$$

$$\overrightarrow{x}'=\begin{bmatrix}16\\5\end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(16,5)$ (toward the first quadrant), which means that the phase portrait must look something like

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■ 6. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 4 \\ -1 & -2-\lambda \end{bmatrix}$$

and the characteristic equation

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$$(2-\lambda)(-2-\lambda)-(4)(-1)=0$$

$$\lambda^2=0$$

$$\lambda=0,\ 0$$

gives the Eigenvalues $\lambda_1=\lambda_2=0$, and the associated Eigenvector

$$\overrightarrow{k_1}=\begin{bmatrix}2\\-1\end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1}=(2,-1)$ lies along the line $y=(-1/2)x$, so we'll sketch this line in our phase portrait. The Eigenvalue associated with $\overrightarrow{k_1}=(2,-1)$ is $\lambda=0$, which means we need to find a second Eigenvector $\overrightarrow{p_1}$.

$$(A-\lambda I)\overrightarrow{p_1}=\overrightarrow{k_1}$$

$$\begin{bmatrix}2&4\\-1&-2\end{bmatrix}\begin{bmatrix}p_1\\p_2\end{bmatrix}=\begin{bmatrix}2\\-1\end{bmatrix}$$

Turn this matrix back into a system of equations.

$$2p_1+4p_2=2$$

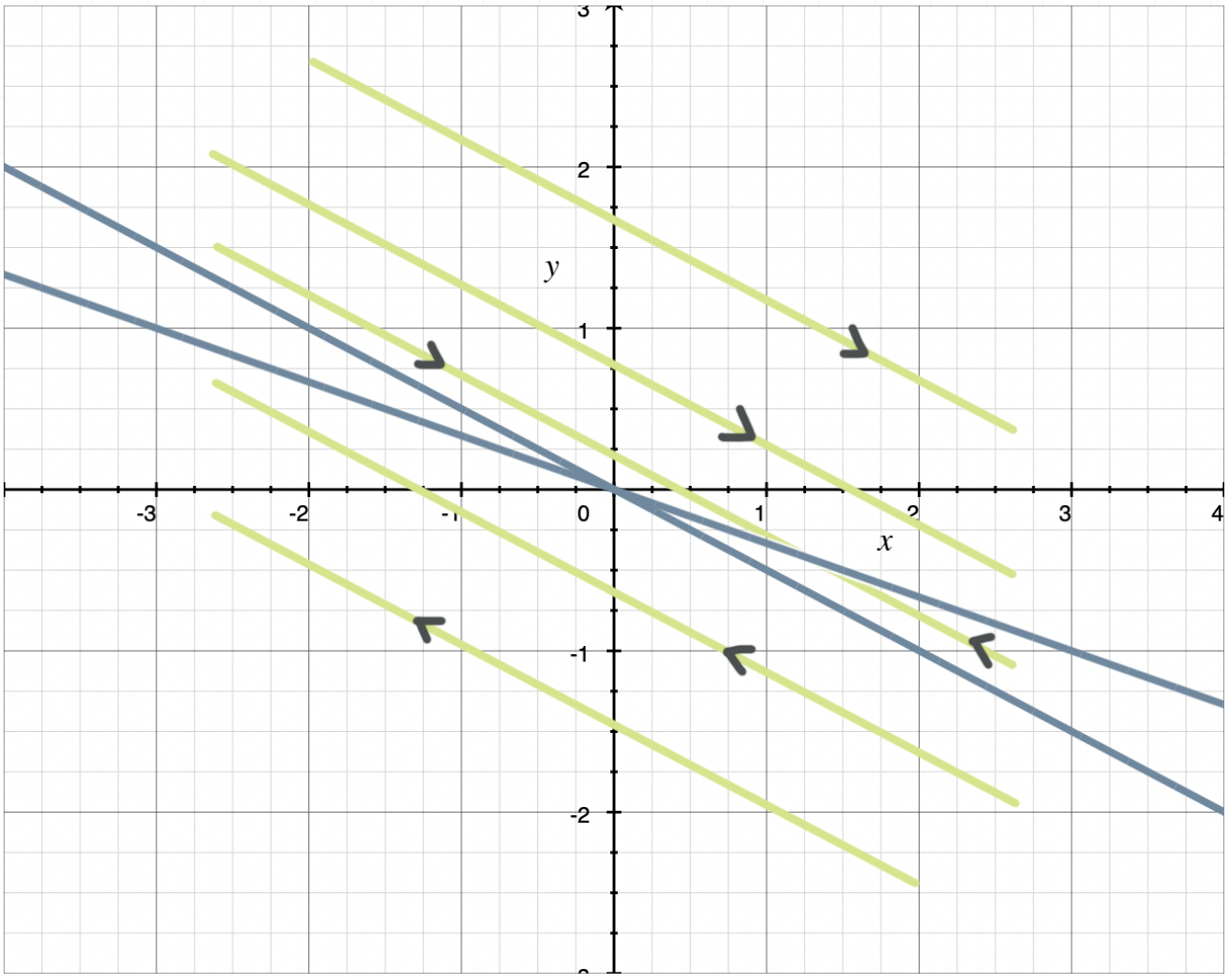
$$-p_1-2p_2=-1$$

The system simplifies to just $p_1+2p_2=1$. If we choose $p_1=3$ and $p_2=-1$, we get

$$\overrightarrow{p_1}=\begin{bmatrix}3\\-1\end{bmatrix}$$

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The trajectories are parallel to $\overrightarrow{k_1} = (2, -1)$, and equilibrium exists at every point along $\overrightarrow{p_1} = (3, -1)$.



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PHASE PORTRAITS FOR COMPLEX EIGENVALUES

■ 1. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 5 & 1 \\ -5 & 2 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 5 - \lambda & 1 \\ -5 & 2 - \lambda \end{bmatrix}$$

so the characteristic equation gives the Eigenvalues for the system.

$$(5 - \lambda)(2 - \lambda) - (1)(-5) = 0$$

$$\lambda^2 - 7\lambda + 15 = 0$$

$$\lambda = \frac{7 \pm \sqrt{11}i}{2}$$

These are Eigenvalues with a positive real part, $\alpha = 7/2 > 0$, which means we're dealing with an unstable spiral that repels all trajectories.

If we test the vector $\vec{x} = (1,0)$ in the matrix equation associated with the system, we get

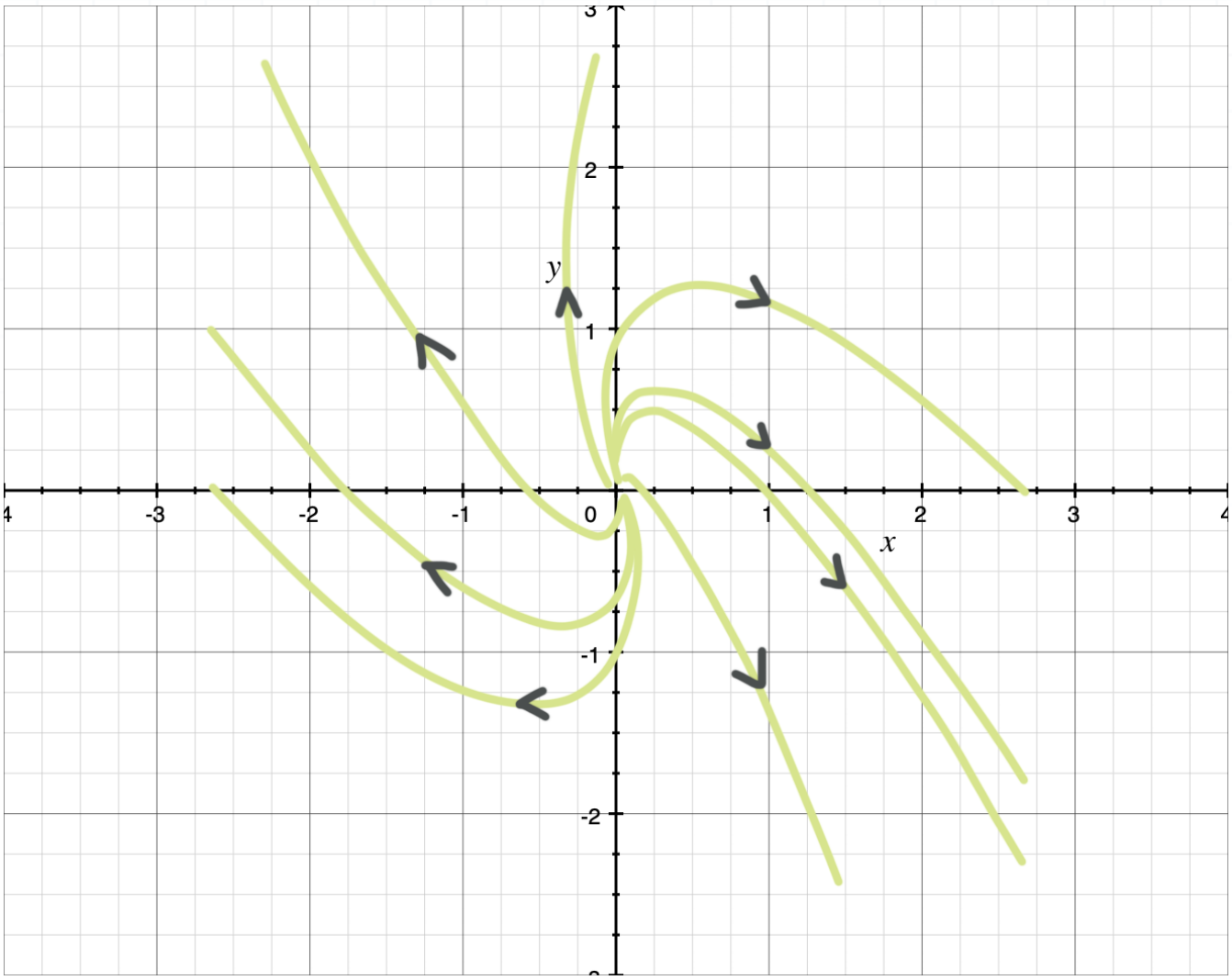
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$$\vec{x}' = \begin{bmatrix} 5 & 1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{x}' = \begin{bmatrix} 5 \\ -5 \end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(5, -5)$ (into the fourth quadrant), which means that the phase portrait must look something like



2. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 1 & -3 \\ 2 & 2 \end{bmatrix}$$

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Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & -3 \\ 2 & 2 - \lambda \end{bmatrix}$$

so the characteristic equation gives the Eigenvalues for the system.

$$(1 - \lambda)(2 - \lambda) - (-3)(2) = 0$$

$$\lambda^2 - 3\lambda + 8 = 0$$

$$\lambda = \frac{3 \pm \sqrt{23}i}{2}$$

These are Eigenvalues with a positive real part, $\alpha = 3/2 > 0$, which means we're dealing with an unstable spiral that repels all trajectories.

If we test the vector $\vec{x} = (1,0)$ in the matrix equation associated with the system, we get

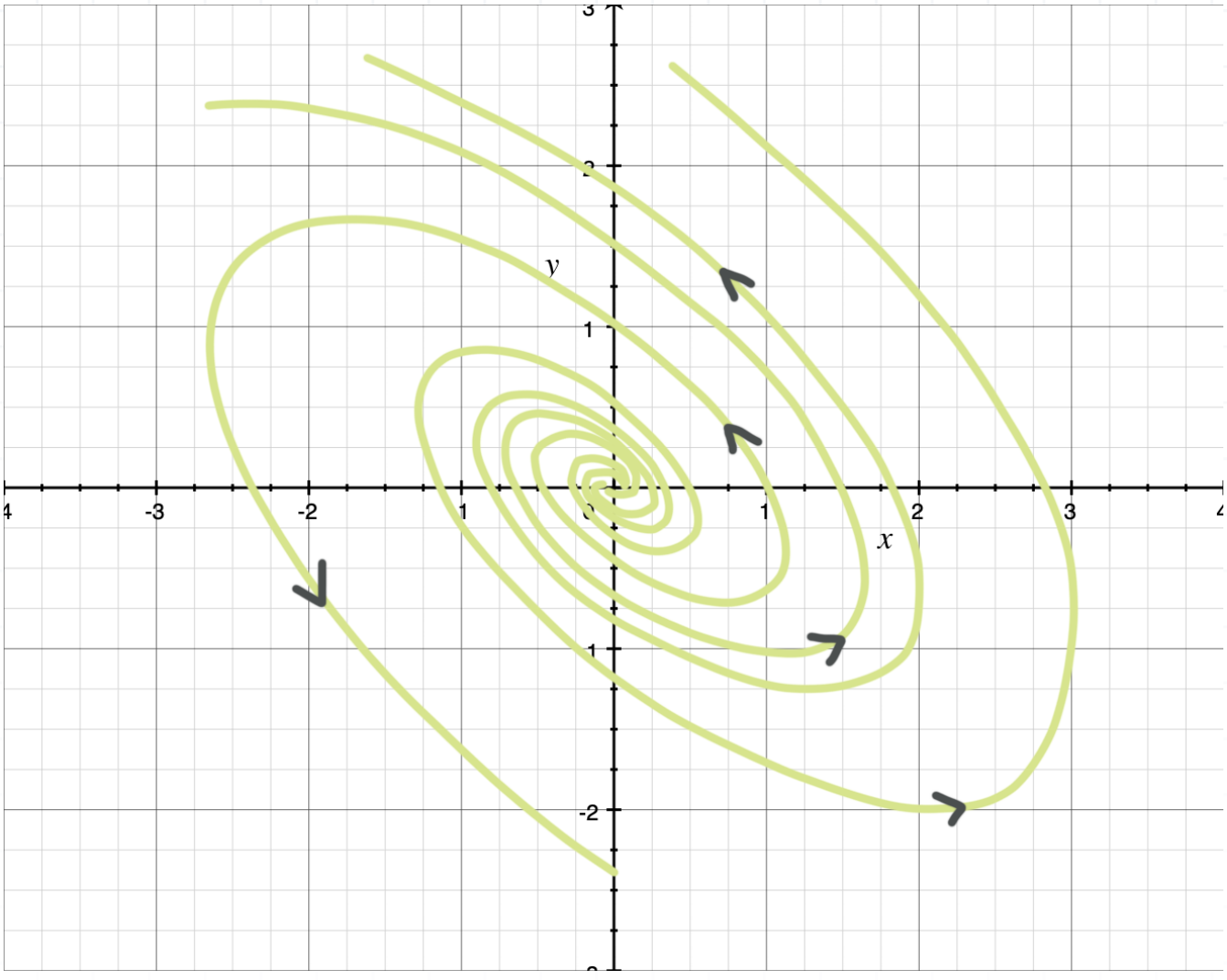
$$\vec{x}' = \begin{bmatrix} 1 & -3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{x}' = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(1,2)$ (into the first quadrant), which means that the phase portrait must look something like

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■ 3. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 2 & -1 \\ 10 & -4 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & -1 \\ 10 & -4 - \lambda \end{bmatrix}$$

so the characteristic equation gives the Eigenvalues for the system.

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$$(2-\lambda)(-4-\lambda)-(-1)(10)=0$$

$$\lambda^2+2\lambda+2=0$$

$$\lambda=-1\pm i$$

These are Eigenvalues with a negative real part, $\alpha=-1<0$, which means we're dealing with an asymptotically stable spiral that attracts all trajectories.

If we test the vector $\overrightarrow{x}=(1,0)$ in the matrix equation associated with the system, we get

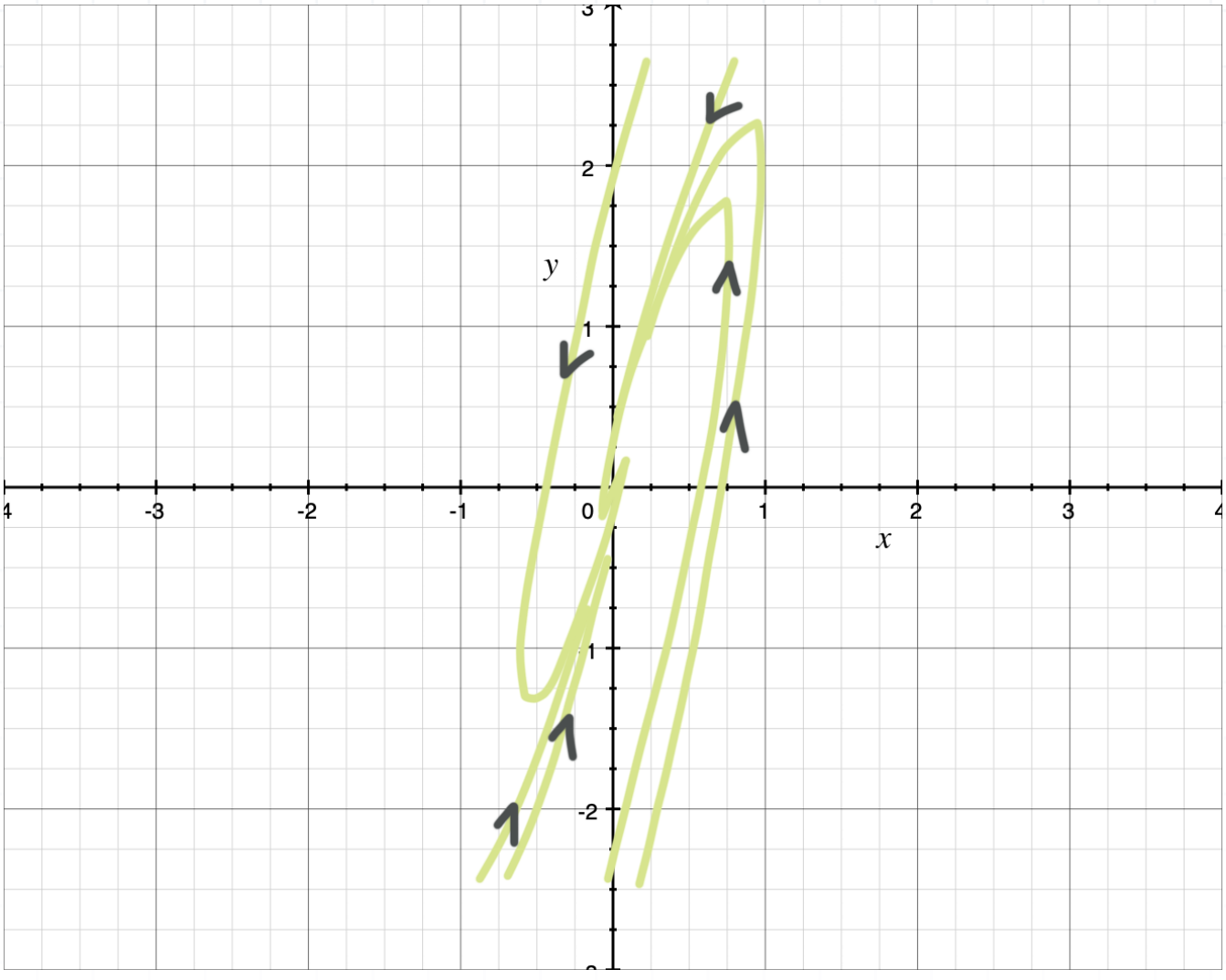
$$\overrightarrow{x}'=\begin{bmatrix}2 & -1\\ 10 & -4\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}$$

$$\overrightarrow{x}'=\begin{bmatrix}2\\10\end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(2,10)$ (into the first quadrant), which means that the phase portrait must look something like

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■ 4. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -2 & 2 \\ -1 & 0 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} -2 - \lambda & 2 \\ -1 & -\lambda \end{bmatrix}$$

so the characteristic equation gives the Eigenvalues for the system.

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$$(-2-\lambda)(-\lambda)-(2)(-1)=0$$

$$\lambda^2+2\lambda+2=0$$

$$\lambda=-1\pm i$$

These are Eigenvalues with a negative real part, $\alpha=-1<0$, which means we're dealing with an asymptotically stable spiral that attracts all trajectories.

If we test the vector $\vec{x}=(1,0)$ in the matrix equation associated with the system, we get

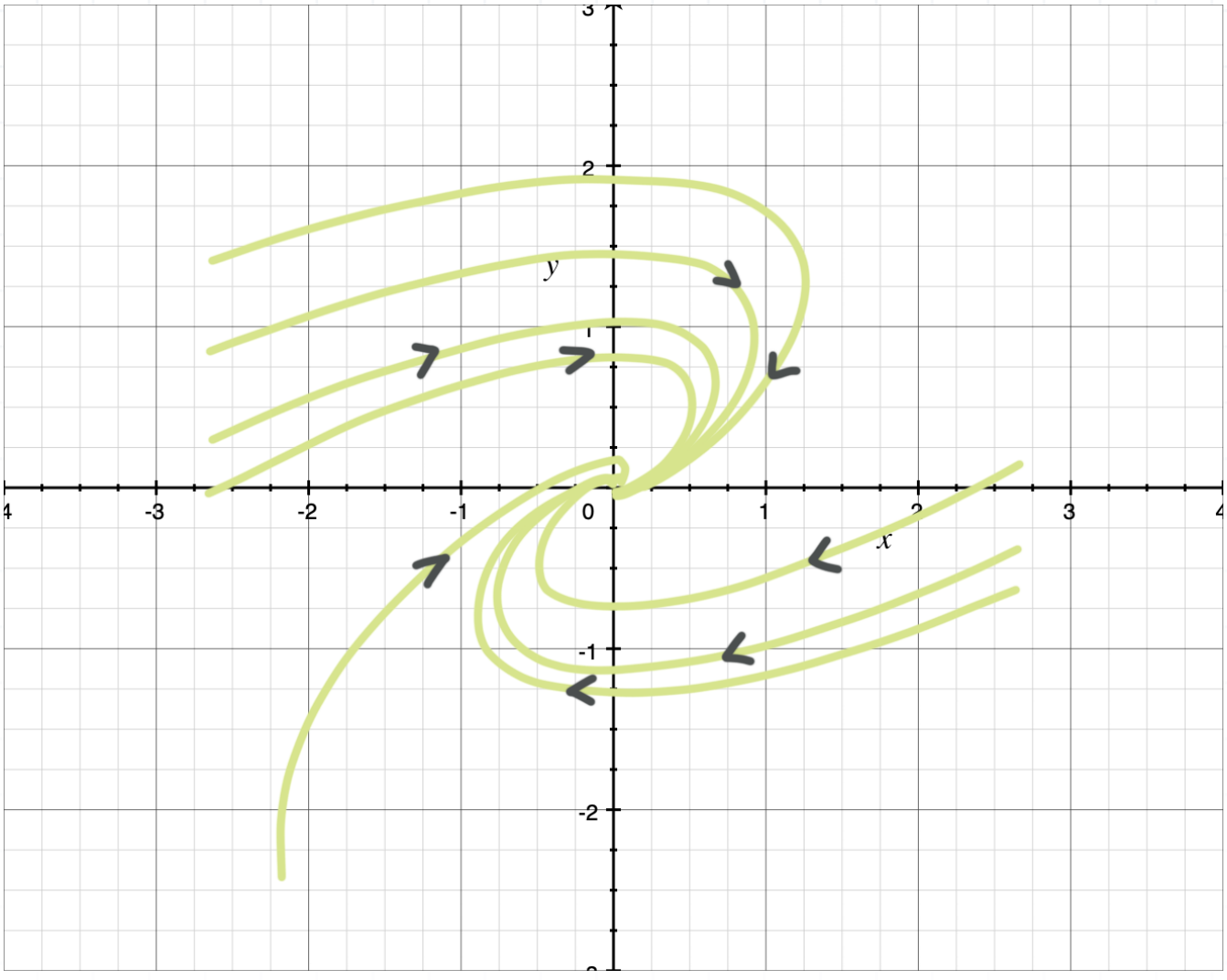
$$\vec{x}'=\begin{bmatrix}-2 & 2\\ -1 & 0\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}$$

$$\vec{x}'=\begin{bmatrix}-2\\-1\end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(-2,-1)$ (into the third quadrant), which means that the phase portrait must look something like

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■ 5. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -2 & -4 \\ 2 & 2 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} -2 - \lambda & -4 \\ 2 & 2 - \lambda \end{bmatrix}$$

so the characteristic equation gives the Eigenvalues for the system.

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$$(-2-\lambda)(2-\lambda)-(-4)(2)=0$$

$$\lambda^2+4=0$$

$$\lambda=\pm 2i$$

These are Eigenvalues with no real part, $\alpha=0$, which means we're dealing with a stable center that neither repels nor attracts trajectories.

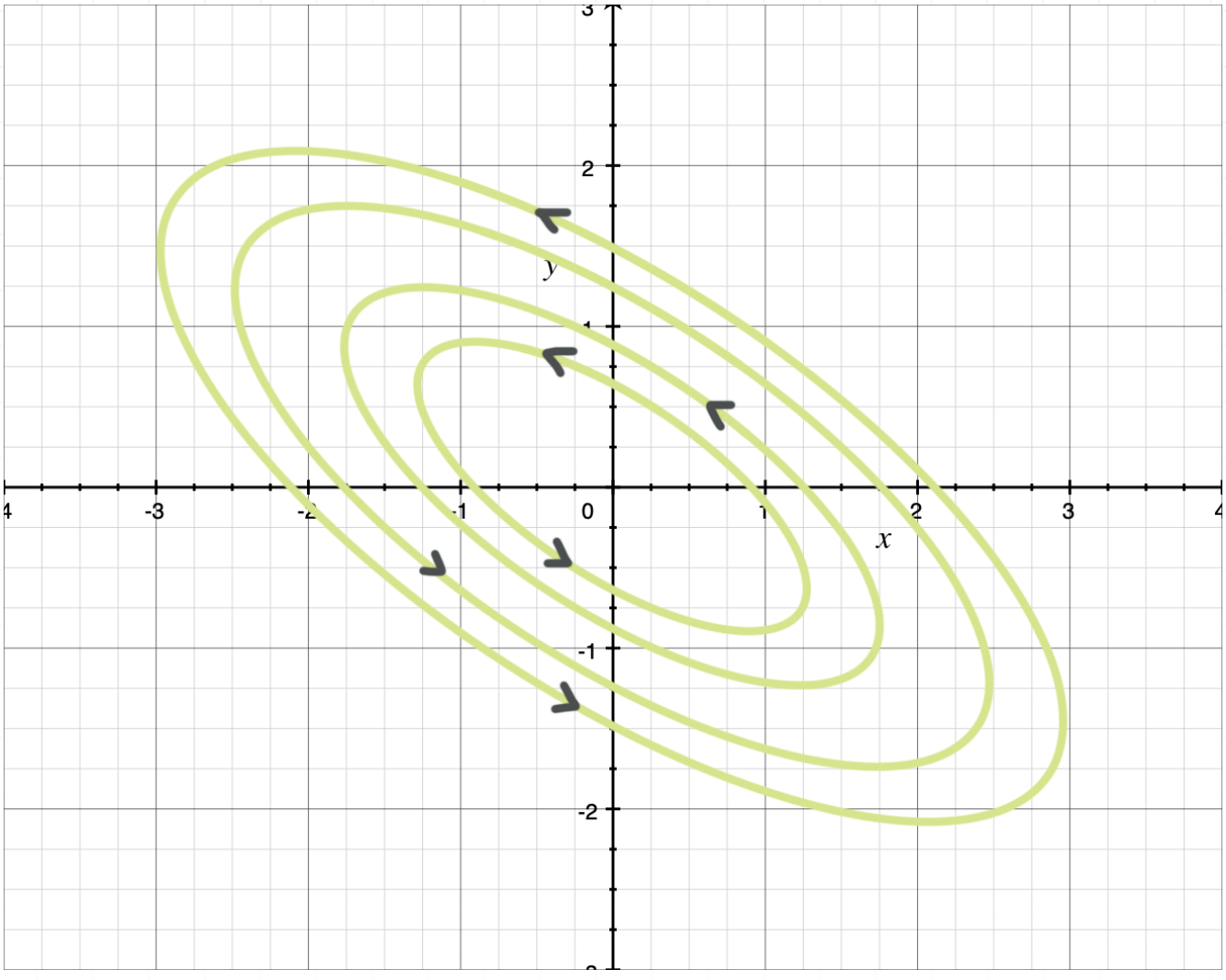
If we test the vector $\vec{x}=(1,0)$ in the matrix equation associated with the system, we get

$$\vec{x}'=\begin{bmatrix}-2 & -4\\ 2 & 2\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}$$

$$\vec{x}'=\begin{bmatrix}-2\\2\end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(-2,2)$ (into the second quadrant), which means that the phase portrait must look something like

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6. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 1 & 2 \\ -13 & -1 \end{bmatrix}$$

Solution:

The matrix $A - \lambda I$ is

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ -13 & -1-\lambda \end{bmatrix}$$

so the characteristic equation gives the Eigenvalues for the system.

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$$(1-\lambda)(-1-\lambda)-(2)(-13)=0$$

$$\lambda^2+25=0$$

$$\lambda=\pm 5i$$

These are Eigenvalues with no real part, $\alpha=0$, which means we're dealing with a stable center that neither repels nor attracts trajectories.

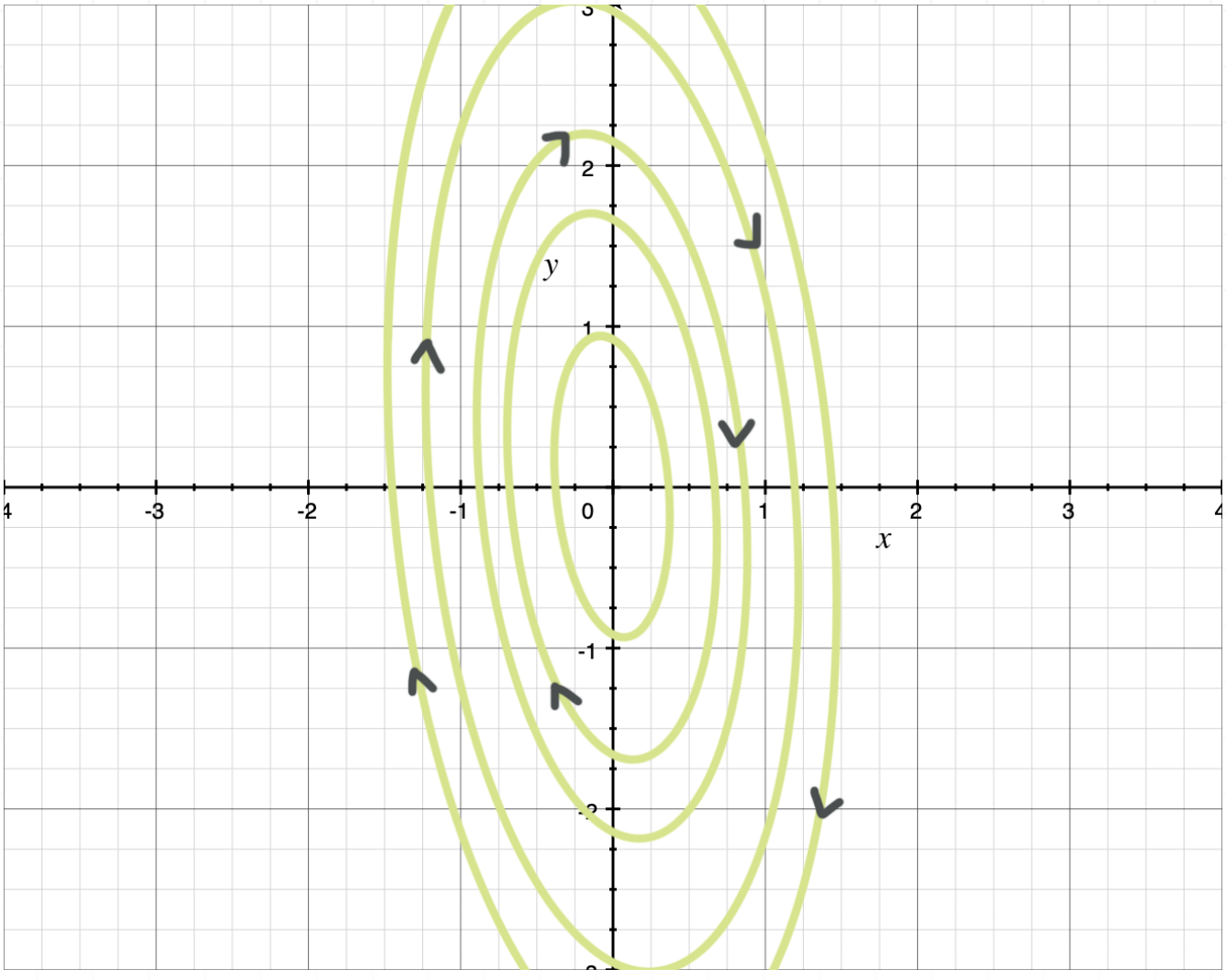
If we test the vector $\vec{x}=(1,0)$ in the matrix equation associated with the system, we get

$$\vec{x}'=\begin{bmatrix}1 & 2\\ -13 & -1\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix}$$

$$\vec{x}'=\begin{bmatrix}1\\-13\end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(1,-13)$ (into the fourth quadrant), which means that the phase portrait must look something like

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UNDETERMINED COEFFICIENTS FOR NONHOMOGENEOUS SYSTEMS

■ 1. Use undetermined coefficients to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} \cos t \\ 2 \sin t \end{bmatrix}$$

Solution:

The determinant $|A - \lambda I|$ is

$$\begin{vmatrix} 2-\lambda & 1 \\ -1 & -\lambda \end{vmatrix}$$

$$(2-\lambda)(-\lambda) - (1)(-1)$$

$$\lambda^2 - 2\lambda + 1$$

Solve the characteristic equation.

$$\lambda^2 - 2\lambda + 1 = 0$$

$$(\lambda - 1)^2 = 0$$

$$\lambda_{1,2} = 1,\ 1$$

Then for the Eigenvalues $\lambda_1 = \lambda_2 = 1$, we find

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$$A - 1I = \begin{bmatrix} 2-1 & 1 \\ -1 & -1 \end{bmatrix}$$

$$A - I = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

Put the matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

If we turn this matrix back into a system of equations, we get

$$k_1 + k_2 = 0$$

$$k_1 = -k_2$$

Choose $k_2 = -1$ to get $k_1 = 1$, which gives the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{t}$$

Because we only find one Eigenvector for the two Eigenvalues $\lambda_1 = \lambda_2 = 1$, we have to use $\overrightarrow{k_1} = (1, -1)$ to find a second solution.

$$(A - \lambda I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

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$$\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$p_1 + p_2 = 1$$

$$-p_1 - p_2 = -1$$

If we choose $p_2 = 0$, we find $p_1 = 1$, and we get the Eigenvector

$$\overrightarrow{p_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and the solution vector

$$\overrightarrow{x_2} = \overrightarrow{k_1} te^{\lambda_1 t} + \overrightarrow{p_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} te^t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1 \overrightarrow{x_1} + c_2 \overrightarrow{x_2}$$

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} te^t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t \right)$$

Now we can turn to finding the particular solution. We'll rewrite the forcing function vector as

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$$F = \begin{bmatrix} \cos t \\ 2\sin t \end{bmatrix}$$

$$F = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cos t + \begin{bmatrix} 0 \\ 2 \end{bmatrix} \sin t$$

We want a particular solution in the same form, so our guess will be

$$\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \cos t + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \sin t$$

$$\overrightarrow{x_p}' = \begin{bmatrix} -a_1 \\ -a_2 \end{bmatrix} \sin t + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \cos t$$

Plugging this into the matrix equation representing the system, we get

$$\begin{bmatrix} -a_1 \\ -a_2 \end{bmatrix} \sin t + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \cos t = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \left[\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \cos t + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \sin t \right] + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cos t + \begin{bmatrix} 0 \\ 2 \end{bmatrix} \sin t$$

$$\begin{bmatrix} -a_1 \sin t + b_1 \cos t \\ -a_2 \sin t + b_2 \cos t \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \cos t + b_1 \sin t \\ a_2 \cos t + b_2 \sin t \end{bmatrix} + \begin{bmatrix} \cos t \\ 2\sin t \end{bmatrix}$$

$$\begin{bmatrix} -a_1 \sin t + b_1 \cos t \\ -a_2 \sin t + b_2 \cos t \end{bmatrix} = \begin{bmatrix} 2a_1 \cos t + 2b_1 \sin t + a_2 \cos t + b_2 \sin t \\ -a_1 \cos t - b_1 \sin t \end{bmatrix} + \begin{bmatrix} \cos t \\ 2\sin t \end{bmatrix}$$

$$\begin{bmatrix} -a_1 \sin t + b_1 \cos t \\ -a_2 \sin t + b_2 \cos t \end{bmatrix} = \begin{bmatrix} (2a_1 + a_2 + 1)\cos t + (2b_1 + b_2)\sin t \\ -a_1 \cos t + (2 - b_1)\sin t \end{bmatrix}$$

Breaking this equation into a system of equations gives

$$-a_1 \sin t + b_1 \cos t = (2a_1 + a_2 + 1)\cos t + (2b_1 + b_2)\sin t$$

$$-a_2 \sin t + b_2 \cos t = -a_1 \cos t + (2 - b_1)\sin t$$

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These equations can each be broken into its own system.

$$-a_1 = 2b_1 + b_2 \qquad -a_2 = 2 - b_1$$

$$b_1 = 2a_1 + a_2 + 1 \qquad b_2 = -a_1$$

Solving these systems together gives $\vec{a} = (a_1, a_2) = (1/2, -2)$ and $\vec{b} = (b_1, b_2) = (0, -1/2)$. Therefore, the particular solution is

$$\vec{x_p} = \begin{bmatrix} \frac{1}{2} \\ -2 \end{bmatrix} \cos t + \begin{bmatrix} 0 \\ -\frac{1}{2} \end{bmatrix} \sin t$$

Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

$$\vec{x} = \vec{x_c} + \vec{x_p}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} te^t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t \right) + \begin{bmatrix} \frac{1}{2} \\ -2 \end{bmatrix} \cos t + \begin{bmatrix} 0 \\ -\frac{1}{2} \end{bmatrix} \sin t$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} te^t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t \right) + \begin{bmatrix} \frac{1}{2} \cos t \\ -2 \cos t - \frac{1}{2} \sin t \end{bmatrix}$$

■ 2. Use undetermined coefficients to find the general solution to the initial value problem.

$$\vec{x}' = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} 2e^{3t} \\ t^2 \end{bmatrix}$$

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$$\vec{x}(0) = \begin{bmatrix} \frac{1}{4} \\ \frac{3}{4} \end{bmatrix}$$

Solution:

The determinant $|A - \lambda I|$ is

$$\begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix}$$

$$(-\lambda)(-\lambda) - 1(1)$$

$$\lambda^2 - 1$$

Solve the characteristic equation.

$$\lambda^2 - 1 = 0$$

$$\lambda_1 = 1 \qquad \lambda_2 = -1$$

Then for these Eigenvalues, $\lambda_1 = 1$ and $\lambda_2 = -1$, we find

$$A - 1I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \qquad A + 1I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

If we turn these back into systems of equations, we get

$$k_1 - k_2 = 0 \qquad k_1 + k_2 = 0$$

$$k_1 = k_2 \qquad k_1 = -k_2$$

If we choose $k_2 = 1$ in the first system as $k_2 = -1$ in the second system, we get the Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

and the solution vectors

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$$

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t}$$

Now we can turn to finding the particular solution. We'll rewrite the forcing function vector as

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$$F = \begin{bmatrix} 2e^{3t} \\ t^2 \end{bmatrix}$$

$$F = \begin{bmatrix} 2 \\ 0 \end{bmatrix} e^{3t} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} t^2$$

We want a particular solution in the same form, so our guess will be

$$\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} e^{3t} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} t^2 + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} t + \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

$$\overrightarrow{x_p}' = \begin{bmatrix} 3a_1 \\ 3a_2 \end{bmatrix} e^{3t} + \begin{bmatrix} 2b_1 \\ 2b_2 \end{bmatrix} t + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Plugging this into the matrix equation representing the system, we get

$$\begin{bmatrix} 3a_1 \\ 3a_2 \end{bmatrix} e^{3t} + \begin{bmatrix} 2b_1 \\ 2b_2 \end{bmatrix} t + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \left[\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} e^{3t} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} t^2 + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} t + \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \right]$$

$$+ \begin{bmatrix} 2 \\ 0 \end{bmatrix} e^{3t} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} t^2$$

$$\begin{bmatrix} 3a_1e^{3t} + 2b_1t + c_1 \\ 3a_2e^{3t} + 2b_2t + c_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a_1e^{3t} + b_1t^2 + c_1t + d_1 \\ a_2e^{3t} + b_2t^2 + c_2t + d_2 \end{bmatrix} + \begin{bmatrix} 2e^{3t} \\ t^2 \end{bmatrix}$$

$$\begin{bmatrix} 3a_1e^{3t} + 2b_1t + c_1 \\ 3a_2e^{3t} + 2b_2t + c_2 \end{bmatrix} = \begin{bmatrix} a_2e^{3t} + b_2t^2 + c_2t + d_2 + 2e^{3t} \\ a_1e^{3t} + b_1t^2 + c_1t + d_1 + t^2 \end{bmatrix}$$

Breaking this equation into a system of two equations gives

$$3a_1e^{3t} + 2b_1t + c_1 = (a_2 + 2)e^{3t} + b_2t^2 + c_2t + d_2$$

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$$3a_2e^{3t} + 2b_2t + c_2 = a_1e^{3t} + (b_1 + 1)t^2 + c_1t + d_1$$

These equations can each be broken into its own system.

$$3a_1 = a_2 + 2 \qquad 3a_2 = a_1$$

$$b_2 = 0 \qquad b_1 + 1 = 0$$

$$2b_1 = c_2 \qquad 2b_2 = c_1$$

$$c_1 = d_2 \qquad c_2 = d_1$$

Putting these results together gives $\vec{a} = (a_1, a_2) = (3/4, 1/4)$, $\vec{b} = (b_1, b_2) = (-1, 0)$, $\vec{c} = (c_1, c_2) = (0, -2)$, and $\vec{d} = (d_1, d_2) = (-2, 0)$. Therefore, the particular solution is

$$\vec{x_p} = \begin{bmatrix} \frac{3}{4} \\ \frac{1}{4} \end{bmatrix} e^{3t} + \begin{bmatrix} -1 \\ 0 \end{bmatrix} t^2 + \begin{bmatrix} 0 \\ -2 \end{bmatrix} t + \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

$$\vec{x_p} = \begin{bmatrix} \frac{3}{4}e^{3t} - t^2 - 2 \\ \frac{1}{4}e^{3t} - 2t \end{bmatrix}$$

Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

$$\vec{x} = \vec{x_c} + \vec{x_p}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + \begin{bmatrix} \frac{3}{4}e^{3t} - t^2 - 2 \\ \frac{1}{4}e^{3t} - 2t \end{bmatrix}$$

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Plug in the initial condition,

$$\begin{bmatrix} \frac{1}{4} \\ \frac{3}{4} \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} \frac{3}{4} - 2 \\ \frac{1}{4} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{4} \\ \frac{3}{4} \end{bmatrix} = \begin{bmatrix} c_1 + c_2 + \frac{3}{4} - 2 \\ c_1 - c_2 + \frac{1}{4} \end{bmatrix}$$

then break the matrix equation into a system of equations.

$$c_1 + c_2 + \frac{3}{4} - 2 = \frac{1}{4}$$

$$c_1 - c_2 + \frac{1}{4} = \frac{3}{4}$$

Solving the system gives $c_1 = 1$ and $c_2 = 1/2$, which means the solution to the system is

$$\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} e^{-t} + \begin{bmatrix} \frac{3}{4}e^{3t} - t^2 - 2 \\ \frac{1}{4}e^{3t} - 2t \end{bmatrix}$$

3. Use undetermined coefficients to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ 2 & -1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} 6e^{4t} \\ \cos(3t) \\ 1 \end{bmatrix}$$

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Solution:

The determinant $|A - \lambda I|$ is

$$\begin{vmatrix} 1-\lambda & -1 & 1 \\ 1 & 1-\lambda & -1 \\ 2 & -1 & -\lambda \end{vmatrix}$$

$$(1-\lambda)\begin{vmatrix} 1-\lambda & -1 \\ -1 & -\lambda \end{vmatrix} - (-1)\begin{vmatrix} 1 & -1 \\ 2 & -\lambda \end{vmatrix} + 1\begin{vmatrix} 1 & 1-\lambda \\ 2 & -1 \end{vmatrix}$$

$$(1-\lambda)((1-\lambda)(-\lambda) - (-1)(-1)) + (1(-\lambda) - (-1)(2)) + (1(-1) - (1-\lambda)(2))$$

$$(1-\lambda)(\lambda^2 - \lambda - 1) - \lambda + 2 + 2\lambda - 3$$

$$-\lambda^3 + 2\lambda^2 + \lambda - 2$$

$$-\lambda^2(\lambda - 2) + (\lambda - 2)$$

Solve the characteristic equation.

$$(\lambda - 2)(-\lambda^2 + 1) = 0$$

$$\lambda_1 = 2 \qquad \lambda_2 = -1 \qquad \lambda_3 = 1$$

Then for these Eigenvalues, $\lambda_1 = 2$, $\lambda_2 = -1$, and $\lambda_3 = 1$, we find

$$\begin{bmatrix} -1 & -1 & 1 \\ 1 & -1 & -1 \\ 2 & -1 & -2 \end{bmatrix} \qquad \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 2 & -1 & 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ 2 & -1 & -1 \end{bmatrix}$$

Put these matrices into reduced row-echelon form.

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$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & -1 \\ 2 & -1 & -2 \end{bmatrix} \quad \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 2 & -1 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ 2 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -2 & 0 \\ 0 & -3 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \\ 0 & -5 & 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -\frac{3}{5} \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & \frac{1}{5} \\ 0 & 1 & -\frac{3}{5} \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn these back into systems of equations, we get

$$k_1 - k_3 = 0 \quad k_1 + \frac{1}{5}k_3 = 0 \quad k_1 - k_3 = 0$$

$$k_2 = 0 \quad k_2 - \frac{3}{5}k_3 = 0 \quad k_2 - k_3 = 0$$

These systems simplify to

$$k_1 = k_3 \quad k_1 = -\frac{1}{5}k_3 \quad k_1 = k_3$$

$$k_2 = 0 \quad k_2 = \frac{3}{5}k_3 \quad k_2 = k_3$$

From these systems, we find the Eigenvectors

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$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} -1 \\ 3 \\ 5 \end{bmatrix} \qquad \overrightarrow{k_3} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

and then the solution vectors

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t} \qquad \overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t} \qquad \overrightarrow{x_2} = \begin{bmatrix} -1 \\ 3 \\ 5 \end{bmatrix} e^{-t} \qquad \overrightarrow{x_3} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1 \overrightarrow{x_1} + c_2 \overrightarrow{x_2} + c_3 \overrightarrow{x_3}$$

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} -1 \\ 3 \\ 5 \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{t}$$

Now we can turn to finding the particular solution. We'll rewrite the forcing function vector as

$$F = \begin{bmatrix} 6e^{4t} \\ \cos(3t) \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} e^{4t} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \cos(3t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

We want a particular solution in the same form, so our guess will be

$$\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} e^{4t} + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \cos(3t) + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \sin(3t) + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

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$$\overrightarrow{x_p}' = \begin{bmatrix} 4a_1 \\ 4a_2 \\ 4a_3 \end{bmatrix} e^{4t} + \begin{bmatrix} -3b_1 \\ -3b_2 \\ -3b_3 \end{bmatrix} \sin(3t) + \begin{bmatrix} 3c_1 \\ 3c_2 \\ 3c_3 \end{bmatrix} \cos(3t)$$

Plugging these into the matrix equation representing the system, we get

$$\begin{bmatrix} 4a_1 \\ 4a_2 \\ 4a_3 \end{bmatrix} e^{4t} + \begin{bmatrix} -3b_1 \\ -3b_2 \\ -3b_3 \end{bmatrix} \sin(3t) + \begin{bmatrix} 3c_1 \\ 3c_2 \\ 3c_3 \end{bmatrix} \cos(3t)$$

$$= \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ 2 & -1 & 0 \end{bmatrix} \left(\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} e^{4t} + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \cos(3t) + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} \sin(3t) + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \right)$$

$$+ \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} e^{4t} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \cos t + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 4a_1e^{4t} - 3b_1\sin(3t) + 3c_1\cos(3t) \\ 4a_2e^{4t} - 3b_2\sin(3t) + 3c_2\cos(3t) \\ 4a_3e^{4t} - 3b_3\sin(3t) + 3c_3\cos(3t) \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ 2 & -1 & 0 \end{bmatrix} \begin{bmatrix} a_1e^{4t} + b_1\cos(3t) + c_1\sin(3t) + d_1 \\ a_2e^{4t} + b_2\cos(3t) + c_2\sin(3t) + d_2 \\ a_3e^{4t} + b_3\cos(3t) + c_3\sin(3t) + d_3 \end{bmatrix}$$

$$+ \begin{bmatrix} 6e^{4t} \\ \cos(3t) \\ 1 \end{bmatrix}$$

Breaking this equation into a system of three equations gives

$$4a_1e^{4t} - 3b_1\sin(3t) + 3c_1\cos(3t) = (a_1 - a_2 + a_3 + 6)e^{4t} + (b_1 - b_2 + b_3)\cos(3t)$$

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$$+(c_1 - c_2 + c_3)\sin(3t) + (d_1 - d_2 + d_3)$$

$$4a_2e^{4t} - 3b_2\sin(3t) + 3c_2\cos(3t) = (a_1 + a_2 - a_3)e^{4t} + (b_1 + b_2 - b_3 + 1)\cos(3t)$$

$$+(c_1 + c_2 - c_3)\sin(3t) + (d_1 + d_2 - d_3)$$

$$4a_3e^{4t} - 3b_3\sin(3t) + 3c_3\cos(3t) = (2a_1 - a_2)e^{4t} + (2b_1 - b_2)\cos(3t)$$

$$+(2c_1 - c_2)\sin(3t) + (2d_1 - d_2 + 1)$$

These equations can each be broken into its own system.

$$4a_1 = a_1 - a_2 + a_3 + 6 \qquad 4a_2 = a_1 + a_2 - a_3 \qquad 4a_3 = 2a_1 - a_2$$

$$-3b_1 = c_1 - c_2 + c_3 \qquad -3b_2 = c_1 + c_2 - c_3 \qquad -3b_3 = 2c_1 - c_2$$

$$3c_1 = b_1 - b_2 + b_3 \qquad 3c_2 = b_1 + b_2 - b_3 + 1 \qquad 3c_3 = 2b_1 - b_2$$

$$0 = d_1 - d_2 + d_3 \qquad 0 = d_1 + d_2 - d_3 \qquad 0 = 2d_1 - d_2 + 1$$

Putting these results together gives $\vec{a} = (a_1, a_2, a_3) = (11/5, 2/5, 1)$, $\vec{b} = (b_1, b_2, b_3) = (7/130, -1/10, 7/130)$, $\vec{c} = (c_1, c_2, c_3) = (9/130, 3/10, 9/130)$, and $\vec{d} = (d_1, d_2, d_3) = (0, 1, 1)$. Therefore, the particular solution is

$$\vec{x_p} = \begin{bmatrix} \frac{11}{5} \\ \frac{2}{5} \\ 1 \end{bmatrix} e^{4t} + \begin{bmatrix} \frac{7}{130} \\ -\frac{1}{10} \\ \frac{7}{130} \end{bmatrix} \cos(3t) + \begin{bmatrix} \frac{9}{130} \\ \frac{3}{10} \\ \frac{9}{130} \end{bmatrix} \sin(3t) + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

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$$\vec{x} = \vec{x_c} + \vec{x_p}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} -1 \\ 3 \\ 5 \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{t}$$

$$+ \begin{bmatrix} \frac{11}{5} \\ \frac{2}{5} \\ 1 \end{bmatrix} e^{4t} + \begin{bmatrix} \frac{7}{130} \\ -\frac{1}{10} \\ \frac{7}{130} \end{bmatrix} \cos(3t) + \begin{bmatrix} \frac{9}{130} \\ \frac{3}{10} \\ \frac{9}{130} \end{bmatrix} \sin(3t) + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

■ 4. Use undetermined coefficients to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix} \vec{x} + \begin{bmatrix} -e^t \\ 3t^2 \\ 2e^{-t} \end{bmatrix}$$

Solution:

The determinant $|A - \lambda I|$ is

$$\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix}$$

$$(3-\lambda)\begin{vmatrix} 5-\lambda & -1 \\ -1 & 3-\lambda \end{vmatrix} - (-1)\begin{vmatrix} -1 & -1 \\ 1 & 3-\lambda \end{vmatrix} + 1\begin{vmatrix} -1 & 5-\lambda \\ 1 & -1 \end{vmatrix}$$

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$$(3-\lambda)((5-\lambda)(3-\lambda)-(-1)(-1))+((-1)(3-\lambda)-(-1)(1))+((-1)(-1)-(5-\lambda)(1))$$

$$(3-\lambda)(\lambda^2-8\lambda+14)+\lambda-2+\lambda-4$$

$$(3-\lambda)(\lambda^2-8\lambda+14)-2(3-\lambda)$$

$$(3-\lambda)(\lambda^2-8\lambda+12)$$

Solve the characteristic equation.

$$(3-\lambda)(\lambda-6)(\lambda-2)=0$$

$$\lambda_1=2 \qquad \lambda_2=3 \qquad \lambda_3=6$$

Then for these Eigenvalues, $\lambda_1=2$, $\lambda_2=3$, and $\lambda_3=6$, we find

$$\begin{bmatrix}1&-1&1\\-1&3&-1\\1&-1&1\end{bmatrix} \qquad \begin{bmatrix}0&-1&1\\-1&2&-1\\1&-1&0\end{bmatrix} \qquad \begin{bmatrix}-3&-1&1\\-1&-1&-1\\1&-1&-3\end{bmatrix}$$

Put these three matrices into reduced row-echelon form.

$$\begin{bmatrix}1&-1&1\\-1&3&-1\\1&-1&1\end{bmatrix} \qquad \begin{bmatrix}0&-1&1\\-1&2&-1\\1&-1&0\end{bmatrix} \qquad \begin{bmatrix}-3&-1&1\\-1&-1&-1\\1&-1&-3\end{bmatrix}$$

$$\begin{bmatrix}1&-1&1\\0&2&0\\0&0&0\end{bmatrix} \qquad \begin{bmatrix}1&-1&0\\0&-1&1\\-1&2&-1\end{bmatrix} \qquad \begin{bmatrix}1&-1&-3\\-3&-1&1\\-1&-1&-1\end{bmatrix}$$

$$\begin{bmatrix}1&-1&1\\0&1&0\\0&0&0\end{bmatrix} \qquad \begin{bmatrix}1&-1&0\\0&1&-1\\0&1&-1\end{bmatrix} \qquad \begin{bmatrix}1&-1&-3\\0&-4&-8\\0&-2&-4\end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & -1 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn these matrices back into systems of equations, we get

$$k_1 + k_3 = 0 \qquad k_1 - k_3 = 0 \qquad k_1 - k_3 = 0$$

$$k_2 = 0 \qquad k_2 - k_3 = 0 \qquad k_2 + 2k_3 = 0$$

Solving these systems gives the Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_3} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

and the solution vectors

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t} \qquad \overrightarrow{x_3} = \overrightarrow{k_3} e^{\lambda_3 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{2t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{3t} \qquad \overrightarrow{x_3} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} e^{6t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

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$$\overrightarrow{x_c} = c_1 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} e^{6t}$$

Now we can turn to finding the particular solution. We'll rewrite the forcing function vector as

$$F = \begin{bmatrix} -e^t \\ 3t^2 \\ 2e^{-t} \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} t^2 + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} e^{-t}$$

We want a particular solution in the same form, so our guess will be

$$\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} e^t + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} t^2 + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} t + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} e^{-t}$$

$$\overrightarrow{x_p}' = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} e^t + \begin{bmatrix} 2b_1 \\ 2b_2 \\ 2b_3 \end{bmatrix} t + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + \begin{bmatrix} -e_1 \\ -e_2 \\ -e_3 \end{bmatrix} e^{-t}$$

Plugging these into the matrix equation representing the system, we get

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} e^t + \begin{bmatrix} 2b_1 \\ 2b_2 \\ 2b_3 \end{bmatrix} t + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + \begin{bmatrix} -e_1 \\ -e_2 \\ -e_3 \end{bmatrix} e^{-t}$$

$$= \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix} \left[\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} e^t + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} t^2 + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} t + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} e^{-t} \right]$$

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$$+\begin{bmatrix}-1\\0\\0\end{bmatrix}e^{t}+\begin{bmatrix}0\\3\\0\end{bmatrix}t^{2}+\begin{bmatrix}0\\0\\2\end{bmatrix}e^{-t}$$

$$\begin{bmatrix}a_1e^{t}+2b_1t+c_1-e_1e^{-t}\\a_2e^{t}+2b_2t+c_2-e_2e^{-t}\\a_3e^{t}+2b_3t+c_3-e_3e^{-t}\end{bmatrix}=\begin{bmatrix}3&-1&1\\-1&5&-1\\1&-1&3\end{bmatrix}\begin{bmatrix}a_1e^{t}+b_1t^{2}+c_1t+d_1+e_1e^{-t}\\a_2e^{t}+b_2t^{2}+c_2t+d_2+e_2e^{-t}\\a_3e^{t}+b_3t^{2}+c_3t+d_3+e_3e^{-t}\end{bmatrix}$$

$$+\begin{bmatrix}-e^{t}\\3t^{2}\\2e^{-t}\end{bmatrix}$$

Breaking this equation into a system of three equations gives

$$a_1e^{t}+2b_1t+c_1-e_1e^{-t}=(3a_1-a_2+a_3-1)e^{t}+(3b_1-b_2+b_3)t^{2}$$

$$+(3c_1-c_2+c_3)t+(3d_1-d_2+d_3)+(3e_1-e_2+e_3)e^{-t}$$

$$a_2e^{t}+2b_2t+c_2-e_2e^{-t}=(-a_1+5a_2-a_3)e^{t}+(-b_1+5b_2-b_3+3)t^{2}$$

$$+(-c_1+5c_2-c_3)t+(-d_1+5d_2-d_3)+(-e_1+5e_2-e_3)e^{-t}$$

$$a_3e^{t}+2b_3t+c_3-e_3e^{-t}=(a_1-a_2+3a_3)e^{t}+(b_1-b_2+3b_3)t^{2}$$

$$+(c_1-c_2+3c_3)t+(d_1-d_2+3d_3)+(e_1-e_2+3e_3+2)e^{-t}$$

These equations can each be broken into its own system.

$$a_1=3a_1-a_2+a_3-1 \qquad a_2=-a_1+5a_2-a_3 \qquad a_3=a_1-a_2+3a_3$$

$$0=3b_1-b_2+b_3 \qquad 0=-b_1+5b_2-b_3+3 \qquad 0=b_1-b_2+3b_3$$

$$2b_1=3c_1-c_2+c_3 \qquad 2b_2=-c_1+5c_2-c_3 \qquad 2b_3=c_1-c_2+3c_3$$

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$$c_1 = 3d_1 - d_2 + d_3 \qquad c_2 = -d_1 + 5d_2 - d_3 \qquad c_3 = d_1 - d_2 + 3d_3$$

$$-e_1 = 3e_1 - e_2 + e_3 \qquad -e_2 = -e_1 + 5e_2 - e_3 \qquad -e_3 = e_1 - e_2 + 3e_3 + 2$$

Putting these results together gives $\vec{a} = (a_1, a_2, a_3) = (7/10, 1/10, -3/10)$, $\vec{b} = (b_1, b_2, b_3) = (-1/6, -2/3, -1/6)$, $\vec{c} = (c_1, c_2, c_3) = (-1/6, -1/3, -1/6)$, $\vec{d} = (d_1, d_2, d_3) = (-7/108, -5/54, -7/108)$, and $\vec{e} = (e_1, e_2, e_3) = (5/42, -1/14, -23/42)$. Therefore, the particular solution is

$$\vec{x_p} = \begin{bmatrix} \frac{7}{10} \\ \frac{1}{10} \\ -\frac{3}{10} \end{bmatrix} e^t + \begin{bmatrix} -\frac{1}{6} \\ -\frac{2}{3} \\ -\frac{1}{6} \end{bmatrix} t^2 + \begin{bmatrix} -\frac{1}{6} \\ -\frac{1}{3} \\ -\frac{1}{6} \end{bmatrix} t + \begin{bmatrix} -\frac{7}{108} \\ -\frac{5}{54} \\ -\frac{7}{108} \end{bmatrix} + \begin{bmatrix} \frac{5}{42} \\ -\frac{1}{14} \\ -\frac{23}{42} \end{bmatrix} e^{-t}$$

Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

$$\vec{x} = \vec{x_c} + \vec{x_p}$$

$$\vec{x} = c_1 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} e^{6t}$$

$$+ \begin{bmatrix} \frac{7}{10} \\ \frac{1}{10} \\ -\frac{3}{10} \end{bmatrix} e^t + \begin{bmatrix} -\frac{1}{6} \\ -\frac{2}{3} \\ -\frac{1}{6} \end{bmatrix} t^2 + \begin{bmatrix} -\frac{1}{6} \\ -\frac{1}{3} \\ -\frac{1}{6} \end{bmatrix} t + \begin{bmatrix} -\frac{7}{108} \\ -\frac{5}{54} \\ -\frac{7}{108} \end{bmatrix} + \begin{bmatrix} \frac{5}{42} \\ -\frac{1}{14} \\ -\frac{23}{42} \end{bmatrix} e^{-t}$$

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■ 5. Use the method of undetermined coefficients to find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix} \vec{x} + \begin{bmatrix} t + e^t \\ 2t^2 + 1 \end{bmatrix}$$

Solution:

The determinant $|A - \lambda I|$ is

$$|A - \lambda I| = \begin{vmatrix} 3 - \lambda & 0 \\ 1 & 4 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (3 - \lambda)(4 - \lambda) - (0)(1)$$

$$|A - \lambda I| = (3 - \lambda)(4 - \lambda)$$

Solve the characteristic equation.

$$(3 - \lambda)(4 - \lambda) = 0$$

$$\lambda_1 = 3 \qquad \lambda_2 = 4$$

Then for these Eigenvalues, $\lambda_1 = 3$ and $\lambda_2 = 4$, we find

$$A - 3I = \begin{bmatrix} 3 - 3 & 0 \\ 1 & 4 - 3 \end{bmatrix} \qquad A - 4I = \begin{bmatrix} 3 - 4 & 0 \\ 1 & 4 - 4 \end{bmatrix}$$

$$A - 3I = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \qquad A - 4I = \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix}$$

Put these matrices into reduced row-echelon form.

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$$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \qquad \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

If we turn these back into systems of equations, we get

$$k_1 + k_2 = 0 \qquad k_1 = 0$$

$$k_1 = -k_2$$

For the first system, we choose $(k_1, k_2) = (1, -1)$. And for the second system, we choose $(k_1, k_2) = (0,1)$. We find the Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and the solution vectors

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{3t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{4t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1 \overrightarrow{x_1} + c_2 \overrightarrow{x_2}$$

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$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{4t}$$

Now we can turn to finding the particular solution. We'll rewrite the forcing function vector as

$$F = \begin{bmatrix} t + e^t \\ 2t^2 + 1 \end{bmatrix}$$

$$F = \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} 0 \\ 2 \end{bmatrix} t^2 + \begin{bmatrix} 1 \\ 0 \end{bmatrix} t + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

We want a particular solution in the same form, so our guess will be

$$\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} e^t + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} t^2 + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} t + \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

$$\overrightarrow{x_p}' = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} e^t + \begin{bmatrix} 2b_1 \\ 2b_2 \end{bmatrix} t + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Plugging these into the matrix equation representing the system, we get

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} e^t + \begin{bmatrix} 2b_1 \\ 2b_2 \end{bmatrix} t + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix} \left[\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} e^t + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} t^2 + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} t + \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \right]$$

$$+ \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} 0 \\ 2 \end{bmatrix} t^2 + \begin{bmatrix} 1 \\ 0 \end{bmatrix} t + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} a_1 e^t + 2b_1 t + c_1 \\ a_2 e^t + 2b_2 t + c_2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} a_1 e^t + b_1 t^2 + c_1 t + d_1 \\ a_2 e^t + b_2 t^2 + c_2 t + d_2 \end{bmatrix} + \begin{bmatrix} t + e^t \\ 2t^2 + 1 \end{bmatrix}$$

Breaking this equation into a system of two equations gives

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$$a_1e^t + 2b_1t + c_1 = 3(a_1e^t + b_1t^2 + c_1t + d_1) + 0(a_2e^t + b_2t^2 + c_2t + d_2) + t + e^t$$

$$a_2e^t + 2b_2t + c_2 = 1(a_1e^t + b_1t^2 + c_1t + d_1) + 4(a_2e^{3t} + b_2t^2 + c_2t + d_2) + 2t^2 + 1$$

which simplifies to

$$a_1e^t + 2b_1t + c_1 = (3a_1 + 1)e^t + 3b_1t^2 + (3c_1 + 1)t + 3d_1$$

$$a_2e^t + 2b_2t + c_2 = (a_1 + 4a_2)e^t + (b_1 + 4b_2 + 2)t^2 + (c_1 + 4c_2)t + (d_1 + 4d_2 + 1)$$

These equations can each be broken into its own system.

$$a_1 = 3a_1 + 1 \qquad a_2 = a_1 + 4a_2$$

$$0 = 3b_1 \qquad 0 = b_1 + 4b_2 + 2$$

$$2b_1 = 3c_1 + 1 \qquad 2b_2 = c_1 + 4c_2$$

$$c_1 = 3d_1 \qquad c_2 = d_1 + 4d_2 + 1$$

Solving these eight equations as a system gives $(a_1, a_2) = (-1/2, 1/6)$, $(b_1, b_2) = (0, -1/2)$, $(c_1, c_2) = (-1/3, -1/6)$, and $(d_1, d_2) = (-1/9, -19/72)$. Therefore, the particular solution is

$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{6} \end{bmatrix} e^t + \begin{bmatrix} 0 \\ -\frac{1}{2} \end{bmatrix} t^2 + \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{6} \end{bmatrix} t + \begin{bmatrix} -\frac{1}{9} \\ -\frac{19}{72} \end{bmatrix}$$

Now we can rewrite the particular solution as one vector.

$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{1}{2}e^t - \frac{1}{3}t - \frac{1}{9} \\ \frac{1}{6}e^t - \frac{1}{2}t^2 - \frac{1}{6}t - \frac{19}{72} \end{bmatrix}$$

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Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

$$\vec{x} = \vec{x_c} + \vec{x_p}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{4t} + \begin{bmatrix} -\frac{1}{2}e^t - \frac{1}{3}t - \frac{1}{9} \\ \frac{1}{6}e^t - \frac{1}{2}t^2 - \frac{1}{6}t - \frac{19}{72} \end{bmatrix}$$

■ 6. Use undetermined coefficients to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 \\ -4 & 4 & 0 \\ -2 & 1 & 2 \end{bmatrix} \vec{x} + \begin{bmatrix} 3e^t \\ 3t + 2e^{3t} \\ t^3 - 4e^t \end{bmatrix}$$

Solution:

The determinant $|A - \lambda I|$ is

$$\begin{vmatrix} -\lambda & 1 & 0 \\ -4 & 4-\lambda & 0 \\ -2 & 1 & 2-\lambda \end{vmatrix}$$

$$(-\lambda)\begin{vmatrix} 4-\lambda & 0 \\ 1 & 2-\lambda \end{vmatrix} - 1\begin{vmatrix} -4 & 0 \\ -2 & 2-\lambda \end{vmatrix} + 0\begin{vmatrix} -4 & 4-\lambda \\ -2 & 1 \end{vmatrix}$$

$$(-\lambda)((4-\lambda)(2-\lambda) - (0)(1)) - ((-4)(2-\lambda) - (0)(-2)) + 0$$

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$$(-\lambda)(4-\lambda)(2-\lambda)+4(2-\lambda)$$

$$(2-\lambda)(\lambda^2-4\lambda+4)$$

Solve the characteristic equation.

$$(2-\lambda)(\lambda^2-4\lambda+4)=0$$

$$(2-\lambda)(\lambda-2)(\lambda-2)=0$$

$$\lambda_1=\lambda_2=\lambda_3=2$$

Then for these Eigenvalues, $\lambda_1=\lambda_2=\lambda_3=2$, we find

$$A-2I=\begin{bmatrix}-2 & 1 & 0\\ -4 & 4-2 & 0\\ -2 & 1 & 2-2\end{bmatrix}$$

$$A-2I=\begin{bmatrix}-2 & 1 & 0\\ -4 & 2 & 0\\ -2 & 1 & 0\end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix}-2 & 1 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}\rightarrow\begin{bmatrix}1 & -\frac{1}{2} & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

If we turn this matrix back into a system of equations, we get

$$k_1-\frac{1}{2}k_2=0$$

$$k_1=\frac{1}{2}k_2$$

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So we can choose $k_2 = 2$ and $k_3 = 0$ to get

$$\vec{k_1} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

We could also choose $k_2 = 0$ and $k_3 = 1$ to get

$$\vec{k_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

So we can say that the Eigenvalue $\lambda_1 = \lambda_2 = \lambda_3 = 2$ produces only two linearly independent Eigenvectors.

$$\vec{x_1} = \vec{k_1} e^{\lambda_1 t} \qquad \vec{x_2} = \vec{k_2} e^{\lambda_2 t}$$

$$\vec{x_1} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} e^{2t} \qquad \vec{x_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t}$$

Because we find two Eigenvectors for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = 2$, we have to find a third solution.

$$(A - \lambda_1 I)\vec{p_1} = a_1\vec{k_1} + a_2\vec{k_2}$$

$$\begin{bmatrix} -2 & 1 & 0 \\ -4 & 2 & 0 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ 2a_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ a_2 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$-2p_1 + p_2 = a_1$$

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$$-4p_1 + 2p_2 = 2a_1$$

$$-2p_1 + p_2 = a_2$$

We can choose $p_1 = -1$ with $p_2 = 0$ and $p_3 = 0$ to get the Eigenvector

$$\overrightarrow{p_1} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

Then the solution vector will be

$$\overrightarrow{x_3} = \overrightarrow{p_1}e^{\lambda_1 t} + (a_1\overrightarrow{k_1} + a_2\overrightarrow{k_2})te^{\lambda_1 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{2t} + \left(2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) te^{2t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{2t} + \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} te^{2t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{2t} + \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} te^{2t} \right)$$

Now we can turn to finding the particular solution. We'll rewrite the forcing function vector as

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$$F = \begin{bmatrix} 3e^t \\ 3t + 2e^{3t} \\ t^3 - 4e^t \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -4 \end{bmatrix} e^t + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} e^{3t} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} t^3 + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} t$$

We want a particular solution in the same form, so our guess will be

$$\overrightarrow{x_p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} e^t + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} e^{3t} + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} t^3 + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} t^2 + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} t + \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}$$

$$\overrightarrow{x_p}' = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} e^t + \begin{bmatrix} 3b_1 \\ 3b_2 \\ 3b_3 \end{bmatrix} e^{3t} + \begin{bmatrix} 3c_1 \\ 3c_2 \\ 3c_3 \end{bmatrix} t^2 + \begin{bmatrix} 2d_1 \\ 2d_2 \\ 2d_3 \end{bmatrix} t + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

Plugging this into the matrix equation representing the system, we get

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} e^t + \begin{bmatrix} 3b_1 \\ 3b_2 \\ 3b_3 \end{bmatrix} e^{3t} + \begin{bmatrix} 3c_1 \\ 3c_2 \\ 3c_3 \end{bmatrix} t^2 + \begin{bmatrix} 2d_1 \\ 2d_2 \\ 2d_3 \end{bmatrix} t + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ -4 & 4 & 0 \\ -2 & 1 & 2 \end{bmatrix} \left(\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} e^t + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} e^{3t} + \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} t^3 + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} t^2 + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} t + \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \right)$$

$$+ \begin{bmatrix} 3 \\ 0 \\ -4 \end{bmatrix} e^t + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} e^{3t} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} t^3 + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} t$$

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$$\begin{bmatrix} a_1e^t+3b_1e^{3t}+3c_1t^2+2d_1t+e_1 \\ a_2e^t+3b_2e^{3t}+3c_2t^2+2d_2t+e_2 \\ a_3e^t+3b_3e^{3t}+3c_3t^2+2d_3t+e_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -4 & 4 & 0 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} a_1e^t+b_1e^{3t}+c_1t^3+d_1t^2+e_1t+f_1 \\ a_2e^t+b_2e^{3t}+c_2t^3+d_2t^2+e_2t+f_2 \\ a_3e^t+b_3e^{3t}+c_3t^3+d_3t^2+e_3t+f_3 \end{bmatrix}$$

$$+\begin{bmatrix} 3e^t \\ 3t+2e^{3t} \\ t^3-4e^t \end{bmatrix}$$

Breaking this equation into a system of three equations gives

$$a_1e^t+3b_1e^{3t}+3c_1t^2+2d_1t+e_1=(a_2+3)e^t+b_2e^{3t}+c_2t^3+d_2t^2+e_2t+f_2$$

$$a_2e^t+3b_2e^{3t}+3c_2t^2+2d_2t+e_2=(4a_2-4a_1)e^t+(2+4b_2-4b_1)e^{3t}$$

$$+(4c_2-4c_1)t^3+(4d_2-4d_1)t^2+(3+4e_2-4e_1)t+(4f_2-4f_1)$$

$$a_3e^t+3b_3e^{3t}+3c_3t^2+2d_3t+e_3=(-2a_1+a_2+2a_3-4)e^t+(-2b_1+b_2+2b_3)e^{3t}$$

$$+(-2c_1+c_2+2c_3+1)t^3+(-2d_1+d_2+2d_3)t^2$$

$$+(-2e_1+e_2+2e_3)t+(-2f_1+f_2+2f_3)$$

These equations can each be broken into its own system.

$$a_1=a_2+3 \qquad a_2=4a_2-4a_1 \qquad a_3=-2a_1+a_2+2a_3-4$$

$$3b_1=b_2 \qquad 3b_2=2+4b_2-4b_1 \qquad 3b_3=-2b_1+b_2+2b_3$$

$$0=c_2 \qquad 0=4c_2-4c_1 \qquad 0=-2c_1+c_2+2c_3+1$$

$$3c_1=d_2 \qquad 3c_2=4d_2-4d_1 \qquad 3c_3=-2d_1+d_2+2d_3$$

$$2d_1=e_2 \qquad 2d_2=3+4e_2-4e_1 \qquad 2d_3=-2e_1+e_2-2e_3$$

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$$e_1 = f_2 \qquad e_2 = 4f_2 - 4f_1 \qquad e_3 = -2f_1 + f_2 + 2f_3$$

Putting these results together gives $\vec{a} = (a_1, a_2, a_3) = (-9, -12, -2)$, $\vec{b} = (b_1, b_2, b_3) = (2,6,2)$, $\vec{c} = (c_1, c_2, c_3) = (0,0, -1/2)$, $\vec{d} = (d_1, d_2, d_3) = (0,0, -3/4)$, $\vec{e} = (e_1, e_2, e_3) = (3/4,0,0)$, and $\vec{f} = (f_1, f_2, f_3) = (3/4,3/4,3/8)$. Therefore, the particular solution becomes

$$\vec{x_p} = \begin{bmatrix} -9 \\ -12 \\ -2 \end{bmatrix} e^t + \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} e^{3t} + \begin{bmatrix} 0 \\ 0 \\ -\frac{1}{2} \end{bmatrix} t^3 + \begin{bmatrix} 0 \\ 0 \\ -\frac{3}{4} \end{bmatrix} t^2 + \begin{bmatrix} \frac{3}{4} \\ 0 \\ 0 \end{bmatrix} t + \begin{bmatrix} \frac{3}{4} \\ \frac{3}{4} \\ \frac{3}{8} \end{bmatrix}$$

Summing the complementary and particular solutions gives the general solution to the nonhomogeneous system of differential equations.

$$\vec{x} = \vec{x_c} + \vec{x_p}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t} + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{2t} + \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} te^{2t} \right)$$

$$+ \begin{bmatrix} -9 \\ -12 \\ -2 \end{bmatrix} e^t + \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} e^{3t} + \begin{bmatrix} 0 \\ 0 \\ -\frac{1}{2} \end{bmatrix} t^3 + \begin{bmatrix} 0 \\ 0 \\ -\frac{3}{4} \end{bmatrix} t^2 + \begin{bmatrix} \frac{3}{4} \\ 0 \\ 0 \end{bmatrix} t + \begin{bmatrix} \frac{3}{4} \\ \frac{3}{4} \\ \frac{3}{8} \end{bmatrix}$$

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VARIATION OF PARAMETERS FOR NONHOMOGENEOUS SYSTEMS

■ 1. Use the method of variation of parameters to find the general solution to the system, given the complementary solution.

$$\vec{x}' = A\vec{x} + \begin{bmatrix} e^{-3t} \\ 2e^{t} + 1 \end{bmatrix}$$

$$\vec{x_c} = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 2 \\ 7 \end{bmatrix} e^{5t}$$

Solution:

We already have the complementary solution, so we'll use the solution vectors $\vec{x_1}$ and $\vec{x_2}$ to form $\Phi(t)$.

$$\Phi(t) = \begin{bmatrix} e^{2t} & 2e^{5t} \\ 3e^{2t} & 7e^{5t} \end{bmatrix}$$

Find the inverse $\Phi^{-1}(t)$ by changing $[\Phi(t)\,|\,I]$ into $[I\,|\,\Phi^{-1}(t)]$.

$$\left[\begin{array}{cc|cc} e^{2t} & 2e^{5t} & 1 & 0 \\ 3e^{2t} & 7e^{5t} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 2e^{3t} & e^{-2t} & 0 \\ 3e^{2t} & 7e^{5t} & 0 & 1 \end{array}\right]$$

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$$\left[\begin{array}{cc|cc} 1 & 2e^{3t} & e^{-2t} & 0 \\ 0 & e^{5t} & -3 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 2e^{3t} & e^{-2t} & 0 \\ 0 & 1 & -3e^{-5t} & e^{-5t} \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 7e^{-2t} & -2e^{-2t} \\ 0 & 1 & -3e^{-5t} & e^{-5t} \end{array}\right]$$

The inverse matrix is

$$\Phi^{-1}(t) = \begin{bmatrix} 7e^{-2t} & -2e^{-2t} \\ -3e^{-5t} & e^{-5t} \end{bmatrix}$$

so we'll integrate the product of the inverse matrix and $F(t)$ to get

$$\int \Phi^{-1}(t)F(t)\ dt$$

$$\int \begin{bmatrix} 7e^{-2t} & -2e^{-2t} \\ -3e^{-5t} & e^{-5t} \end{bmatrix} \begin{bmatrix} e^{-3t} \\ 2e^{t}+1 \end{bmatrix}\ dt$$

$$\int \begin{bmatrix} 7e^{-5t}-4e^{-t}-2e^{-2t} \\ -3e^{-8t}+2e^{-4t}+e^{-5t} \end{bmatrix}\ dt$$

$$\begin{bmatrix} -\frac{7}{5}e^{-5t}+4e^{-t}+e^{-2t} \\ \frac{3}{8}e^{-8t}-\frac{1}{2}e^{-4t}-\frac{1}{5}e^{-5t} \end{bmatrix}$$

Then the particular solution is given by

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$$\overrightarrow{x_p} = \Phi(t) \int \Phi^{-1}(t) F(t)\ dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} & 2e^{5t} \\ 3e^{2t} & 7e^{5t} \end{bmatrix} \begin{bmatrix} -\frac{7}{5}e^{-5t} + 4e^{-t} + e^{-2t} \\ \frac{3}{8}e^{-8t} - \frac{1}{2}e^{-4t} - \frac{1}{5}e^{-5t} \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{7}{5}e^{-3t} + 4e^{t} + 1 + \frac{3}{4}e^{-3t} - e^{t} - \frac{2}{5} \\ -\frac{21}{5}e^{-3t} + 12e^{t} + 3 + \frac{21}{8}e^{-3t} - \frac{7}{2}e^{t} - \frac{7}{5} \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} -\frac{13}{20}e^{-3t} + 3e^{t} + \frac{3}{5} \\ -\frac{63}{40}e^{-3t} + \frac{17}{2}e^{t} + \frac{8}{5} \end{bmatrix}$$

$$\overrightarrow{x_p} = -\frac{1}{20} \begin{bmatrix} 13 \\ \frac{63}{2} \end{bmatrix} e^{-3t} + \begin{bmatrix} 3 \\ \frac{17}{2} \end{bmatrix} e^{t} + \frac{1}{5} \begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

Then the general solution is

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 2 \\ 7 \end{bmatrix} e^{5t} - \frac{1}{20} \begin{bmatrix} 13 \\ \frac{63}{2} \end{bmatrix} e^{-3t} + \begin{bmatrix} 3 \\ \frac{17}{2} \end{bmatrix} e^{t} + \frac{1}{5} \begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

2. Find the general solution to the system of differential equations.

$$\overrightarrow{x}' = \begin{bmatrix} 6 & -8 \\ 4 & -6 \end{bmatrix} \overrightarrow{x} + \begin{bmatrix} 2t + 1 \\ e^{t} \end{bmatrix}$$

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Solution:

We need to start by solving the corresponding homogeneous system,

$$\vec{x}' = \begin{bmatrix} 6 & -8 \\ 4 & -6 \end{bmatrix} \vec{x}$$

so we'll find $|A - \lambda I|$.

$$|A - \lambda I| = \left| \begin{bmatrix} 6 & -8 \\ 4 & -6 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 6 & -8 \\ 4 & -6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 6 - \lambda & -8 \\ 4 & -6 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = \lambda^2 - 36 + 32 = \lambda^2 - 4$$

$$|A - \lambda I| = (\lambda + 2)(\lambda - 2)$$

Solve the characteristic equation for the Eigenvalues of the system.

$$(\lambda + 2)(\lambda - 2) = 0$$

$$\lambda = 2, \ -2$$

Then for these Eigenvalues, $\lambda_1 = 2$ and $\lambda_2 = -2$, we find

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$$A - 2I = \begin{bmatrix} 4 & -8 \\ 4 & -8 \end{bmatrix} \qquad A + 2I = \begin{bmatrix} 8 & -8 \\ 4 & -4 \end{bmatrix}$$

Put these matrices into reduced row-echelon form.

$$\begin{bmatrix} 4 & -8 \\ 4 & -8 \end{bmatrix} \qquad \begin{bmatrix} 8 & -8 \\ 4 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ 4 & -8 \end{bmatrix} \qquad \begin{bmatrix} 1 & -1 \\ 4 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

If we turn each matrix back into a system of equations, we get

$$k_1 - 2k_2 = 0 \qquad k_1 - k_2 = 0$$

$$k_1 = 2k_2 \qquad k_1 = k_2$$

For the first system, we choose $(k_1, k_2) = (2,1)$, and for the second system, we choose $(k_1, k_2) = (1,1)$, and we get the Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then the solution vectors are

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{2t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-2t}$$

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Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\vec{x_c} = c_1\vec{x_1} + c_2\vec{x_2}$$

$$\vec{x_c} = c_1\begin{bmatrix}2\\1\end{bmatrix}e^{2t} + c_2\begin{bmatrix}1\\1\end{bmatrix}e^{-2t}$$

Now we can turn to finding the particular solution. We'll use the solution vectors $\vec{x_1}$ and $\vec{x_2}$ to form $\Phi(t)$,

$$\Phi(t) = \begin{bmatrix}2e^{2t} & e^{-2t}\\ e^{2t} & e^{-2t}\end{bmatrix}$$

then we'll find its inverse $\Phi^{-1}(t)$ by changing $[\Phi(t)\,|\,I]$ into $[I\,|\,\Phi^{-1}(t)]$.

$$\left[\begin{array}{cc|cc}2e^{2t} & e^{-2t} & 1 & 0\\ e^{2t} & e^{-2t} & 0 & 1\end{array}\right]$$

$$\left[\begin{array}{cc|cc}1 & \frac{1}{2}e^{-4t} & \frac{1}{2}e^{-2t} & 0\\ e^{2t} & e^{-2t} & 0 & 1\end{array}\right]$$

$$\left[\begin{array}{cc|cc}1 & \frac{1}{2}e^{-4t} & \frac{1}{2}e^{-2t} & 0\\ 0 & \frac{1}{2}e^{-2t} & -\frac{1}{2} & 1\end{array}\right]$$

$$\left[\begin{array}{cc|cc}1 & \frac{1}{2}e^{-4t} & \frac{1}{2}e^{-2t} & 0\\ 0 & 1 & -e^{2t} & 2e^{2t}\end{array}\right]$$

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$$\left[\begin{array}{cc|cc} 1 & 0 & e^{-2t} & -e^{-2t} \\ 0 & 1 & -e^{2t} & 2e^{2t} \end{array}\right]$$

The inverse matrix is

$$\Phi^{-1}(t) = \begin{bmatrix} e^{-2t} & -e^{-2t} \\ -e^{2t} & 2e^{2t} \end{bmatrix}$$

so we'll integrate the product of the inverse matrix and $F(t)$ to get

$$\int \Phi^{-1}(t)F(t)\ dt$$

$$\int \begin{bmatrix} e^{-2t} & -e^{-2t} \\ -e^{2t} & 2e^{2t} \end{bmatrix} \begin{bmatrix} 2t+1 \\ e^{t} \end{bmatrix}\ dt$$

$$\int \begin{bmatrix} 2te^{-2t} + e^{-2t} - e^{-t} \\ -2te^{2t} - e^{2t} + 2e^{3t} \end{bmatrix}\ dt$$

$$\begin{bmatrix} -te^{-2t} - e^{-2t} + e^{-t} \\ -te^{2t} + \frac{2}{3}e^{3t} \end{bmatrix}$$

Then the particular solution is given by

$$\overrightarrow{x_p} = \Phi(t) \int \Phi^{-1}(t)F(t)\ dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} 2e^{2t} & e^{-2t} \\ e^{2t} & e^{-2t} \end{bmatrix} \begin{bmatrix} -te^{-2t} - e^{-2t} + e^{-t} \\ -te^{2t} + \frac{2}{3}e^{3t} \end{bmatrix}$$

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$$\overrightarrow{x_p} = \begin{bmatrix} -2t - 2 + 2e^t - t + \frac{2}{3}e^t \\ -t - 1 + e^t - t + \frac{2}{3}e^t \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} -3 \\ -2 \end{bmatrix} t + \begin{bmatrix} -2 \\ -1 \end{bmatrix} + \begin{bmatrix} \frac{8}{3} \\ \frac{5}{3} \end{bmatrix} e^t$$

Then the general solution is

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-2t} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} t - \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{3}e^t \begin{bmatrix} 8 \\ 5 \end{bmatrix}$$

3. Solve the system of differential equations.

$$\overrightarrow{x}' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \overrightarrow{x} + \begin{bmatrix} \sin t + \cos t \\ \sin t \end{bmatrix}$$

Solution:

We need to start by solving the corresponding homogeneous system,

$$\overrightarrow{x}' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \overrightarrow{x}$$

so we'll find $|A - \lambda I|$.

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$$|A-\lambda I| = \left| \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A-\lambda I| = \left| \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A-\lambda I| = \begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix}$$

$$|A-\lambda I| = \lambda^2 + 1$$

Solve the characteristic equation for the Eigenvalues of the system.

$$\lambda^2 + 1 = 0$$

$$\lambda = \pm i$$

Then for these Eigenvalues, $\lambda_1 = i$ and $\lambda_2 = -i$, we find

$$A - iI = \begin{bmatrix} -i & 1 \\ -1 & -i \end{bmatrix}$$

Put the matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & i \\ -1 & -i \end{bmatrix}$$

$$\begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix}$$

Rewriting the matrix as a system of equations gives

$$k_1 + ik_2 = 0$$

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$$k_1 = -ik_2$$

We'll choose $k_1 = i$ and $k_2 = -1$ to get the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} i \\ -1 \end{bmatrix}$$

$$\overrightarrow{k_1} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} i$$

Then the solutions to the system are

$$\overrightarrow{x_1} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \cos t - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \sin t = \begin{bmatrix} -\sin t \\ -\cos t \end{bmatrix}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cos t + \begin{bmatrix} 0 \\ -1 \end{bmatrix} \sin t = \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$$

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} -\sin t \\ -\cos t \end{bmatrix} + c_2 \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix}$$

Now we can turn to finding the particular solution. We'll use the solution vectors $\overrightarrow{x_1}$ and $\overrightarrow{x_2}$ to form $\Phi(t)$,

$$\Phi(t) = \begin{bmatrix} -\sin t & \cos t \\ -\cos t & -\sin t \end{bmatrix}$$

then we'll find its inverse $\Phi^{-1}(t)$ by changing $[\Phi(t)\,|\,I]$ into $[I\,|\,\Phi^{-1}(t)]$.

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$$\left[\begin{array}{cc|cc} -\sin t & \cos t & 1 & 0 \\ -\cos t & -\sin t & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & -\cot t & -\frac{1}{\sin t} & 0 \\ -\cos t & -\sin t & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & -\cot t & -\frac{1}{\sin t} & 0 \\ 0 & -\frac{1}{\sin t} & -\frac{\cos t}{\sin t} & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & -\cot t & -\frac{1}{\sin t} & 0 \\ 0 & 1 & \cos t & -\sin t \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 0 & -\sin t & -\cos t \\ 0 & 1 & \cos t & -\sin t \end{array}\right]$$

The inverse matrix is

$$\Phi^{-1}(t) = \begin{bmatrix} -\sin t & -\cos t \\ \cos t & -\sin t \end{bmatrix}$$

so we'll integrate the product of the inverse matrix and $F(t)$ to get

$$\int \Phi^{-1}(t) F(t)\, dt$$

$$\int \begin{bmatrix} -\sin t & -\cos t \\ \cos t & -\sin t \end{bmatrix} \begin{bmatrix} \sin t + \cos t \\ \sin t \end{bmatrix} dt$$

$$\int \begin{bmatrix} -\sin^2 t - 2\sin t \cos t \\ \sin t \cos t + \cos^2 t - \sin^2 t \end{bmatrix} dt$$

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$$\int \begin{bmatrix} -\frac{1}{2} + \frac{1}{2}\cos(2t) - \sin(2t) \\ \frac{1}{2}\sin(2t) + \cos(2t) \end{bmatrix} dt$$

$$\begin{bmatrix} -\frac{t}{2} + \frac{1}{4}\sin(2t) + \frac{1}{2}\cos(2t) \\ -\frac{1}{4}\cos(2t) + \frac{1}{2}\sin(2t) \end{bmatrix}$$

Then the particular solution is given by

$$\vec{x_p} = \Phi(t) \int \Phi^{-1}(t) F(t)\ dt$$

$$\vec{x_p} = \begin{bmatrix} -\sin t & \cos t \\ -\cos t & -\sin t \end{bmatrix} \begin{bmatrix} -\frac{t}{2} + \frac{1}{4}\sin(2t) + \frac{1}{2}\cos(2t) \\ -\frac{1}{4}\cos(2t) + \frac{1}{2}\sin(2t) \end{bmatrix}$$

$$\vec{x_p} = \begin{bmatrix} \frac{t}{2}\sin t - \frac{1}{4}\sin(2t)\sin t - \frac{1}{2}\cos(2t)\sin t - \frac{1}{4}\cos t\cos(2t) + \frac{1}{2}\cos t\sin(2t) \\ \frac{t}{2}\cos t - \frac{1}{4}\sin(2t)\cos t - \frac{1}{2}\cos(2t)\cos t + \frac{1}{4}\cos(2t)\sin t - \frac{1}{2}\sin(2t)\sin t \end{bmatrix}$$

$$\vec{x_p} = \frac{1}{4}\begin{bmatrix} 2t - \sin(2t) - 2\cos(2t) \\ \cos(2t) - 2\sin(2t) \end{bmatrix} \sin t + \frac{1}{4}\begin{bmatrix} -\cos(2t) + 2\sin(2t) \\ 2t - \sin(2t) - 2\cos(2t) \end{bmatrix} \cos t$$

Then the general solution is

$$\vec{x} = \vec{x_c} + \vec{x_p}$$

$$\vec{x} = c_1 \begin{bmatrix} -\sin t \\ -\cos t \end{bmatrix} + c_2 \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix} + \frac{1}{4}\begin{bmatrix} 2t - \sin(2t) - 2\cos(2t) \\ \cos(2t) - 2\sin(2t) \end{bmatrix} \sin t$$

$$+\frac{1}{4}\begin{bmatrix} -\cos(2t) + 2\sin(2t) \\ 2t - \sin(2t) - 2\cos(2t) \end{bmatrix} \cos t$$

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■ 4. Solve the system of differential equations.

$$x_1' = -5x_1 + 12x_2 + e^t$$

$$x_2' = -4x_1 + 9x_2 + 3e^{-t}$$

Solution:

We need to start by solving the corresponding homogeneous system,

$$\vec{x}' = \begin{bmatrix} -5 & 12 \\ -4 & 9 \end{bmatrix} \vec{x}$$

so we'll find $|A - \lambda I|$.

$$|A - \lambda I| = \left| \begin{bmatrix} -5 & 12 \\ -4 & 9 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} -5 & 12 \\ -4 & 9 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} -5 - \lambda & 12 \\ -4 & 9 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = \lambda^2 - 4\lambda - 45 + 48 = (\lambda - 1)(\lambda - 3)$$

Solve the characteristic equation for the Eigenvalues of the system.

$$(\lambda - 1)(\lambda - 3) = 0$$

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$$\lambda = 1,\ 3$$

Then for these Eigenvalues, $\lambda_1 = 1$ and $\lambda_2 = 3$, we find

$$A - I = \begin{bmatrix} -6 & 12 \\ -4 & 8 \end{bmatrix} \qquad A - 3I = \begin{bmatrix} -8 & 12 \\ -4 & 6 \end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix} -6 & 12 \\ -4 & 8 \end{bmatrix} \qquad \begin{bmatrix} -8 & 12 \\ -4 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ -4 & 8 \end{bmatrix} \qquad \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 2 & -3 \\ 0 & 0 \end{bmatrix}$$

If we turn these back into systems of equations, we get

$$k_1 - 2k_2 = 0 \qquad 2k_1 - 3k_2 = 0$$

$$k_1 = 2k_2 \qquad 2k_1 = 3k_2$$

For the first system, we choose $(k_1, k_2) = (2,1)$ and from the second system, we choose $(k_1, k_2) = (3,2)$, which gives the Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

and the solution vectors

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

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$$\vec{x_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^t \qquad \vec{x_2} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{3t}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\vec{x_c} = c_1\vec{x_1} + c_2\vec{x_2}$$

$$\vec{x_c} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{3t}$$

Now we can turn to finding the particular solution. We'll use the solution vectors $\vec{x_1}$ and $\vec{x_2}$ to form $\Phi(t)$,

$$\Phi(t) = \begin{bmatrix} 2e^t & 3e^{3t} \\ e^t & 2e^{3t} \end{bmatrix}$$

then we'll find its inverse $\Phi^{-1}(t)$ by changing $[\Phi(t)\,|\,I]$ into $[I\,|\,\Phi^{-1}(t)]$.

$$\left[\begin{array}{cc|cc} 2e^t & 3e^{3t} & 1 & 0 \\ e^t & 2e^{3t} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & \frac{3}{2}e^{2t} & \frac{1}{2}e^{-t} & 0 \\ e^t & 2e^{3t} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & \frac{3}{2}e^{2t} & \frac{1}{2}e^{-t} & 0 \\ 0 & \frac{1}{2}e^{3t} & -\frac{1}{2} & 1 \end{array}\right]$$

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$$\left[\begin{array}{cc|cc} 1 & \frac{3}{2}e^{2t} & \frac{1}{2}e^{-t} & 0 \\ 0 & 1 & -e^{-3t} & 2e^{-3t} \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 2e^{-t} & -3e^{-t} \\ 0 & 1 & -e^{-3t} & 2e^{-3t} \end{array}\right]$$

The inverse matrix is

$$\Phi^{-1}(t) = \begin{bmatrix} 2e^{-t} & -3e^{-t} \\ -e^{-3t} & 2e^{-3t} \end{bmatrix}$$

so we'll integrate the product of the inverse matrix and $F(t)$ to get

$$\int \Phi^{-1}(t)F(t)\ dt$$

$$\int \begin{bmatrix} 2e^{-t} & -3e^{-t} \\ -e^{-3t} & 2e^{-3t} \end{bmatrix} \begin{bmatrix} e^{t} \\ 3e^{-t} \end{bmatrix}\ dt$$

$$\int \begin{bmatrix} 2 - 9e^{-2t} \\ -e^{-2t} + 6e^{-4t} \end{bmatrix}\ dt$$

$$\begin{bmatrix} 2t + \frac{9}{2}e^{-2t} \\ \frac{1}{2}e^{-2t} - \frac{3}{2}e^{-4t} \end{bmatrix}$$

Then the particular solution is given by

$$\overrightarrow{x_p} = \Phi(t) \int \Phi^{-1}(t)F(t)\ dt$$

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$$\overrightarrow{x_p} = \begin{bmatrix} 2e^t & 3e^{3t} \\ e^t & 2e^{3t} \end{bmatrix} \begin{bmatrix} 2t + \frac{9}{2}e^{-2t} \\ \frac{1}{2}e^{-2t} - \frac{3}{2}e^{-4t} \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} 4te^t + 9e^{-t} + \frac{3}{2}e^t - \frac{9}{2}e^{-t} \\ 2te^t + \frac{9}{2}e^{-t} + e^t - 3e^{-t} \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} te^t + \begin{bmatrix} \frac{3}{2} \\ 1 \end{bmatrix} e^t + \begin{bmatrix} \frac{9}{2} \\ \frac{3}{2} \end{bmatrix} e^{-t}$$

$$\overrightarrow{x_p} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} te^t + \begin{bmatrix} \frac{3}{2} \\ 1 \end{bmatrix} e^t + \frac{3}{2} \begin{bmatrix} 3 \\ 1 \end{bmatrix} e^{-t}$$

Then the general solution is

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{3t} + \begin{bmatrix} 4 \\ 2 \end{bmatrix} te^t + \begin{bmatrix} \frac{3}{2} \\ 1 \end{bmatrix} e^t + \frac{3}{2} \begin{bmatrix} 3 \\ 1 \end{bmatrix} e^{-t}$$

5. Solve the system of differential equations.

$$x_1' = x_1 + 3x_2 + 1 + e^{2t}$$

$$x_2' = x_2 + e^t$$

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Solution:

We need to start by solving the corresponding homogeneous system,

$$\vec{x}' = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \vec{x}$$

so we'll find $|A - \lambda I|$.

$$|A - \lambda I| = \left| \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 3 \\ 0 & 1 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (1 - \lambda)^2 - 0 = (1 - \lambda)^2$$

Solve the characteristic equation for the Eigenvalues of the system.

$$(1 - \lambda)^2 = 0$$

$$\lambda = 1$$

Then for the Eigenvalue $\lambda = 1$, we find

$$A - I = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}$$

Put the matrix into reduced row-echelon form.

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$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

If we turn this matrix back into a system of equations, we get just $k_2 = 0$, so we'll choose $k_1 = 1$ with $k_2 = 0$ to find the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Then we'll use this Eigenvector to find a second solution vector.

$$\begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

This matrix equation gives the system of equations $3p_2 = 1$ and $p_1 = 0$, and then the solution vector

$$\overrightarrow{p} = \begin{bmatrix} 0 \\ \frac{1}{3} \end{bmatrix}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ \frac{1}{3} \end{bmatrix} e^t \right)$$

Now we can turn to finding the particular solution. We'll use the solution vectors $\overrightarrow{x_1}$ and $\overrightarrow{x_2}$ to form $\Phi(t)$,

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$$\Phi(t) = \begin{bmatrix} e^t & te^t \\ 0 & \frac{1}{3}e^t \end{bmatrix}$$

then we'll find its inverse $\Phi^{-1}(t)$ by changing $[\Phi(t)\,|\,I]$ into $[I\,|\,\Phi^{-1}(t)]$.

$$\left[\begin{array}{cc|cc} e^t & te^t & 1 & 0 \\ 0 & \frac{1}{3}e^t & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & t & e^{-t} & 0 \\ 0 & \frac{1}{3}e^t & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & t & e^{-t} & 0 \\ 0 & 1 & 0 & 3e^{-t} \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 0 & e^{-t} & -3te^{-t} \\ 0 & 1 & 0 & 3e^{-t} \end{array}\right]$$

The inverse matrix is

$$\Phi^{-1}(t) = \begin{bmatrix} e^{-t} & -3te^{-t} \\ 0 & 3e^{-t} \end{bmatrix}$$

so we'll integrate the product of the inverse matrix and $F(t)$ to get

$$\int \Phi^{-1}(t)F(t)\,dt$$

$$\int \begin{bmatrix} e^{-t} & -3te^{-t} \\ 0 & 3e^{-t} \end{bmatrix} \begin{bmatrix} e^{2t}+1 \\ e^t \end{bmatrix}\,dt$$

$$\int \begin{bmatrix} e^t + e^{-t} - 3t \\ 3 \end{bmatrix}\,dt$$

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$$\begin{bmatrix} e^t - e^{-t} - \frac{3t^2}{2} \\ 3t \end{bmatrix}$$

Then the particular solution is given by

$$\overrightarrow{x_p} = \Phi(t) \int \Phi^{-1}(t) F(t)\ dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^t & te^t \\ 0 & \frac{1}{3}e^t \end{bmatrix} \begin{bmatrix} e^t - e^{-t} - \frac{3t^2}{2} \\ 3t \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} - 1 - \frac{3t^2}{2}e^t + 3t^2 e^t \\ te^t \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{2t} - 1 + \frac{3t^2}{2}e^t \\ te^t \end{bmatrix}$$

Then the general solution is

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ \frac{1}{3} \end{bmatrix} e^t \right) + \begin{bmatrix} e^{2t} - 1 + \frac{3t^2}{2}e^t \\ te^t \end{bmatrix}$$

6. Solve the system of differential equations.

$$\overrightarrow{x}' = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \overrightarrow{x} + \begin{bmatrix} t^5 \\ t^3 \end{bmatrix}$$

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Solution:

We need to start by solving the corresponding homogeneous system,

$$\vec{x}' = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \vec{x}$$

so we'll find $|A - \lambda I|$.

$$|A - \lambda I| = \left| \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 2 \\ 0 & -\lambda \end{vmatrix}$$

$$|A - \lambda I| = \lambda^2$$

Solve the characteristic equation for the Eigenvalues of the system.

$$\lambda^2 = 0$$

$$\lambda = 0$$

Then for the Eigenvalue $\lambda = 0$, we find

$$A - (0)I = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$

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Put the matrix into reduced row-echelon form.

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

If we rewrite this matrix as a system of equations, we get just $k_2 = 0$, so we'll choose $k_1 = 1$ with $k_2 = 0$ to get the Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Then we'll use this Eigenvector to find a second solution vector.

$$\begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

This matrix equation gives the system of equations $2p_2 = 1$, so we find $p_2 = 1/2$ and choose $p_1 = 0$ to get the second solution vector

$$\overrightarrow{p} = \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix}$$

Therefore, the complementary solution to the nonhomogeneous system, which is also the general solution to the associated homogeneous system, will be

$$\overrightarrow{x_c} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \left(\begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} t \right)$$

Now we can turn to finding the particular solution. We'll use the solution vectors $\overrightarrow{x_1}$ and $\overrightarrow{x_2}$ to form $\Phi(t)$,

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$$\Phi(t) = \begin{bmatrix} 1 & t \\ 0 & \frac{1}{2} \end{bmatrix}$$

then we'll find its inverse $\Phi^{-1}(t)$ by changing $[\Phi(t)\,|\,I]$ into $[I\,|\,\Phi^{-1}(t)]$.

$$\left[\begin{array}{cc|cc} 1 & t & 1 & 0 \\ 0 & \frac{1}{2} & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & t & 1 & 0 \\ 0 & 1 & 0 & 2 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & -2t \\ 0 & 1 & 0 & 2 \end{array}\right]$$

The inverse matrix is

$$\Phi^{-1}(t) = \begin{bmatrix} 1 & -2t \\ 0 & 2 \end{bmatrix}$$

so we'll integrate the product of the inverse matrix and $F(t)$ to get

$$\int \Phi^{-1}(t)F(t)\,dt$$

$$\int \begin{bmatrix} 1 & -2t \\ 0 & 2 \end{bmatrix} \begin{bmatrix} t^5 \\ t^3 \end{bmatrix}\,dt$$

$$\int \begin{bmatrix} t^5 - 2t^4 \\ 2t^3 \end{bmatrix}\,dt$$

$$\begin{bmatrix} \frac{1}{6}t^6 - \frac{2}{5}t^5 \\ \frac{1}{2}t^4 \end{bmatrix}$$

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Then the particular solution is given by

$$\overrightarrow{x_p} = \Phi(t) \int \Phi^{-1}(t) F(t)\ dt$$

$$\overrightarrow{x_p} = \begin{bmatrix} 1 & t \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{6}t^6 - \frac{2}{5}t^5 \\ \frac{1}{2}t^4 \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} \frac{1}{6}t^6 + \frac{1}{10}t^5 \\ \frac{1}{4}t^4 \end{bmatrix}$$

Then the general solution is

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \left(\begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} t \right) + \begin{bmatrix} \frac{1}{6}t^6 + \frac{1}{10}t^5 \\ \frac{1}{4}t^4 \end{bmatrix}$$

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THE MATRIX EXPONENTIAL

■ 1. Use an inverse Laplace transform to calculate the matrix exponential.

$$A = \begin{bmatrix} 1 & -2 \\ 3 & -1 \end{bmatrix}$$

Solution:

First, we'll find $sI - A$.

$$sI - A = s\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ 3 & -1 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ 3 & -1 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s-1 & 2 \\ -3 & s+1 \end{bmatrix}$$

Then we'll find the inverse of this matrix, by changing $[sI - A \,|\, I]$ into $[I \,|\, (sI - A)^{-1}]$.

$$\left[\begin{array}{cc|cc} s-1 & 2 & 1 & 0 \\ -3 & s+1 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & \frac{2}{s-1} & \frac{1}{s-1} & 0 \\ -3 & s+1 & 0 & 1 \end{array}\right]$$

$$\to \left[\begin{array}{cc|cc} 1 & \frac{2}{s-1} & \frac{1}{s-1} & 0 \\ 0 & \frac{5+s^2}{s-1} & \frac{3}{s-1} & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & \frac{2}{s-1} & \frac{1}{s-1} & 0 \\ 0 & 1 & \frac{3}{5+s^2} & \frac{s-1}{5+s^2} \end{array}\right]$$

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$$\rightarrow \begin{bmatrix} 1 & 0 & | & \frac{s+1}{5+s^2} & \frac{-2}{5+s^2} \\ 0 & 1 & | & \frac{3}{5+s^2} & \frac{s-1}{5+s^2} \end{bmatrix}$$

Now we can apply the inverse Laplace transform to this inverse matrix,

$$(sI-A)^{-1} = \begin{bmatrix} \frac{s+1}{5+s^2} & \frac{-2}{5+s^2} \\ \frac{3}{5+s^2} & \frac{s-1}{5+s^2} \end{bmatrix} = \begin{bmatrix} \frac{s}{s^2+5} + \frac{1}{s^2+5} & \frac{-2}{s^2+5} \\ \frac{3}{s^2+5} & \frac{s}{s^2+5} - \frac{1}{s^2+5} \end{bmatrix}$$

$$\mathscr{L}^{-1}((sI-A)^{-1}) = \begin{bmatrix} \cos(\sqrt{5}t) + \frac{1}{\sqrt{5}}\sin(\sqrt{5}t) & -\frac{2}{\sqrt{5}}\sin(\sqrt{5}t) \\ \frac{3}{\sqrt{5}}\sin(\sqrt{5}t) & \cos(\sqrt{5}t) - \frac{1}{\sqrt{5}}\sin(\sqrt{5}t) \end{bmatrix}$$

and then this result is the matrix exponential.

$$e^{At} = \begin{bmatrix} \cos(\sqrt{5}t) + \frac{1}{\sqrt{5}}\sin(\sqrt{5}t) & -\frac{2}{\sqrt{5}}\sin(\sqrt{5}t) \\ \frac{3}{\sqrt{5}}\sin(\sqrt{5}t) & \cos(\sqrt{5}t) - \frac{1}{\sqrt{5}}\sin(\sqrt{5}t) \end{bmatrix}$$

■ 2. Use the matrix exponential to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 1 & -1 \\ -3 & 3 \end{bmatrix} \vec{x}$$

Solution:

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First, we'll find $sI - A$.

$$sI - A = s\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ -3 & 3 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ -3 & 3 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s-1 & 1 \\ 3 & s-3 \end{bmatrix}$$

Then we'll find the inverse of this matrix, by changing $[sI - A \,|\, I]$ into $[I \,|\, (sI - A)^{-1}]$.

$$\left[\begin{array}{cc|cc} s-1 & 1 & 1 & 0 \\ 3 & s-3 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & \frac{1}{s-1} & \frac{1}{s-1} & 0 \\ 3 & s-3 & 0 & 1 \end{array}\right]$$

$$\to \left[\begin{array}{cc|cc} 1 & \frac{1}{s-1} & \frac{1}{s-1} & 0 \\ 0 & \frac{s^2-4s}{s-1} & -\frac{3}{s-1} & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & \frac{1}{s-1} & \frac{1}{s-1} & 0 \\ 0 & 1 & \frac{-3}{s(s-4)} & \frac{s-1}{s(s-4)} \end{array}\right]$$

$$\to \left[\begin{array}{cc|cc} 1 & 0 & \frac{3}{4s} + \frac{1}{4(s-4)} & \frac{-1}{s(s-4)} \\ 0 & 1 & \frac{-3}{s(s-4)} & \frac{s-1}{s(s-4)} \end{array}\right]$$

Now we can apply the inverse Laplace transform to this inverse matrix,

$$(sI - A)^{-1} = \begin{bmatrix} \frac{3}{4s} + \frac{1}{4(s-4)} & \frac{-1}{s(s-4)} \\ \frac{-3}{s(s-4)} & \frac{s-1}{s(s-4)} \end{bmatrix} = \begin{bmatrix} \frac{3}{4s} + \frac{1}{4(s-4)} & \frac{1}{4s} - \frac{1}{4(s-4)} \\ \frac{3}{4s} - \frac{3}{4(s-4)} & \frac{1}{4s} + \frac{3}{4(s-4)} \end{bmatrix}$$

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$$\mathscr{L}^{-1}((sI-A)^{-1}) = \begin{bmatrix} \frac{3}{4}+\frac{1}{4}e^{4t} & \frac{1}{4}-\frac{1}{4}e^{4t} \\ \frac{3}{4}-\frac{3}{4}e^{4t} & \frac{1}{4}+\frac{3}{4}e^{4t} \end{bmatrix}$$

and then this result is the matrix exponential.

$$e^{At} = \begin{bmatrix} \frac{3}{4}+\frac{1}{4}e^{4t} & \frac{1}{4}-\frac{1}{4}e^{4t} \\ \frac{3}{4}-\frac{3}{4}e^{4t} & \frac{1}{4}+\frac{3}{4}e^{4t} \end{bmatrix}$$

Now that we have the matrix exponential, we can say that the general solution to the homogeneous system is

$$\vec{x} = e^{At}C$$

$$\vec{x} = \begin{bmatrix} \frac{3}{4}+\frac{1}{4}e^{4t} & \frac{1}{4}-\frac{1}{4}e^{4t} \\ \frac{3}{4}-\frac{3}{4}e^{4t} & \frac{1}{4}+\frac{3}{4}e^{4t} \end{bmatrix} \begin{bmatrix} 4c_1 \\ 4c_2 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} (3c_1+c_2)+(c_1-c_2)e^{4t} \\ (3c_1+c_2)-3(c_1-c_2)e^{4t} \end{bmatrix}$$

$$\vec{x} = (3c_1+c_2)\begin{bmatrix} 1 \\ 1 \end{bmatrix} + (c_1-c_2)\begin{bmatrix} e^{4t} \\ -3e^{4t} \end{bmatrix}$$

Simplifying the constants, we get

$$\vec{x} = c_1\begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2\begin{bmatrix} 1 \\ -3 \end{bmatrix} e^{4t}$$

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■ 3. Use the matrix exponential to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 0 & -1 \\ -3 & 2 \end{bmatrix} \vec{x}$$

Solution:

First, we'll find $sI - A$.

$$sI - A = s\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & -1 \\ -3 & 2 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & -1 \\ -3 & 2 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 1 \\ 3 & s-2 \end{bmatrix}$$

Then we'll find the inverse of this matrix, by changing $[sI - A \,|\, I]$ into $[I \,|\, (sI - A)^{-1}]$.

$$\left[\begin{array}{cc|cc} s & 1 & 1 & 0 \\ 3 & s-2 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & \frac{1}{s} & \frac{1}{s} & 0 \\ 3 & s-2 & 0 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cc|cc} 1 & \frac{1}{s} & \frac{1}{s} & 0 \\ 0 & \frac{s^2-2s-3}{s} & -\frac{3}{s} & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & \frac{1}{s} & \frac{1}{s} & 0 \\ 0 & 1 & \frac{-3}{(s+1)(s-3)} & \frac{s}{(s+1)(s-3)} \end{array}\right]$$

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$$\rightarrow \begin{bmatrix} 1 & 0 & | & \frac{1}{s}+\frac{3}{s(s+1)(s-3)} & \frac{-1}{(s+1)(s-3)} \\ 0 & 1 & | & \frac{-3}{(s+1)(s-3)} & \frac{s}{(s+1)(s-3)} \end{bmatrix}$$

Now we can apply the inverse Laplace transform to this inverse matrix,

$$(sI-A)^{-1} = \begin{bmatrix} \frac{1}{s}+\frac{3}{s(s+1)(s-3)} & \frac{-1}{(s+1)(s-3)} \\ \frac{-3}{(s+1)(s-3)} & \frac{s}{(s+1)(s-3)} \end{bmatrix} = \begin{bmatrix} \frac{\frac{3}{4}}{s+1}+\frac{\frac{1}{4}}{s-3} & \frac{\frac{1}{4}}{s+1}+\frac{-\frac{1}{4}}{s-3} \\ \frac{\frac{3}{4}}{s+1}+\frac{-\frac{3}{4}}{s-3} & \frac{\frac{1}{4}}{s+1}+\frac{\frac{3}{4}}{s-3} \end{bmatrix}$$

$$\mathscr{L}^{-1}((sI-A)^{-1}) = \begin{bmatrix} \frac{3}{4}e^{-t}+\frac{1}{4}e^{3t} & \frac{1}{4}e^{-t}-\frac{1}{4}e^{3t} \\ \frac{3}{4}e^{-t}-\frac{3}{4}e^{3t} & \frac{1}{4}e^{-t}+\frac{3}{4}e^{3t} \end{bmatrix}$$

and then this result is the matrix exponential.

$$e^{At} = \begin{bmatrix} \frac{3}{4}e^{-t}+\frac{1}{4}e^{3t} & \frac{1}{4}e^{-t}-\frac{1}{4}e^{3t} \\ \frac{3}{4}e^{-t}-\frac{3}{4}e^{3t} & \frac{1}{4}e^{-t}+\frac{3}{4}e^{3t} \end{bmatrix}$$

Now that we have the matrix exponential, we can say that the general solution to the homogeneous system is

$$\vec{x} = e^{At}C$$

$$\vec{x} = \begin{bmatrix} \frac{3}{4}e^{-t}+\frac{1}{4}e^{3t} & \frac{1}{4}e^{-t}-\frac{1}{4}e^{3t} \\ \frac{3}{4}e^{-t}-\frac{3}{4}e^{3t} & \frac{1}{4}e^{-t}+\frac{3}{4}e^{3t} \end{bmatrix} \begin{bmatrix} 4c_1 \\ 4c_2 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} (3c_1+c_2)e^{-t}+(c_1-c_2)e^{3t} \\ (3c_1+c_2)e^{-t}-3(c_1-c_2)e^{3t} \end{bmatrix}$$

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$$\vec{x} = (3c_1 + c_2)\begin{bmatrix} e^{-t} \\ e^{-t} \end{bmatrix} + (c_1 - c_2)\begin{bmatrix} e^{3t} \\ -3e^{3t} \end{bmatrix}$$

Simplifying the constants, we get

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ -3 \end{bmatrix} e^{3t}$$

■ 4. Use the matrix exponential to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 6t^2 - te^{-3t} \\ \sin t \end{bmatrix}$$

Solution:

First, we'll find $sI - A$.

$$sI - A = s\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s+2 & 0 \\ 0 & s-1 \end{bmatrix}$$

Then we'll find the inverse of this matrix, by changing $[sI - A \,|\, I]$ into $[I \,|\, (sI - A)^{-1}]$.

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$$\left[\begin{array}{cc|cc} s+2 & 0 & 1 & 0 \\ 0 & s-1 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{s+2} & 0 \\ 0 & 1 & 0 & \frac{1}{s-1} \end{array}\right]$$

Now we can apply the inverse Laplace transform to this inverse matrix,

$$(sI-A)^{-1} = \begin{bmatrix} \frac{1}{s+2} & 0 \\ 0 & \frac{1}{s-1} \end{bmatrix}$$

$$\mathscr{L}^{-1}((sI-A)^{-1}) = \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{t} \end{bmatrix}$$

and then this result is the matrix exponential.

$$e^{At} = \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{t} \end{bmatrix}$$

Now that we have the matrix exponential, we can say that the complementary solution will be

$$\overrightarrow{x_c} = e^{At}C$$

$$\overrightarrow{x_c} = \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$\overrightarrow{x_c} = \begin{bmatrix} c_1 e^{-2t} \\ c_2 e^{t} \end{bmatrix}$$

To find the particular solution, we'll need e^{-As}, which we find by making the substitution $t = s$ into e^{At},

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$$e^{As} = \begin{bmatrix} e^{-2s} & 0 \\ 0 & e^{s} \end{bmatrix}$$

and then calculating the inverse of this resulting e^{As}.

$$\left[\begin{array}{cc|cc} e^{-2s} & 0 & 1 & 0 \\ 0 & e^{s} & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & e^{2s} & 0 \\ 0 & 1 & 0 & e^{-s} \end{array}\right]$$

So e^{-As} is given by this resulting matrix.

$$e^{-As} = \begin{bmatrix} e^{2s} & 0 \\ 0 & e^{-s} \end{bmatrix}$$

We'll find $F(s)$ by substituting $t = s$ into F.

$$F(s) = \begin{bmatrix} 6s^2 - se^{-3s} \\ \sin s \end{bmatrix}$$

Therefore, the particular solution will be

$$\overrightarrow{x_p} = e^{At} \int_{t_0}^{t} e^{-As} F(s)\, ds$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{t} \end{bmatrix} \int_0^t \begin{bmatrix} e^{2s} & 0 \\ 0 & e^{-s} \end{bmatrix} \begin{bmatrix} 6s^2 - se^{-3s} \\ \sin s \end{bmatrix} ds$$

$$\overrightarrow{x_p} = \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{t} \end{bmatrix} \int_0^t \begin{bmatrix} 6s^2 e^{2s} - se^{-s} \\ e^{-s} \sin s \end{bmatrix} ds$$

Now we'll integrate and then evaluate on $[0,t]$.

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$$\overrightarrow{x_p} = \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{t} \end{bmatrix} \begin{bmatrix} e^{2t}\left(3t^2 - 3t + \frac{3}{2}\right) + e^{-t}(t+1) - \frac{5}{2} \\ -\frac{1}{2}e^{-t}(\sin t + \cos t) + \frac{1}{2} \end{bmatrix}$$

$$\overrightarrow{x_p} = \begin{bmatrix} 3t^2 - 3t + \frac{3}{2} + e^{-3t}(t+1) - \frac{5}{2}e^{-2t} \\ -\frac{1}{2}(\sin t + \cos t) + \frac{1}{2}e^{t} \end{bmatrix}$$

Then the general solution is the sum of the complementary and particular solutions.

$$\overrightarrow{x} = \overrightarrow{x_c} + \overrightarrow{x_p}$$

$$\overrightarrow{x} = \begin{bmatrix} c_1 e^{-2t} \\ c_2 e^{t} \end{bmatrix} + \begin{bmatrix} 3t^2 - 3t + \frac{3}{2} + e^{-3t}(t+1) - \frac{5}{2}e^{-2t} \\ -\frac{1}{2}(\sin t + \cos t) + \frac{1}{2}e^{t} \end{bmatrix}$$

■ 5. Use the matrix exponential to find the general solution to the system.

$$\overrightarrow{x}' = \begin{bmatrix} -1 & 0 & 0 \\ -2 & 0 & 3 \\ 3 & -1 & -4 \end{bmatrix} \overrightarrow{x}$$

Solution:

First, we'll find $sI - A$.

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$$sI - A = s\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} - \begin{bmatrix}-1 & 0 & 0\\ -2 & 0 & 3\\ 3 & -1 & -4\end{bmatrix}$$

$$sI - A = \begin{bmatrix}s & 0 & 0\\ 0 & s & 0\\ 0 & 0 & s\end{bmatrix} - \begin{bmatrix}-1 & 0 & 0\\ -2 & 0 & 3\\ 3 & -1 & -4\end{bmatrix}$$

$$sI - A = \begin{bmatrix}s+1 & 0 & 0\\ 2 & s & -3\\ -3 & 1 & s+4\end{bmatrix}$$

Then we'll find the inverse of this matrix, by changing $[sI - A \,|\, I]$ into $[I \,|\, (sI - A)^{-1}]$.

$$\left[\begin{array}{ccc|ccc} s+1 & 0 & 0 & 1 & 0 & 0\\ 2 & s & -3 & 0 & 1 & 0\\ -3 & 1 & s+4 & 0 & 0 & 1\end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s+1} & 0 & 0\\ 2 & s & -3 & 0 & 1 & 0\\ -3 & 1 & s+4 & 0 & 0 & 1\end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s+1} & 0 & 0\\ 0 & s & -3 & -\frac{2}{s+1} & 1 & 0\\ 0 & 1 & s+4 & \frac{3}{s+1} & 0 & 0\end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s+1} & 0 & 0\\ 0 & 1 & s+4 & \frac{3}{s+1} & 0 & 1\\ 0 & s & -3 & -\frac{2}{s+1} & 1 & 0\end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s+1} & 0 & 0\\ 0 & 1 & s+4 & \frac{3}{s+1} & 0 & 1\\ 0 & 0 & -(s+1)(s+3) & -\frac{3s+2}{s+1} & 1 & -s\end{array}\right]$$

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$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s+1} & 0 & 0 \\ 0 & 1 & s+4 & \frac{3}{s+1} & 0 & 1 \\ 0 & 0 & 1 & \frac{3s+2}{(s+1)^2(s+3)} & \frac{-1}{(s+1)(s+3)} & \frac{s}{(s+1)(s+3)} \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s+1} & 0 & 0 \\ 0 & 1 & 0 & \frac{1-2s}{(s+1)^2(s+3)} & \frac{s+4}{(s+1)(s+3)} & \frac{3}{(s+1)(s+3)} \\ 0 & 0 & 1 & \frac{3s+2}{(s+1)^2(s+3)} & \frac{-1}{(s+1)(s+3)} & \frac{s}{(s+1)(s+3)} \end{array}\right]$$

Now we can apply the inverse Laplace transform to this inverse matrix,

$$(sI-A)^{-1} = \begin{bmatrix} \frac{1}{s+1} & 0 & 0 \\ \frac{-2s+1}{(s+1)^2(s+3)} & \frac{s+4}{(s+1)(s+3)} & \frac{3}{(s+1)(s+3)} \\ \frac{3s+2}{(s+1)^2(s+3)} & \frac{-1}{(s+1)(s+3)} & \frac{s}{(s+1)(s+3)} \end{bmatrix}$$

$$(sI-A)^{-1} = \begin{bmatrix} \frac{1}{s+1} & 0 & 0 \\ \frac{3}{2(s+1)^2} - \frac{7}{4(s+1)} + \frac{7}{4(s+3)} & \frac{3}{2(s+1)} - \frac{1}{2(s+3)} & \frac{3}{2(s+1)} - \frac{3}{2(s+3)} \\ -\frac{1}{2(s+1)^2} + \frac{7}{4(s+1)} - \frac{7}{4(s+3)} & -\frac{1}{2(s+1)} + \frac{1}{2(s+3)} & -\frac{1}{2(s+1)} + \frac{3}{2(s+3)} \end{bmatrix}$$

$$\mathscr{L}^{-1}((sI-A)^{-1}) = \begin{bmatrix} e^{-t} & 0 & 0 \\ \frac{3}{2}te^{-t} - \frac{7}{4}e^{-t} + \frac{7}{4}e^{-3t} & \frac{3}{2}e^{-t} - \frac{1}{2}e^{-3t} & \frac{3}{2}e^{-t} - \frac{3}{2}e^{-3t} \\ -\frac{1}{2}te^{-t} + \frac{7}{4}e^{-t} - \frac{7}{4}e^{-3t} & -\frac{1}{2}e^{-t} + \frac{1}{2}e^{-3t} & -\frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t} \end{bmatrix}$$

and then this result is the matrix exponential.

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$$e^{At} = \begin{bmatrix} e^{-t} & 0 & 0 \\ \frac{3}{2}te^{-t} - \frac{7}{4}e^{-t} + \frac{7}{4}e^{-3t} & \frac{3}{2}e^{-t} - \frac{1}{2}e^{-3t} & \frac{3}{2}e^{-t} - \frac{3}{2}e^{-3t} \\ -\frac{1}{2}te^{-t} + \frac{7}{4}e^{-t} - \frac{7}{4}e^{-3t} & -\frac{1}{2}e^{-t} + \frac{1}{2}e^{-3t} & -\frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t} \end{bmatrix}$$

Now that we have the matrix exponential, we can say that the general solution will be

$$\vec{x} = \begin{bmatrix} e^{-t} & 0 & 0 \\ \frac{3}{2}te^{-t} - \frac{7}{4}e^{-t} + \frac{7}{4}e^{-3t} & \frac{3}{2}e^{-t} - \frac{1}{2}e^{-3t} & \frac{3}{2}e^{-t} - \frac{3}{2}e^{-3t} \\ -\frac{1}{2}te^{-t} + \frac{7}{4}e^{-t} - \frac{7}{4}e^{-3t} & -\frac{1}{2}e^{-t} + \frac{1}{2}e^{-3t} & -\frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$\vec{x_c} = \begin{bmatrix} c_1e^{-t} \\ \frac{3}{2}c_1te^{-t} + \left(-\frac{7}{4}c_1 + \frac{3}{2}c_2 + \frac{3}{2}c_3\right)e^{-t} + \left(\frac{7}{4}c_1 - \frac{1}{2}c_2 - \frac{3}{2}c_3\right)e^{-3t} \\ -\frac{1}{2}c_1te^{-t} + \left(\frac{7}{4}c_1 - \frac{1}{2}c_2 - \frac{1}{2}c_3\right)e^{-t} + \left(-\frac{7}{4}c_+\frac{1}{2}c_2 + \frac{3}{2}c_3\right)e^{-3t} \end{bmatrix}$$

■ 6. Use the matrix exponential to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 4 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ e^{3t} \\ 6t^2 \end{bmatrix}$$

Solution:

First, we'll find $sI - A$.

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$$sI - A = s\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} - \begin{bmatrix}0 & 1 & 4\\ 0 & 0 & -2\\ 0 & 0 & 0\end{bmatrix}$$

$$sI - A = \begin{bmatrix}s & 0 & 0\\ 0 & s & 0\\ 0 & 0 & s\end{bmatrix} - \begin{bmatrix}0 & 1 & 4\\ 0 & 0 & -2\\ 0 & 0 & 0\end{bmatrix}$$

$$sI - A = \begin{bmatrix}s & -1 & -4\\ 0 & s & 2\\ 0 & 0 & s\end{bmatrix}$$

Then we'll find the inverse of this matrix, by changing $[sI - A \,|\, I]$ into $[I \,|\, (sI - A)^{-1}]$.

$$\left[\begin{array}{ccc|ccc} s & -1 & -4 & 1 & 0 & 0\\ 0 & s & 2 & 0 & 1 & 0\\ 0 & 0 & s & 0 & 0 & 1\end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{s} & -\frac{4}{s} & \frac{1}{s} & 0 & 0\\ 0 & 1 & \frac{2}{s} & 0 & \frac{1}{s} & 0\\ 0 & 0 & 1 & 0 & 0 & \frac{1}{s}\end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{s} & 0 & \frac{1}{s} & 0 & \frac{4}{s^2}\\ 0 & 1 & 0 & 0 & \frac{1}{s} & -\frac{2}{s^2}\\ 0 & 0 & 1 & 0 & 0 & \frac{1}{s}\end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{s} & \frac{1}{s^2} & \frac{-2+4s}{s^3}\\ 0 & 1 & 0 & 0 & \frac{1}{s} & -\frac{2}{s^2}\\ 0 & 0 & 1 & 0 & 0 & \frac{1}{s}\end{array}\right]$$

Now we can apply the inverse Laplace transform to this inverse matrix,

$$(sI - A)^{-1} = \begin{bmatrix}\frac{1}{s} & \frac{1}{s^2} & -\frac{2}{s^3} + \frac{4}{s^2}\\ 0 & \frac{1}{s} & -\frac{2}{s^2}\\ 0 & 0 & \frac{1}{s}\end{bmatrix}$$

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$$\mathscr{L}^{-1}((sI-A)^{-1}) = \begin{bmatrix} 1 & t & -t^2+4t \\ 0 & 1 & -2t \\ 0 & 0 & 1 \end{bmatrix}$$

and then this result is the matrix exponential.

$$e^{At} = \begin{bmatrix} 1 & t & -t^2+4t \\ 0 & 1 & -2t \\ 0 & 0 & 1 \end{bmatrix}$$

Now that we have the matrix exponential, we can say that the complementary solution will be

$$\vec{x_c} = e^{At}C$$

$$\vec{x_c} = \begin{bmatrix} 1 & t & -t^2+4t \\ 0 & 1 & -2t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$\vec{x_c} = \begin{bmatrix} c_1 + c_2 t - c_3 t^2 + 4c_3 t \\ c_2 - 2c_3 t \\ c_3 \end{bmatrix}$$

To find the particular solution, we'll need e^{-As}, which we find by making the substitution $t = s$ into e^{At},

$$e^{As} = \begin{bmatrix} 1 & s & -s^2+4s \\ 0 & 1 & -2s \\ 0 & 0 & 1 \end{bmatrix}$$

and then calculating the inverse of this resulting e^{As}.

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$$\left[\begin{array}{ccc|ccc} 1 & s & -s^2+4s & 1 & 0 & 0 \\ 0 & 1 & -2s & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & s & 0 & 1 & 0 & s^2-4s \\ 0 & 1 & 0 & 0 & 1 & 2s \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -s & -s^2-4s \\ 0 & 1 & 0 & 0 & 1 & 2s \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

So e^{-As} is given by this resulting matrix.

$$e^{-As} = \begin{bmatrix} 1 & -s & -s^2-4s \\ 0 & 1 & 2s \\ 0 & 0 & 1 \end{bmatrix}$$

We'll find $F(s)$ by substituting $t = s$ into F.

$$F(s) = \begin{bmatrix} 0 \\ e^{3s} \\ 6s^2 \end{bmatrix}$$

Therefore, the particular solution will be

$$\overrightarrow{x_p} = e^{At} \int_{t_0}^{t} e^{-As} F(s)\, ds$$

$$\overrightarrow{x_p} = \begin{bmatrix} 1 & t & -t^2+4t \\ 0 & 1 & -2t \\ 0 & 0 & 1 \end{bmatrix} \int_0^t \begin{bmatrix} 1 & -s & -s^2-4s \\ 0 & 1 & 2s \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ e^{3s} \\ 6s^2 \end{bmatrix} \, ds$$

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$$\vec{x_p} = \begin{bmatrix} 1 & t & -t^2+4t \\ 0 & 1 & -2t \\ 0 & 0 & 1 \end{bmatrix} \int_0^t \begin{bmatrix} -se^{3s} - 6s^4 - 24s^3 \\ e^{3s} + 12s^3 \\ 6s^2 \end{bmatrix} ds$$

Now we'll integrate and then evaluate on $[0,t]$.

$$\vec{x_p} = \begin{bmatrix} 1 & t & -t^2+4t \\ 0 & 1 & -2t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -\frac{6t^5}{5} - 6t^4 + \frac{1}{9}e^{3t}(1-3t) - \frac{1}{9} \\ \frac{1}{3}(9t^4 + e^{3t} - 1) \\ 2t^3 \end{bmatrix}$$

$$\vec{x_p} = \begin{bmatrix} -\frac{t^5}{5} + 2t^4 - \frac{1}{3}t + \frac{1}{9}e^{3t} - \frac{1}{9} \\ -t^4 + \frac{1}{3}e^{3t} - \frac{1}{3} \\ 2t^3 \end{bmatrix}$$

Then the general solution is the sum of the complementary and particular solutions.

$$\vec{x} = \vec{x_c} + \vec{x_p}$$

$$\vec{x} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + \begin{bmatrix} c_2 + 4c_3 \\ -2c_3 \\ 0 \end{bmatrix} t + \begin{bmatrix} -c_3 \\ 0 \\ 0 \end{bmatrix} t^2 + \begin{bmatrix} -\frac{t^5}{5} + 2t^4 - \frac{1}{3}t + \frac{1}{9}e^{3t} - \frac{1}{9} \\ -t^4 + \frac{1}{3}e^{3t} - \frac{1}{3} \\ 2t^3 \end{bmatrix}$$

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